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Subject: Research Methodology

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Research Methodology



Research Methodology

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Description of Research Project Components'

Components of Full Research Project		
Chapter	Description	
1	Introduction	+ References The components of a Proposal
2	Literature Review	
3	Methodology	
4	Results	
5	Discussion	
6	Conclusion and Recommendations	
--	References	
--	Appendices	

Main Components of the Chapter 1

Title of Chapter 1: Introduction



Main Components of the Proposal

Chapter 1: Introduction	
Section	Description
1.1	Background (Based on Literature Review)
1.2	Statement of the Problem (Justification)
1.3	Significance (Importance of the study)
1.4	Research Question
1.5	Objectives; Main and Specific
1.6	Hypotheses
1.7	Framework of the study (Optional)
1.8	Operational definitions (Preferred)

1

Introduction to Health Research

Definition

- Research is a scientific inquiry to learn new facts, test ideas, etc. It is the systematic collection, analysis and interpretation of data to generate new knowledge and answer a certain question or solve a problem.
- Characteristics of research:
  - It demands a clear statement of the problem
  - It requires a plan (it is not aimlessly “ looking” for something in the hope that you will come across a solution)
  - It builds on existing data, using both positive and negative findings
  - New data should be collected as required and be organized in such a way that they answer the research question(s)

Types of research: Basic vs. Applied

1. Basic research is necessary to generate new knowledge and technologies for major unresolved health problems.
2. .. Applied research is necessary to identify priority problems and design and evaluate policies and programs that will deliver the greatest health benefit, making optimal use of available resources.

- Qualitative research is concerned with finding the answers to questions which begin with:
  - Why? How? In What way?
- Quantitative research is concerned with questions about:
  - How much? How many? How often? To what extent?

Quantitative vs. Qualitative

1. Quantitative: Early forms of research originated in the natural sciences, such as biology, chemistry, physics, geology, etc. and were concerned with investigating things which we could observe and measure in some way.
   - Other researchers can make observations and measurements objectively and repeatedly.
2. Qualitative: Research in the social sciences, such as psychology, sociology, anthropology, etc. They were interested in studying human behavior and the social world inhabited by humans.
   - Qualitative research is concerned with developing explanations of social phenomena.

Importance of Qualitative Research

- Public health problems are complex, not only because of their multicausality but also as a result of new and emerging domestic and international health problems.
- Social, economic, political, ethnic, environmental, and genetic factors all are associated with today's public health concerns.
- Consequently, public health practitioners and researchers recognize the need for multiple approaches to understanding problems and developing effective interventions that address contemporary public health issues.
- Qualitative methods fill a gap in the public health toolbox; they help us understand behaviors, attitudes, perceptions, and culture in a way that quantitative methods alone cannot.

Differences between Qualitative and Quantitative Research

	Quantitative	Qualitative
Research definition	The researcher attempts to understand some larger reality by isolating and measuring components of that reality without regard to their contextual setting.	The researcher attempts to understand some larger reality by examining it in a holistic way or by examining components of that reality within their contextual setting
General framework	To confirm hypothesis. Structured methods of data collection – questionnaires, structured observations	Explore phenomena. Semi-structured methods e.g. in-depth interviews, focus groups discussion etc.
Analytical objectives	To quantify variation To predict causal relationships To describe characteristics of a population	To describe variation To describe and explain relationship To describe individual experiences
Data format	Numerical	Textual

Data Collection for Qualitative Research

In-depth interviews	Focus Group Discussion
• Sampling the subjects	• Taps into human tendencies
• Other Methods: observations, documents, digital footprints, etc.	
• schedule/ guide • Ethical issues • Types of interview questions (how/what) • Record • Transcribe	perceptions are developed through interaction with other people • Uses of FGD • Size varies: 4-6 people; 6–12 people; per 1 group • Homogenous vs. Heterogenous • Familiar faces vs. strangers • Shared vs. diverse experiences • Organized vs. free flow

Qualitative Research

Advantages	Disadvantages
• In-depth examination of a phenomenon • Uses subjective information • Not limited to rigidly definable variables • Examine complex questions that can be impossible with quantitative methods • Deal with value-laden questions • Explore new areas of research • Build new theories	• Subjectivity leads to procedural problems • Replicability is very difficult • Researcher bias is built in and unavoidable • In-depth, comprehensive approach to data gathering limits scope • Labour intensive, expensive • Not understood well by 'classical' researchers

Mixed Method Research: Types of design

- Triangulation
  - Obtain different but complementary data on the same topic
  - Need for both approaches to provide a more complete picture
- Embedded
  - One data set provides a supportive, secondary role in a study based primarily on the other data type
  - To enhance the study with a second source of data
- Explanatory
  - 2-phase: qualitative data helps explain or build upon initial quantitative results
  - To explain the quantitative results
- Exploratory
  - The results of the qualitative data help develop the quantitative method
  - To first explore qualitatively

Groups of Health Research

1. Essential health research: Much of this research falls under the category of health service research, but depending on the situation, it may also involve clinical research and the development of technology.
2. Clinical research: This group of topics ranges from studies of the prevention and diagnosis of diseases through new methods of treatment to problems of care and rehabilitation.
3. Biomedical research is the most basic part of health research, demanding more resources, facilities, and skilled investigators. Its results are often of universal importance.

Health systems research – 1

- It is concerned with improving the health of a community by enhancing the efficiency and effectiveness of the health system as an integral part of the overall process of socioeconomic development.
- The institutional arrangements within which the health-seeking and health-promoting behavior may include:
  1. individual, family and the community,
  2. health care services,
  3. Health-related sector such as education and agriculture,
  4. the international sectors.

Health systems research – 2

To support decision-making at all levels that can improve the functioning of the health system, some of the essential features are summarized as follows:

1. Health systems research should focus on priority problems
2. An integrated multidisciplinary approach is required
3. The research should be participatory in nature
4. Research must be timely
5. Emphasis should be placed on comparatively simple, short-term research designs that will likely yield practical results.

2

Topic Selection

Topic Selection

- If the answer to the research question is evident, we are dealing with a management problem that may be solved without further research.
- Several research questions could be presented that may be posed at the various levels of the health system.
- Whether a problem requires research depends on three conditions:
  1. There should be a perceived difference or discrepancy between what it is and what it should be
  2. The reason(s) for this difference should be unclear (so that it makes sense to develop a research question), and
  3. There should be more than one possible and plausible answer to the question (or solution to the problem).

Criteria for prioritizing problems for research

- Each problem proposed for research has to be judged according to certain guidelines or criteria.
  - There may be several ideas to choose from.
- Before deciding on a research topic, each proposed topic must be compared with all other options.
- Those responsible for the health status of the community should be involved in selecting and analyzing the research problem.
  - This would include health service managers, healthcare workers, community leaders, and researchers.

Criteria and scales for selecting a research topic – 1

- Relevance:
  - The topic you choose should be a priority problem. Questions to be asked include:

a) Is the problem significant or widespread

b) Who is affected?

c)How severe is the problem?

- Scales for rating relevance:
  - 1 = Not relevant
  - 2 = Relevant
  - 3 = very relevant

Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 2

- Avoidance of duplication:
  - If the topic has been researched, the results should be reviewed to explore whether major questions that deserve further investigation remain unanswered.
  - If not, another topic should be chosen.

- Scales for rating avoidance of duplication:
  - 1 = Sufficient information already available
  - 2 = Some information available, but major issues not covered
  - 3 = No sound information available on which to base problem-solving

Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 3

- Feasibility:
  - Consider the complexity of the problem and the resources you will require to carry out the study
  - Thought should be given first to personnel, time, equipment and locally available money.

- Scales for rating feasibility:
  - 1 = Study not feasible considering available resources
  - 2 = Study feasible considering available resources
  - 3 = Study very feasible considering available resources

  Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 4

- Political acceptability:
  - It is advisable to research a topic that has the interest and support of the authorities.
  - This will facilitate the smooth conduct of the research and increase the chance that the results of the study will be implemented.

- Scales for rating political acceptability:
  - 1 = Topic not acceptable
  - 2 = Topic somewhat acceptable
  - 3 = Topic fully acceptable

  Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 5

- Applicability of possible results and recommendations
  - Is it likely that the recommendations from the study will be applied?

- Scales for rating applicability :
  - 1 = No chance of recommendations being implemented
  - 2 = Some chance of recommendations being implemented
  - 3 = Good chance of recommendations being implemented

Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 6

- Urgency of data needed:
  - How urgent are the results needed to make a decision?
  - Which research should be done first, and which can be done late?

- Scales for rating urgency of data needed:
  - 1 = Information not urgently needed
  - 2 = Information could be used, but a delay of some months would be acceptable
  - 3 = Data very urgently needed for decision-making

Rating Scale: 1 = Low; 2 = Medium; 3 = High

Criteria and scales for selecting a research topic – 7

- Ethical acceptability:
  - We should always consider the possibility that we may harm others while conducting research.

- Scales for rating ethical acceptability:
  - 1 = Major ethical problems
  - 2 = Minor ethical problems
  - 3 = No ethical problems

  Rating Scale: 1 = Low; 2 = Medium; 3 = High

N.B. The above rating should be based on existing data, not on assumptions.

Rating Sheet (Total score should be ≥ 14)

Criteria for selecting topic	Proposed Health Topic		
	Problem I	Problem II	Problem
Relevance			
Avoidance of			
Feasibility			
Political acceptability			
Applicability			
Urgency of data			
Ethical acceptability			
Total			
Rating Scale: 1 = Low; 2 = Medium; 3 = High			

Select the TITLE of the study

- Complete relevant and informative
- It contains:
  - Name of disease or condition under investigation
  - The population of the study
  - Terms that telegraph the purpose of contents of the research
  - Design terms, such as cross-sectional, case-control, clinical trial, prevalence, risk factors or combination of two.
- You have to avoid:
  - Unnecessary words and too many details
  - Vague or confusing words
  - Undefined abbreviation
- National Institute of Health (NIH) required: Not exceeding 20 words.

3

Analysis and Statement of The Problem

After identifying, selecting and analyzing the problem, the next major section in a research proposal is “ ”

Statement of the Problem

- A statement of the problem is used in research as a claim outlining the issue addressed by a study. A good research problem should address an existing gap in knowledge in the field and lead to further research.
- The ultimate goal of a statement of the problem is to transform a generalized problem (something that bothers you, a perceived lack) into a targeted, well-defined problem that can be resolved through focused research and careful decision-making.
- Writing a statement of the problem should help you clearly identify the purpose of your proposed research project. Often, the statement of the problem will also serve as the basis for the introductory section of your final proposal, directing your reader's attention quickly to the issues that your proposed project will address and providing the reader with a concise statement of the proposed project itself.

Analyzing the problem

- A systematic analysis of the problem, completed jointly by the researchers, health workers, managers, and community representatives, is a very crucial step in designing the research because it:

1. Enables those concerned to bring together their knowledge of the problem,
2. Clarifies the problem and the possible factors that may be contributing to it,
3. Facilitates decisions concerning the focus and scope of the research.

- The following are the steps for formulating the problem statement:

1. Why is it important to state and define the problem well?

- Because a clear statement of the problem:
  1. Is the foundation for the further development of the research proposal (research objectives, methodology, work plan, etc);
  2. Makes it easier to find information and reports of similar studies from which your study design can benefit;
  3. Enables the researcher to systematically point out why the proposed research on the problem should be undertaken and what you hope to achieve with the study results.

2. Points that need to be considered for justifying the selected research problem

- A health problem selected to be studied has to be justified in terms of its:
  1. Being a current and existing problem which needs a solution
  2. Being a widely spread problem affecting a target population
  3. Effects on the health service programmes
  4. This is a problem that concerns planners, policy makers, and the communities at large.

3. Information included in the statement of a problem – 1

- A brief description of socioeconomic and cultural characteristics and an overview of health status.
- A more detailed description of the nature of the problem
  - A basic description of the research problem
  - The discrepancy between what is and what should be
  - Its size, distribution, and severity (who is affected, where, since when, etc.)
- A brief description of any solutions that have been tried in the past, how well they have worked, and why further research is needed.

3. Information included in the statement of a problem – 2

An analysis of the significant factors that may influence the problem and a convincing argument that available knowledge is insufficient to answer a specific question and to update the previous knowledge.

A description of the type of information expected to result from the project and how this information will be used to help solve the problem.

If necessary, a short list of definitions of crucial concepts used in the statement of the problem.

Each point included in the problem statement should be referenced.

Steps in Defining Research Problem – 1

1.Start with a general statement of the problem

At this stage, the research problem is stated broadly, taking into consideration scientific or intellectual interest.

A researcher should, at this stage, read thoroughly the area of interest.

Some preliminary surveys might be, and the researcher can also seek expert help. necessary

2.Understand the nature of the problem

This step helps to understand the origin and nature of the problem.

The researcher should analyze what made him raise the topic or discuss the problem with those who know more about it.

Steps in Defining Research Problem – 2

3.Surveying the available literature

This process is also known as literature review where the researcher goes through all the relevant journal, books, newspaper articles, this will help him know if there are certain gaps in the theory, or whether the existing theories applicable to the problem under study are consistent with each other, or whether the findings of different study do not follow a pattern consistent with the theoretical expectation.

4.Developing ideas through discussion

Discussion helps in discovering new ideas and developing different perceptions of the problem being looked at.

Discuss the problem with colleagues and other experts in the same area.

Steps in Defining Research Problem – 3

5. Rephrasing the research problem

Finally, the researcher should rephrase the problem into a working condition.

Once the nature of the problem has been understood, rephrasing the problem should be easy.

By rephrasing, the researcher puts the research problem in as specific a term as possible so that it becomes operational and helps develop the hypothesis.

4

Literature Review

Use of literature review

Literature review gives the chance to review different titles and guide the researcher to new ideas.

It prevents you from duplicating work that has been done before.

It increases your knowledge of the problem you want to study, and this may assist you in refining your "statement of the problem".

Literature review helps the researcher:

- Select the proper study method and avoid mistakes from previous studies.
- To justify the selection of the research project.
- To be familiar with different research methods.

Sources of information

1. Card catalogues of books in libraries
2. Organizations (institutions)
3. Published information (books, journals, etc.)
4. Unpublished documents (studies in related fields, reports, etc.)
5. Computer-based literature searches such as Medline, PubMed, Scopus, Cochrane Library
6. Opinions, beliefs of key persons

References that are identified:

7. Should first be skimmed or read
8. Then summaries of the critical information in each of the references may be recorded on separate index cards
9. The references might be categorized to make retrieving the material and the information easier.

Organizing the information on index cards

Index cards are used to organize information according to the following methods:

Use keywords.

Write a summary of the contents of books or articles that are relevant to one's study.

Analyze briefly the content, with comments such as: how information from that particular study could be used in one's study.

Information obtained from key persons could also be summarized on the index card.

After collecting the required information on index cards, the researcher should decide in which order they want to discuss previous research findings: from global to local, from broader to focused, and from past to current.

Some examples of resources where information could be obtained

Clinic and hospital-based data from routine activity statistics

Local surveys, annual reports

Scientific conferences

Statistics issued at region and district levels

Articles from national and international journals

Internet

Documentation, reports, and raw data from the Ministry of Health, Palestinian Bureau of Central Statistics (PBCS)

Offices, Nongovernmental organizations, etc.

Others.

Importance of Literature Review – 1

The primary purpose of the literature review is to determine what has been already related to the research being studied, to avoid unnecessary and unintentional duplication and to form the framework within which the research findings are to be interpreted.

A review of the literature will reveal what strategies, procedures, and measuring instruments have been found useful in investigating the problem at hand.

This information helps one avoid mistakes that other researchers have made and helps to benefit from other researchers' experiences.

Importance of Literature Review – 2

In most cases, the literature review will suggest other procedures and approaches.

This is beneficial information because a researcher could try out suggested approaches, significantly if they will improve the research study.

It enables the researcher to be familiar with previous studies and thus facilitate the interpretation of the study's results.

Reviewing the literature critically will provide the foundation for your research.

The primary purpose of the literature review is to help you develop a good understanding and insight into relevant previous research and emerging trends.

Importance of Literature Review – 3

The precise purpose of your reading of the literature will depend on the approach you intend to use in your research.

For some research projects, you will use the literature to help you identify theories and ideas and test them using data.

This is known as a deductive approach, in which you develop a theoretical or conceptual framework, which you subsequently test using data.

For another research project, you will plan to explore your data and develop theories from them that you will subsequently relate to the literature.

Importance of Literature Review – 4

This is known as the inductive approach. Although your research still has a clearly defined purpose with the research question(s) and objectives, you do not start with any predetermined or conceptual frameworks.

We believe such an approach cannot be taken without a conceptual knowledge of your subject area.

It is impossible to review every piece of the literature before collecting your data.

The purpose of your literature review is not to summarise everything written on your research topic but to review the most relevant and significant research on your subject.

In conclusion, review of literatures

While reviewing a literature, all what is known about the study topic should be summarized with the relevant references. This review should answer the following:

- How much is known?
- What is not known?
- What should be done based on what is lacking?

Overall, the literature review should be adequate, relevant and critical.

In addition to this, appropriate referencing procedures should always be followed in research proposals and reports.

While reviewing the literature, emphasize both positive and negative findings and avoid any information distortion to suit your study objectives.

Finally, after an exhaustive literature review, summarize the findings and write a coherent discussion by indicating the research gap supporting your study's undertaking.

Steps of reviewing literature

This is a search for published / unpublished work and experts in the area to guide you in planning your study.

The literature review should be incorporated into the introduction, methods, and discussion.

Steps of reviewing the literature:

- Search for current work related to the problem of interest
- Evaluate the obtained articles for relevance, applicability, and critical appraisal
- Identify possible methods others used to study the problem
- Always acknowledge the source of your ideas as references

(See literature review ppt 2)

Theoretical Framework – 1

After conducting the interviews, completing a literature survey, and defining the problem, one is ready to develop a theoretical framework.

A theoretical framework is a conceptual model of how one theory makes logical sense of the relationships among the factors identified as necessary to the problem.

The theoretical framework discusses the interrelationships among the variables deemed integral to the dynamics of the situation being investigated.

Theoretical Framework – 2

Developing such a framework helps us postulate or hypothesize and test certain relationships, thus improving our understanding of the situation's dynamic.

From the theoretical framework, testable hypotheses can be developed to examine whether the theory formulated is valid or not.

Thus, the entire research rests based on the theoretical framework.

A good theoretical research is central to examining the problem under investigation.

Conceptual Framework

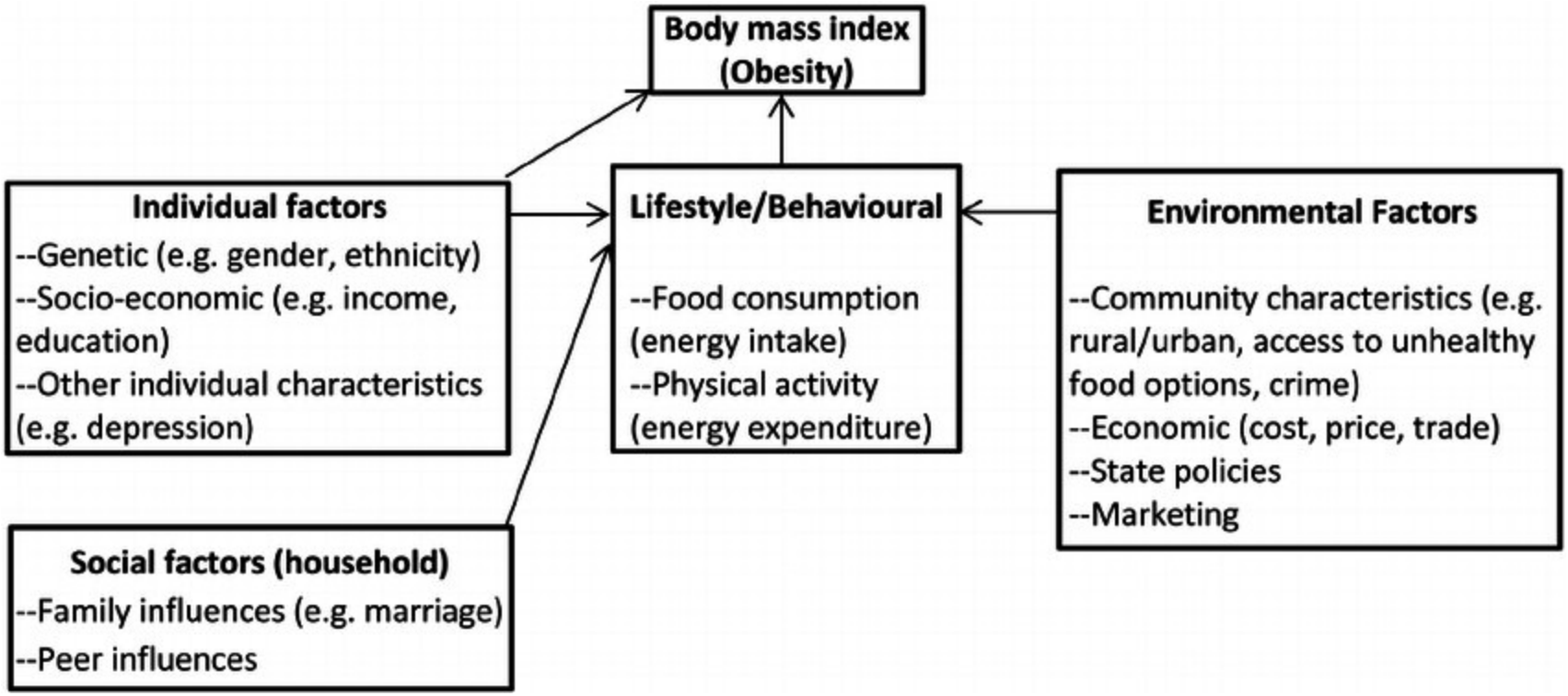
A conceptual framework is defined as:

a set of broad ideas and principles taken from relevant fields of enquiry and used to structure a subsequent presentation.

It explains, either graphically or in narrative form, the main dimensions being studied or their presumed relationships.

It is a framework showing the relationship between the independent and dependent variables.

Example Conceptual Framework



Referencing: in-text citation uses author name and year of publication

Text: A certain level of ROS generation is physiological and required for optimal cellular functioning (Pizzino et al., 2017).

References:

APA: https://youtu.be/qZVIa2sTbpM (PLZ See the video)

Pizzino, G., Irrera, N., Cucinotta, M., Pallio, G., Mannino, F., Arcoraci, V., … Bitto, A. (2017). Oxidative Stress: Harms and Benefits for Human Health. Oxidative Medicine and Cellular Longevity, 2017, 1–13. doi:10.1155/2017/8416763

Harvard:

Pizzino, G. et al., 2017. Oxidative Stress: Harms and Benefits for Human Health. Oxidative Medicine and Cellular Longevity, 2017, pp.1–13. Available at: http://dx.doi.org/10.1155/2017/8416763.

Referencing: in-text citation uses numerical characters

Text: A certain level of ROS generation is physiological and required for optimal cellular functioning [1].

References:

IEEE:

[1] G. Pizzino, N. Irrera, M. Cucinotta, G. Pallio, F. Mannino, V. Arcoraci, F. Squadrito, D. Altavilla, and A. Bitto, “Oxidative Stress: Harms and Benefits for Human Health,” Oxidative Medicine and Cellular Longevity, vol. 2017, pp. 1–13, 2017.

Vancouver:

1. Pizzino G, Irrera N, Cucinotta M, Pallio G, Mannino F, Arcoraci V, et al. Oxidative Stress: Harms and Benefits for Human Health. Oxidative Medicine and Cellular Longevity [Internet]. Hindawi Limited; 2017; 2017:1–13. Available from: http://dx.doi.org/10.1155/2017/8416763.

How to Cite in APA Format (7th edition)

https://www.scribbr.com/category/apa-style/

This citation guide outlines the most important citation guidelines from the 7th edition of the APA Publication Manual (2020).

APA in-text citations

The basics

In-text citations are brief references in the running text that direct readers to the reference entry at the end of the paper. You include them when quoting or paraphrasing someone else's ideas or words to avoid plagiarism.

How to Cite in APA Format (7th edition) cont.

An APA in-text citation consists of the author's last name and the year of publication (also known as the author-date system). If you're citing a specific part of a source, you should also include a locator such as a page number or timestamp. For example: (Smith, 2020, p. 170).

Parenthetical vs. narrative citation

The in-text citation can take two forms: parenthetical and narrative. Both types are generated automatically when citing a source with Scribbr's APA Citation Generator.

Parenthetical citation: According to new research … (Smith, 2020).

Narrative citation: Smith (2020) notes that …

How to Cite in APA Format (7th edition) cont.

Multiple authors and corporate authors

The in-text citation changes slightly when a source has multiple authors or an organization as an author. Pay attention to punctuation and the use of the ampersand (&) symbol.

Author type	Parenthetical citation	Narrative citation
One author	(Smith, 2020)	Smith (2020)
Two authors	(Smith & Jones, 2020)	Smith and Jones (2020)
Three or more authors	(Smith et al., 2020)	Smith et al. (2020)
Organization	(Scribbr, 2020)	Scribbr (2020)

Reference examples

Citing a source starts with choosing the correct reference format. Use Scribbr's Citation Example Generator to learn more about the format for the most common source types. Pay close attention to punctuation, capitalization, and italicization.

Reference entry

Andreff, W., Staudohar, P. D., & Streefkerk, R. (2000). The evolving European model of professional sports finance. Journal of Sports Economics, 1(3), 257–276. https://doi.org/10.1177/152700250000100304

Reference examples cont.

Andreff, W., & Staudohar, P. D. (2000). The evolving European model of professional sports finance. Journal of Sports Economics, 1(3), 257–276. https://doi.org/10.1177/152700250000100304

Organization

Deloitte. (2000). The evolving European model of professional sports finance. Journal of Sports Economics, 1(3), 257–276. https://doi.o

Organization. (Year of publication). Title of article. Title of Journal, Volume(Issue), Pages. https://doi.org/DOIrg/10.1177/152700250000100304

Reference examples cont.

More than three

Parenthetical: (Author's Last Name et al., Year of publication) Narrative: Author's Last Name et al. (Year of publication)

Apply these formatting guidelines to the APA reference page:

Double spacing (within and between references)

Hanging indent of ½ inch

Legible font (e.g. Times New Roman 12 or Arial 11)

Page number in the top proper header

Repositories of research

1. CrossRef: https://search.crossref.org/
   - Search based on the DOI (Digital Object Identifier)
   - Provide different options for in-text citations
2. Google Scholar: https://scholar.google.com/
   - Search based on the keywords (majorly) and DOI
   - Provide different options for in-text citations
3. PubMed: https://pubmed.ncbi.nlm.nih.gov/
   - Search based on the DOI and keywords
   - Provide ONE option for in-text citations
4. Semantic Scholar: https://www.semanticscholar.org/
   - Search based on the keywords only
   - Provide few options for in-text citations

Recommended Citation Software

1. Mendeley Reference Manager
   - Available as free
   - Allows to create self-library (saved online or on the desktop)
   - Accept importing citations from Google Scholar
   - Easy to set and to be used and modifiable
   - https://www.mendeley.com/download-desktop-new/
2. EndNote | The best reference management tool
   - Not free
   - Allows to create self-library (saved on the desktop)
   - Easy to be used (after training) and accept modifications
   - Very Easy to import citations from Google Scholar
   - https://endnote.com/

5

Ethics in Medical Research

Seven requirements for ethical research – 1

1. Societal/scientific value: improve health or increase knowledge
2. Scientific validity: using approved scientific concepts and methods to provide valid and trustworthy data that is scientifically sound and completed by qualified, trained, and experienced individuals
3. Fair subject selection: vulnerable subjects are not targeted for risky research
4. Favorable risk/benefit ratio: minimize risks, enhance potential benefits

Seven requirements for ethical research – 2

4. Respect for subjects: protection of privacy/ confidentiality {identified by number and initial}, ability to withdraw, monitoring of well-being {adverse reactions, emergencies, pregnancy}, informed of new information regarding research
5. Informed consent: provide adequate information to patients for voluntary decision
6. Independent review: review by an external individual not involved in the research

Informed consent

- A process when a person voluntarily confirms his participation in a particular study after being informed of all aspects relevant to the subject's decision to participate
- Adequate explanation given
- Needs to be documented
- Written format, signed, dated
- Given fact sheet
- Questions should be answered
- Given ample time to think over

Consent vs. Permission

Consent	Permission
Legally effective agreement of the subject or the subject's legally authorized representative based on information that is given to the subject or the representative in language that is understandable	Agreement of parent/ guardian to the participation of child/ ward in research

Exceptions to informed consent

- Medical emergencies
- Public health emergencies for treatment or vaccination
  - For isolation, observation or surveillance
  - To any other measures as the authorized officer considers necessary to control the disease
- Incompetence
  - Substitute decision maker
  - The doctor must always act in the best interest of the patient

Consent of subjects – 1

- Consent of subjects-children
  - Normally, children should not be involved in research that can be carried out on adults
  - Indispensable to diseases of childhood and conditions that are susceptible to children
  - Get the permission of parents/guardian: After a full explanation of aims, possible hazards, discomfort, inconvenience
- Consent of subjects' pregnant and nursing mothers
  - Should not be involved in research that jeopardize the fetus or neonate unless needed to elucidate on problems of pregnancy or lactation

Consent of subjects – 2

- Consent of subjects mentally ill/defective patients
  - Not all psychiatric illnesses are incapable of giving consent; it depends on mental capacity
  - Never involve this person if it can be done to normal people
  - Unless for research of origin and treatment of mental illnesses
  - Need to get the agreement of the immediate family of mentally incapacitated patients. E.g. spouse, parent, adult offspring
- Consent of subjects and other vulnerable groups
  - Willingness to volunteer is due to the expectation of the hierarchical structure or perceived benefits or retaliatory measures
  - Medical/nursing students
  - Members of armed forces, prisoners
  - Incurable patients, refugees, emergency

Consent of subjects – 3

- Proxy consent
  - Permissible for totally non-invasive procedure; not exposed to physical, social or psychological risks
  - Or, impractical to obtain consent
  - Alarm or harmful to the patient – e.g. medical records; Use of education test
- Verbal consent
  - In non-invasive research – e.g., a questionnaire without highly personal material
- Use of human tissues
  - Use of clinical specimens for purposes other than what it was collected for – e.g. organs, tissues, body fluids (blood & urine)
  - Use tissues collected at autopsy – e.g. placenta

Use of animals – 1

- In principle, animals:
  - should not be tortured [training of researchers and lab staff, good housing conditions]
  - should not be given excessive pain or distress,
  - should receive proper method of sacrifice and disposal of the bodies
- Clear benefits or knowledge gained from experiments using animal samples
- Should use alternative methods if possible

Use of animals – 2

- Refinement:
  - Improvement of all aspects of the lifetime experience of animals to reduce suffering and improve welfare
- Reduction:
  - The use of fewer animals without compromising scientific output and the quality of biomedical research and without compromising animal welfare
- Replacement:
  - Use of methods that permit a given scientific purpose to be achieved without the use of animals

Budget

- Identify significant categories for a budget.
- Make reasonable estimates of the expenses in various budget categories.
- List various ways a budget can be reduced, if necessary, without substantially damaging a project.
- Prepare a realistic and appropriate budget for the project proposal being developed.

6

Objectives

All terms of goals, objectives, aims, purpose, hypothesis, or research questions are accepted

Formulation of the research objectives

The formulation of objectives will help us to:

- Focus the study (narrowing it down to essentials)
- Avoid collection of data that are not strictly necessary for understanding and solving the identified problem
- Organize the study in clearly defined parts

The explicit formulation of study objectives is an essential step in the planning of a study.

It is said that "a question well-stated is a question half-answered", but a poorly stated or unstated question is unlikely to be answered at all.

Stating the objectives

We have to make sure that our objectives:

1. Cover the different aspects of the problem and its contributing factors in a coherent way and a logical sequence
2. Are clearly expressed in measurable terms
3. Are realistic considering local conditions
4. Meet the purpose of the study
5. Use action verbs that are specific enough to be measured

Use Action Verbs	Avoid Non-Action Verbs
to determine	to appreciate
to compare	to understand
to verify	to study
to calculate	to believe
to describe	
to find out	
to establish	
to identify	

Essentials before formulating the objectives – 1

1.. You must state your study objectives and determine the general and specific objectives. Investigators must differentiate between the study objectives and the other common terms used to describe the purpose and aim of the study.

a) The purpose of the study whose study objectives "What" do you intend to know or measure?

b) The study aims to determine the desired outcome, such as "Improvement of child immunization coverage."

Essentials before formulating the objectives – 2

3. The term "Research questions" is similar to the study objectives and could replace them.

   The terms "research questions" and "study objectives" could be seen for the same research but preferably not.

4. The term "Research Hypothesis" is commonly used.

   Hypothesis is "A predictive statement about relationships between independent and dependent variables".

   In the text, either it is written separately or under specific objectives.

Objectives in descriptive studies

In descriptive studies, the investigator states the characteristics he wants to measure. These characteristics could be diseases, deaths or disabilities.

The study objective could be stated generally as:

To measure the prevalence of a specific disease in population X at time Y.

Examples:

To determine the infant mortality rate in the Gaza Strip during the year 2000

To measure the incidence rate of Anemia in Rafah district during the year 2000

To determine the serum cholesterol value among schoolchildren in Jerusalem

Specific objectives could answer the research questions in specific groups, such as age, sex, ethnic groups, or special groups.

Objectives in analytic studies

In analytic studies the investigator examines relationships between exposure(s) and outcome(s).

The study objective could be:

1. To examine the association between anemia and parasitic infestations in Jenin preschool children.
2. To define risk factors associated with stroke in the Gaza area.
3. To test the hypothesis that the hospitalization rate is higher among smokers than non-smokers

7

Study Variables

Identifying variables

- What do we want to measure?

1. A variable is a characteristic/event that has different values. It is measurable.
2. This should be linked to your objectives.
3. Use your secondary, tertiary and further generation bubbles to identify areas to be studied (these are your variables).
4. Can they be measured directly or indirectly?
5. . It is important to be realistic about the number of variables you want to measure and justify why certain areas will not be studied.
6. List all variables to be studied (so that you do not miss out on any).

Identification of variables will help the investigator to:

- Specify the essential items for study.
- Figure out their interrelationships and what you need to know about them.
- Avoid unnecessary data collection.
- Ensure that all data that are relevant to the objectives are collected.
- How do we measure it?
  1. Write the definition for your variable. It must be easily understood by all and measurable
  2. Determine the type of variable. Variables can be discrete (e.g. smoking status – smoker/non-smoker) or continuous (e.g. smoking status – number of cigarettes/day).
  3. Define the scale of measurement you want to use for that variable based on the type of variable

Study Variables

- Study variables are a group of characteristics of interest that the investigator will measure.
- Classification of the study variables:

1. Numeric variables: Numbers express numeric variables. Examples of numeric variables are age by years or months, height by centimetres, weight by kilograms and income by Dollars.
2. Categorical variables: They are expressed in two or more categories. These variables include sex (male or female), Absence or presence of diseases, and satisfaction level (No, mild, moderate, high). Numeric variables could be recoded into categorical variables, whereas age in years (numeric) could be recoded into age groups (categorical).
3. Composite variable: These variables are calculated statistically based on two or more variables. For example, adiposity is measured by body mass index.

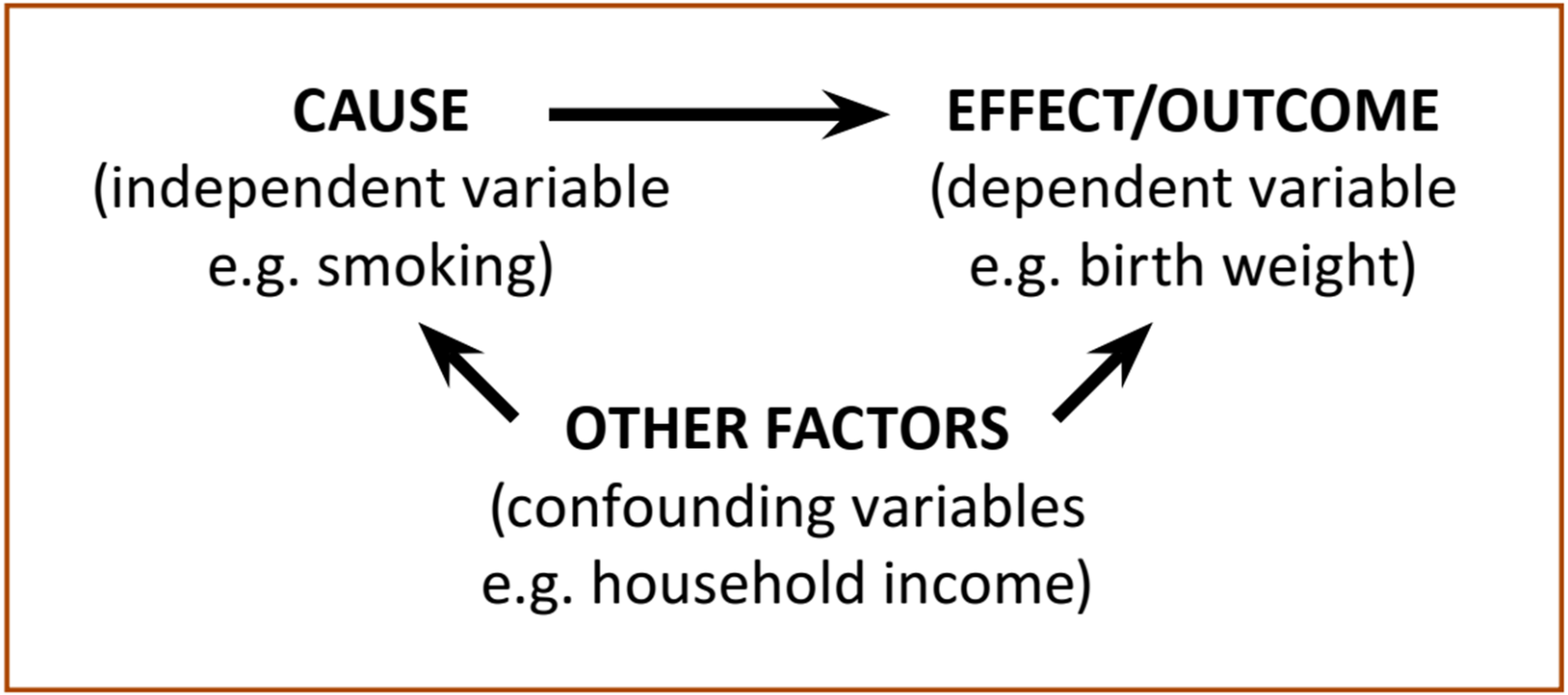
Dependent vs. Independent

- It is essential for investigators to identify correctly dependent and independent study variables.
- The dependent variable is the variable of concern for the investigator that other variables can influence.
- Independent variables are variables that may affect the dependent variable.
  - In a study of risk factors for anaemia in a specific population, anaemia is the dependent variable, while risk factors and other socio-demographic variables are independent variables.
- The dependent variable in one study could be independent in another study.
  - When we study maternal mortality, anaemia is independent, whereas maternal mortality is dependent.

Types of study variables

- Universal or background variables:
  - Examples of these variables are age, sex, parity, ethnic groups, religion, marital status, place of residence and social class.
- Variables related to the study:
  - These are variables specific to specific studies. In descriptive studies, they reflect characteristics examined by the investigator.
  - In analytic studies, variables are examined in terms of both exposure(s) and outcome(s).
- Confounding variables:
  - A variable associated with the problem and a possible cause of the problem is a potential confounding variable.
  - This variable may strengthen or weaken the apparent relationship between the problem and a possible cause.

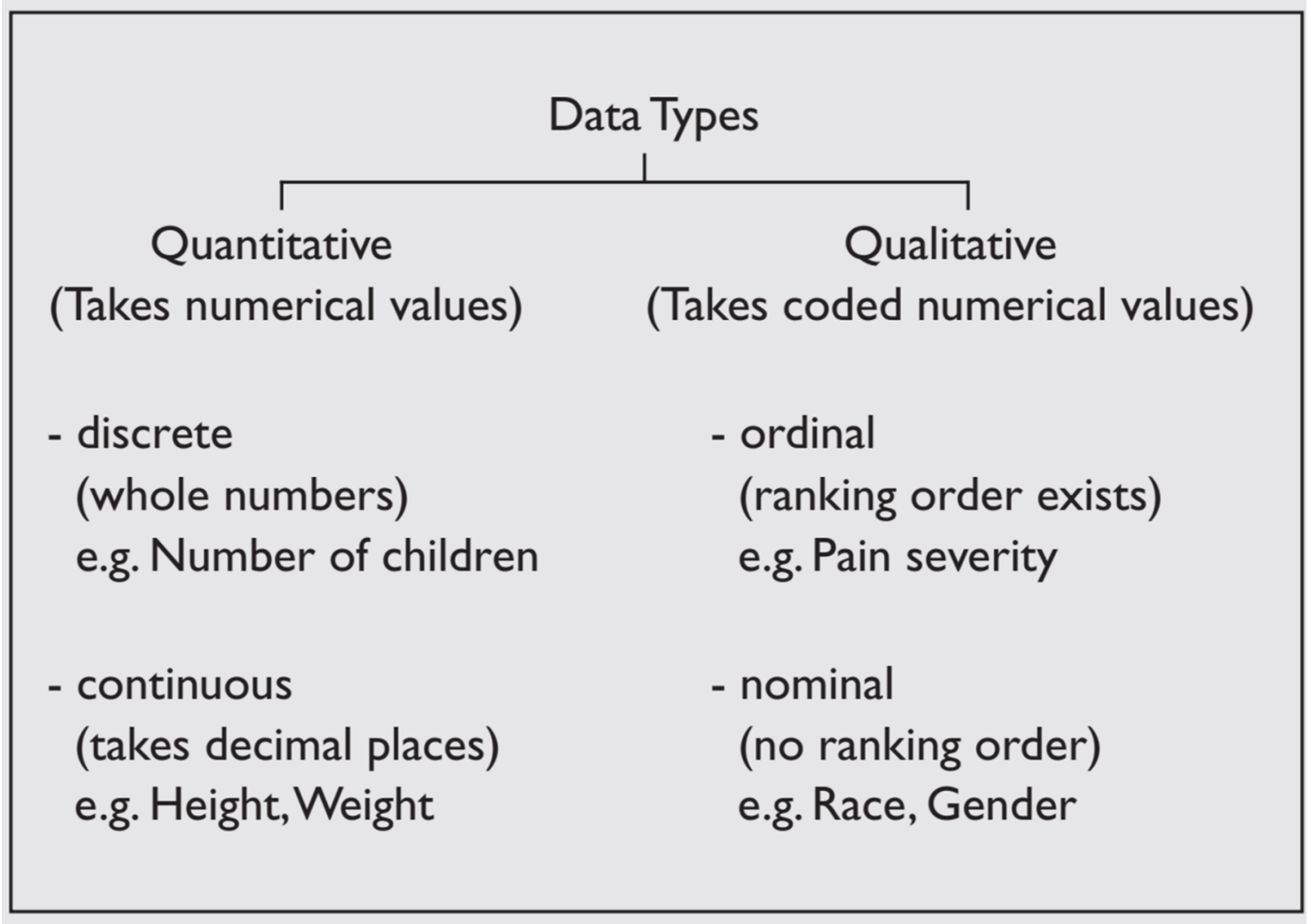
Relationship between groups of variables



Variable definitions

- Conceptual definition:
  - As we conceive the variable, e.g. anaemia is defined as pallor and obesity as overweight.
- Operational definition:
  - It is a more accurate measure of the variable. WHO defines anaemia as a Hemoglobin concentration of less than 11gm per 100 ml.
- Scale of variable measurement:
  - Nominal: [Yes – No], [Male – Female].
  - Ordinal: [mild – moderate – severe]
  - Interval: 0-6, 7-12, 13-18, 19+ (equal intervals)
  - Ratio: Weight measured in kilograms, height in cm, etc.

Data Presentation



Operational variables/definitions – 1

- Operational variables refer to how the researcher defines and measures a specific variable used in the study.
  - For example, if we are concerned with the effect of media violence on aggression, then we need to be very clear about what we mean by the different terms.
- For certain variables, it may not be possible to adequately define the variable or the indicator immediately because further information may be needed for this purpose.
- The researcher may need to review the literature to find out what definitions have been used by other researchers so that s/he can standardize their definitions and thus be able later to compare their findings with those of the different studies.
- In some cases, the opinions of "experts" or community members of health care providers may be needed to define the variable or indicator.

Operational variables/definitions – 2

- Once the variables are selected, each of them should be clarified. There are two aspects to be considered:

1. Clear definition of variables regarding objectively measurable facts (i.e., operational definition).
2. The scale of measurement to be used in data collection.

- The two kinds of definitions are:

3. Conceptual: akin to a dictionary definition.
   - e.g., Obesity may be defined as "Excessive fatness."
4. Operational: heavily influenced by considerations of practicability.
   - e.g., Obesity may be defined as: "a weight, based on weighing in light clothes and without shoes".

Example of a framework for defining variables

Variable	Definition of variable	Scale of
Age	Child's age at admission to hospital,	Months and years
Patient has	Dengue fever confirmed by serology	Yes/No/Not
Social class	Head of the household's primary	Detailed
Hemoglobin	Hemoglobin concentration in capillary blood, measured by haemoglobinometer.	g/dL
	respondent in answer to a question in a structured questionnaire.	classified into social class I – V.

8

Measures of Central Tendency

Central Location and Dispersion

Type of	Central	Dispersion
Normal distribution	Mean	SD
Skewed (Not	Median	IQR

This observation set will be used to facilitate understanding of some essential concepts in measuring central tendency

5, 3, 9, 7, 1, 3, 6, 8, 2, 6, 6

Calculation – 1

- Mode: is the most frequent value = 6 (three times)
- Minimum Value = 1
- Maximum Value = 9
- **Range** = Maximum Value – Minimum Value = 9 – 1 = 8
- **Mean** (Average) = $\frac{Sum\ of\ the\ observation\ values}{Number\ of\ observations}$
- In the previous observation set

  Sum = 56; Number of observations = 11; **Mean = 5.1**

Calculation – 2

- Median: Value that divides a distribution into two equal parts.

1. Arrange the observation by order 1,2,3,3,5,6,6,6,7,8,9.
2. Median = (No. of Observations + 1) / 2
   = (11 +1) / 2 = 6 (that means observation No. 6)
   = Observation No. 6 = 6

- If the number of observations is even:
  2,3,3,5,6,6,6,7,8,9
- Median = (No. of Observations + 1) / 2
  = (10 + 1) / 2 = 5.5
  = (Observation No. 5 + Observation No. 6) / 2
  = (6 + 6) / 2 = 6

Stem and leaf

- Stem and leaf describe three items:
  1. First: Drawing stem and leaf
  2. Second: Five-number summary
  3. Third: Box and whisker plots

- Application of the three items will be on this theoretical example:
- The recorded systolic blood pressure for 13 patients was as follows:

136, 140, 142, 162, 124, 130, 130, 156, 136, 144, 128, 122,132

First: Drawing stem and leaf

1. Look for the Minimum value (122) and the Maximum value (162). Arrange the values by order between Minimum and Maximum values as the following:
2. 122, 124, 128, 130, 130, 132, 136, 136, 140, 142, 144, 156, 162
3. The stem will be: 12, 13, 14, 15, 16
4. Distribute the leaves as the following:
5.

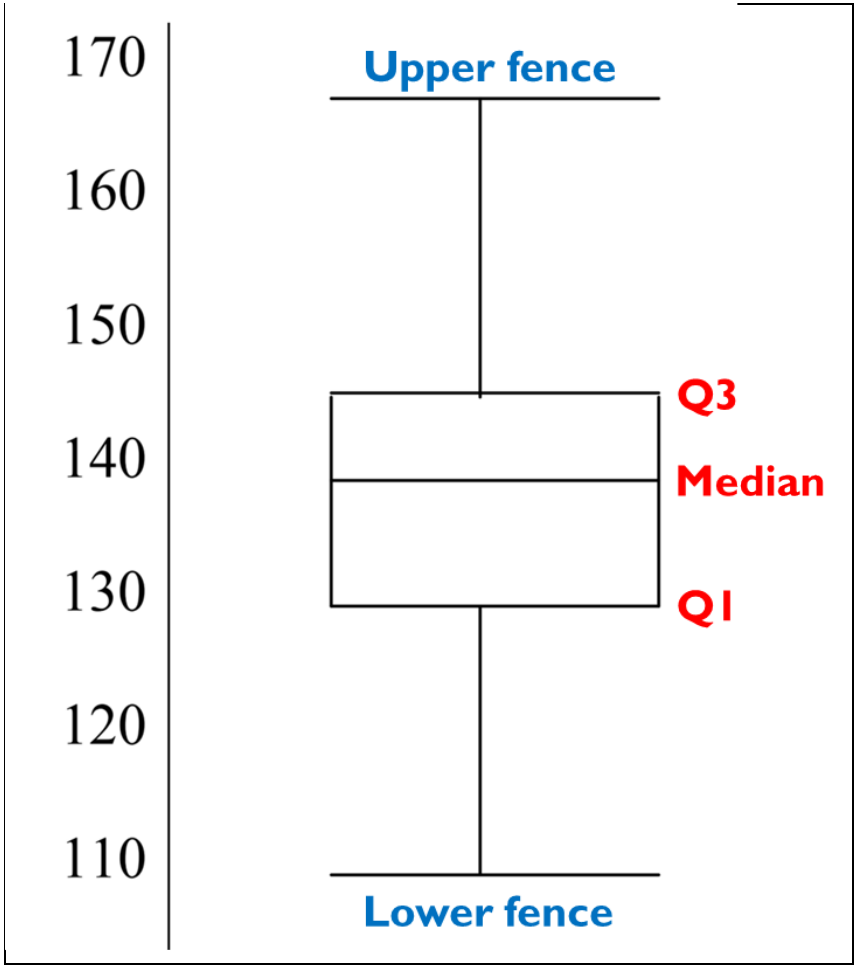
Stem	Leaves
16	2
15	6
14	0 2 4
13	0 0 2 6 6
12	2 4 8

Second: Five number summary

1. Calculate Range = Maximum – Minimum = 162 – 122 = 40.
2. Calculate Median = (n + 1) / 2 = (13 + 1) / 2 = 7th value = 136.
3. Calculate Q1 (lower hinge) = [(n + 1) / 4] = (13 + 1)/4 = 3.5 = (128 + 130) / 2 = 129.
4. Calculate Q3 (upper hinge) = [3(n + 1) / 4] = (3 x 14)/4 = 10.5 = (142 + 144) / 2 = 143.
5. Calculate inter quartile interval (IQR) = Q3 – Q1 = 143 – 129 = 14

Third: Box and whisker plots

- Fence = 1 step out the inter quartile (IQR)
- Calculate step = 1.5 X IQR = 1.5 X 14 = 21
- Upper fence = 143 + 21 = 164
- Lower fence = 129 – 21 = 108



Calculation of Standard Deviation – 1

- This is the formula for Standard Deviation:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

- To calculate the standard deviation of those numbers:

1. Work out the Mean (the simple average of the numbers)
2. Then, for each number, subtract the Mean and square the result
3. Then, work out the mean of those squared differences.
4. Take the square root of that, and we are done!

Example of SD calculation – 1

Set Observations: 9, 2, 5, 4, 12, 7, 8, 11		
1. Mean	= 58 / 8 = 7.25	
2. Subtract	$(9 - 7.25)^2$	= 3.0625
3. Summation	$(2 - 7.25)^2$	= 27.5625 + 5.0625 + 10.5625 + 22.5625 + 0.0625 + 0.5625 + 14.0625 = 83.5
	$(5 - 7.25)^2$	= 5.0625
	$(4 - 7.25)^2$	= 10.5625
	$(12 - 7.25)^2$	= 22.5625
	$(7 - 7.25)^2$	= 0.0625
	$(8 - 7.25)^2$	= 0.5625
	$(11 - 7.25)^2$	= 14.0625

Example of SD calculation – 2

4. Calculate variance Calculate variance	= Summation / No. = 83.5 / 8 = 10.4375
5. Take the square root of variance Take the square root of variance	$= \sqrt{10.4375}$ = 3.23

- Standard Deviation = 3.23
- Central Tendency for the set observation: 9, 5, 4, 2, 7, 8, 11, 7 (If normally distributed)
- Central Tendency: Mean ± SD = 7.25 ± 3.23

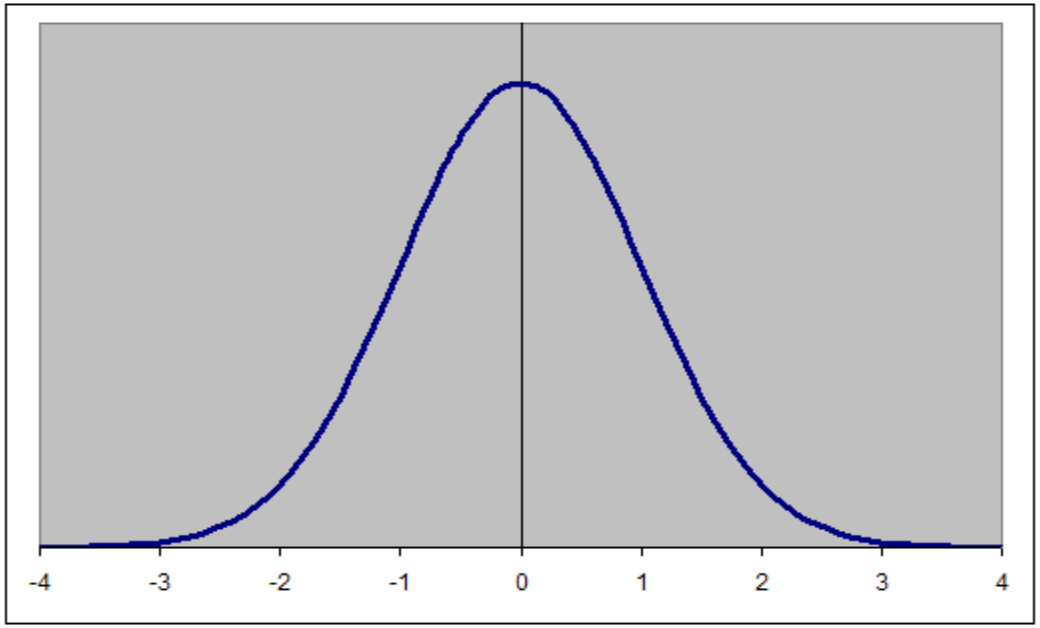
9

Central Limit Theorem

Normal Distribution

The normal distribution is the most used statistical distribution. The principal reasons are:

1. Normality arises naturally in many physical, biological, and social measurement situations.
2. Normality is vital in statistical inference



Normal Moments

The normal distribution is characterized by the mean (μ) and the standard deviation (Sigma).

The mean is a measure of location or center.

The standard deviation is a measure of scale or spread.

The mean can be any value between ± infinity, and the standard deviation must be positive.

Standard Normal Distribution

The standard normal distribution is a unique member of the typical family that has:

a mean of 0, and

a standard deviation of 1.

Z denotes the standard normal random variable.

The standard normal distribution is essential since the probabilities and quantiles of any normal distribution can be computed from the standard normal distribution if μ and sigma are known.

Properties of the normal distribution

The area under the curve bound by one standard deviation is approximately 68%, two standard deviations are approximately 95%, and three standard deviations are approximately 99.7%

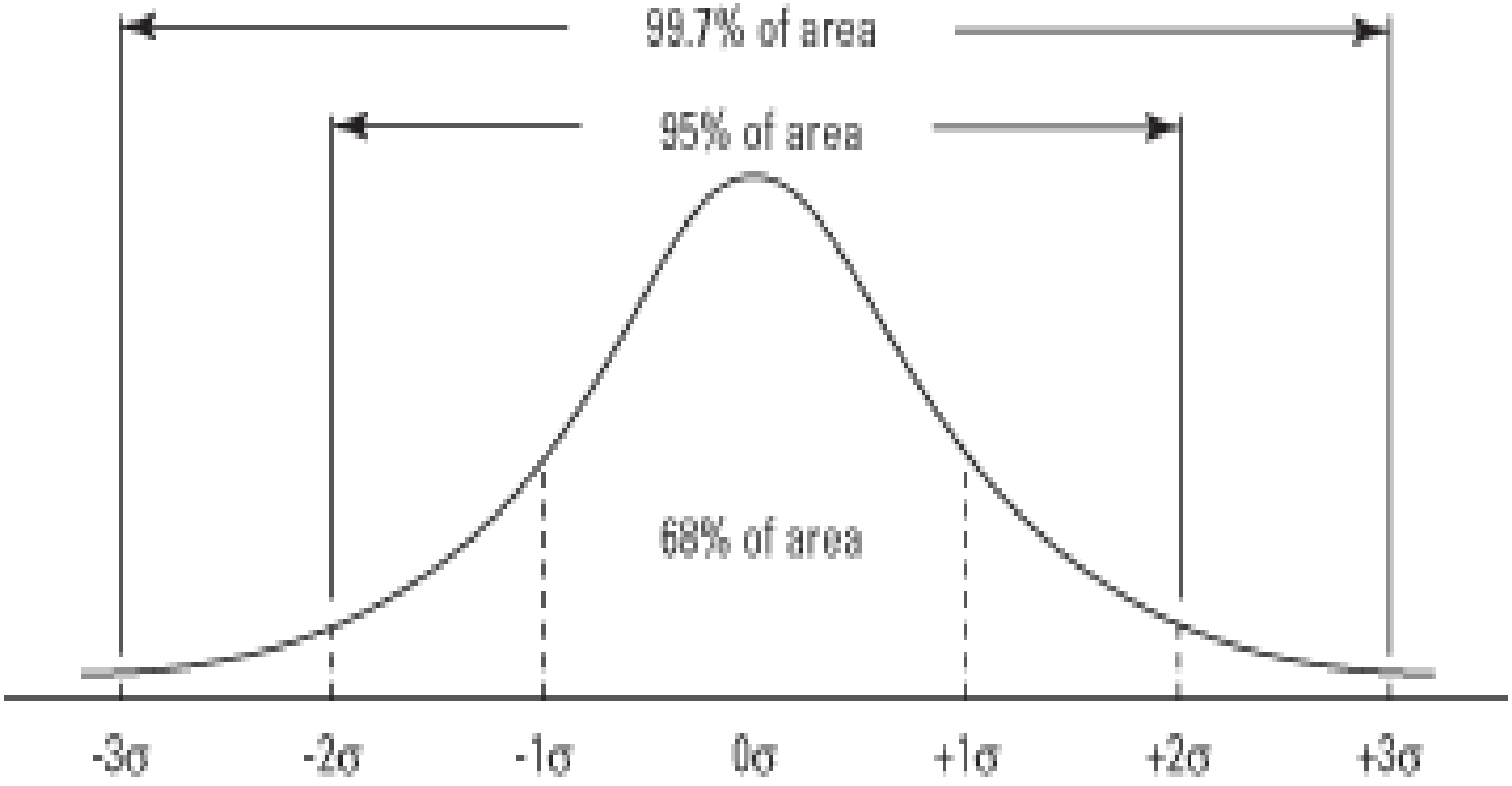
The normal distribution is ultimately determined by the parameters μ and sigma.

, i.e. a different normal distribution is specified for each value of μ and sigma.

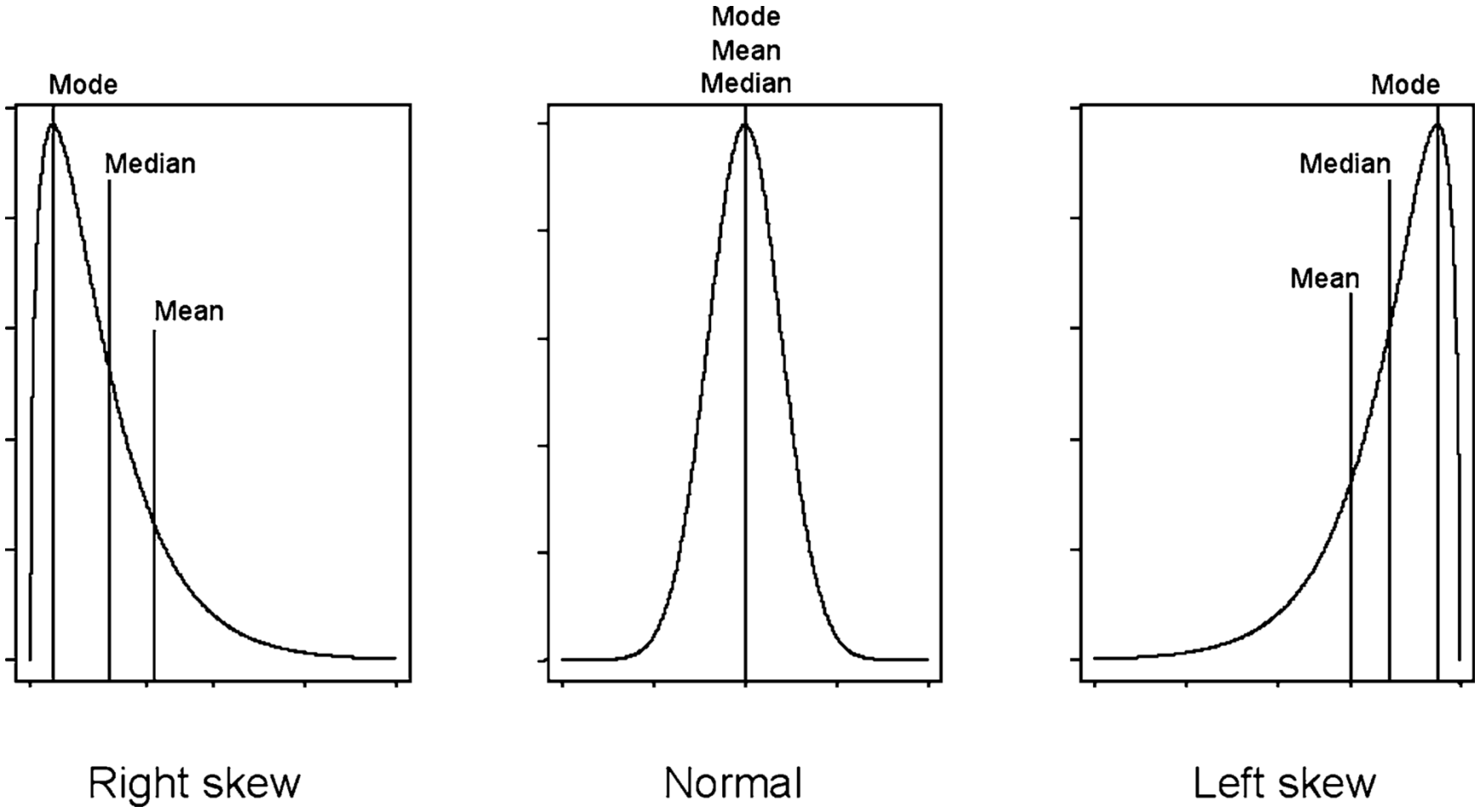
Different values of μ shift the distribution graph along the x-axis.

Different sigma values determine the distribution graph's degree of flatness or peakness.

Area under the normal curve



Distributions of Quantitative Data: Histogram



Standard Error (SE)

- **It measures the extent to which the sample means deviate from the true population mean.**
- The smaller it is, the closer to the true population mean.
- When the sample size is more significant, the SE becomes smaller.
- SE is calculated from the formula:

$$SE = \frac{SD}{\sqrt{n}}$$

Interval Estimates (Confidence Interval) – 1

The accuracy of the normality distribution estimates is dependent on the sampling plan of the study

The assumption that a representative sample is obtained.

if we are allowed to repeat a study (with a fixed sample size) many times, the mean and SD obtained for each study may be different, and from the theory of the Sampling Distribution of the Mean, the mean of all the means of the repeated samples will give us a more precise point estimate for the population mean.

Interval Estimates (Confidence Interval) – 2

In medical research, we do not have this luxury of doing repeated studies (ethical and budget constraints), but from the Central Limit Theorem, with a sample large enough, an interval estimate provides us with a range of scores within which we are confident, usually a 95% Confidence Interval (CI), that the population mean lies within.

Confidence interval

What is the meaning of 95% confidence interval?

This interval represents an interval of parameter values consistent with the data.

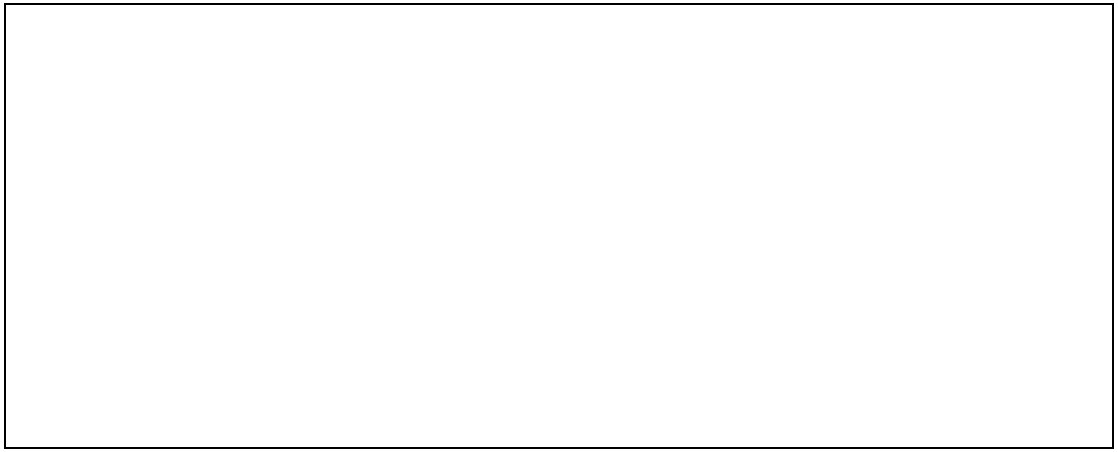
Of all intervals constructed as this one, 95% would contain the actual value of the population parameter.

Confidence interval = Estimate of parameter ± (Reliability coefficient X S.E)

Estimate of parameter indicates Mean.

Reliability coefficient equals 1.96 and is often written as 2.

Example of Confidence Interval



A shown in the Table “an output of a statistical analysis conducted by SPSS for exploration of the age in one of the studies”

The mean age is estimated as 38.06 years. [use 2 instead of 1.96]

95% confidence interval = 38.06 ± (1.96 X 0.98).

The result, as shown in the table that the confidence interval ranges between 36.12 40.01

Hypothesis Testing

Null hypothesis proposes that there is no difference.

It is the basis for the statistical testing.

The null hypothesis is accepted if no difference is found.

If there is a difference, the null hypothesis is rejected, and the alternative hypothesis is accepted.

Alpha: The probability of concluding the groups differ when, in reality, they do not.

Beta: Probability of concluding the groups do not differ when, in reality, they differ.

Power: Probability of detecting a difference between the groups if the groups do, in fact, differ.

Test of Significance – 1

Hypothesis: A statement of belief used to evaluate population values.

Null Hypothesis: A claim that there is no difference between the hypothesized values.

Alternative Hypothesis: A claim that disagrees with the null hypothesis. If the null hypothesis is rejected, we accept the alternative hypothesis.

Test Statistics: A statistic used to determine the relative position of the hypothesized value from the mean of its distribution.

Significant Level: The level corresponding to the critical region area. This area is usually small. When a test statistic falls in this area, the result is considered significant at the α level.

Test of Significance – 2

P-value: The area in the distribution's tail beyond the test statistic's value.

If P-value ≤ α, we reject the null hypothesis.

If P-value > α, we can't reject the null hypothesis.

One Tailed versus Two-Tailed Tests:

If you are interested in determining whether the sample mean is significantly different from a given population mean, you would perform a two-tailed test since the deviation (X – μ) could be either positive or negative.

If you were interested in whether the sample mean is significantly larger than the population mean, you would perform a one-tailed test. Likewise, you would go to the one-tailed test for the sample mean smaller than μ.

Example of Hypotheses

Research Question: Does the BMI differ between male and female students at Al-Azhar University?

Objective: To find out the differences in the BMI between male and female students at Al-Azhar University.

Hypotheses:

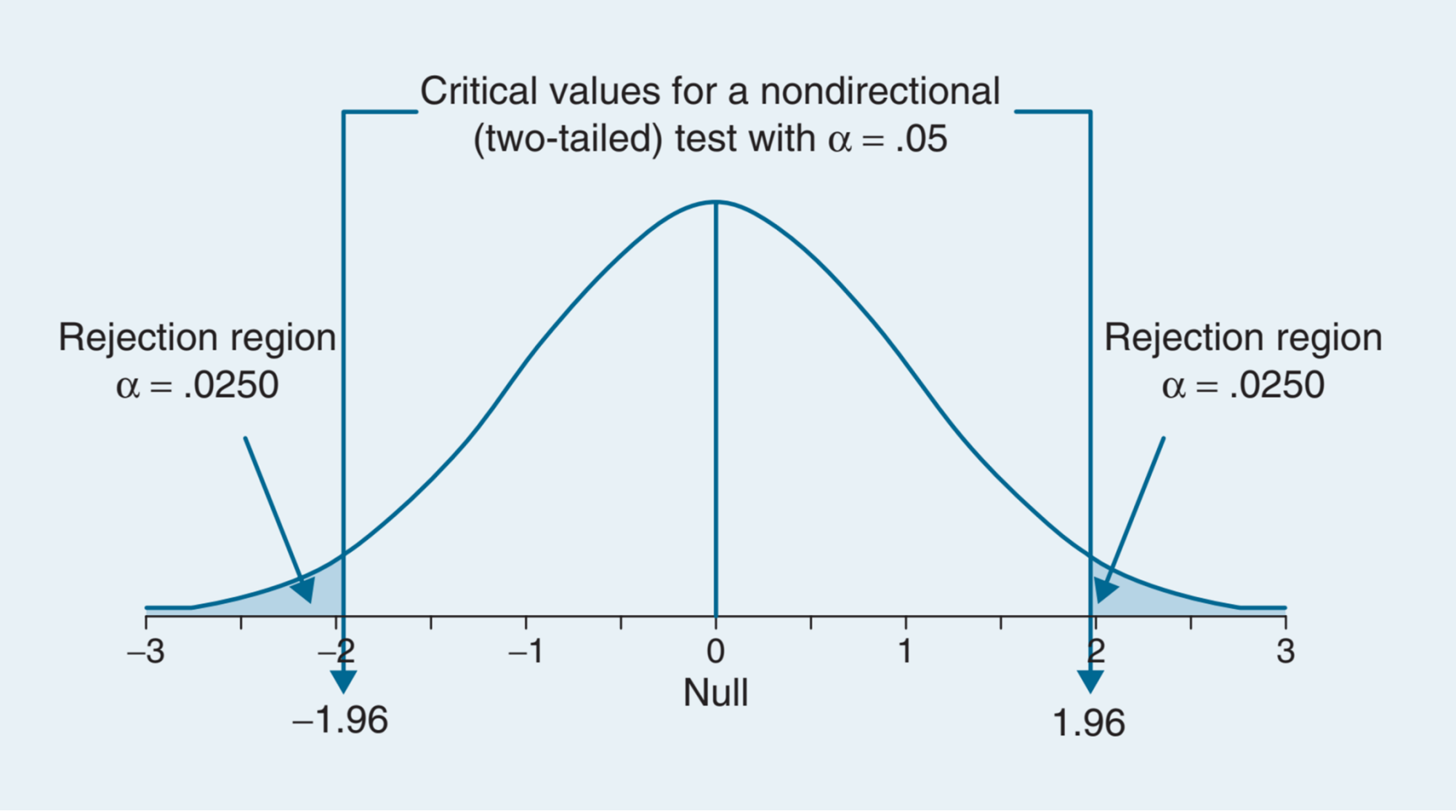
1. Null; H_0: Mean of Males' BMI = Mean of Females' BMI.
2. Alternative (Two-Tailed):

   H_1: Mean of Male's BMI ≠ Mean of Female's BMI.
3. Alternative (One-Tailed):

   a) H_1: Mean of Males' BMI > Mean of Females' BMI.

   b) H_1: Mean of Males' BMI < Mean of Females' BMI.

The critical values (±1.96) for a non-directional (two-tailed) test with α = 0.05 level of significance



Type I and Type II Errors

Significance testing is a method for assessing whether a result is likely to be due to chance or to some real effect.

It cannot prove that it is one or the other:

1. Type I Error: is the rejection of a true null hypothesis
   known as a "false positive" finding or conclusion.
2. Type II Error: the non-rejection of a false null hypothesis
   known as a "false negative" finding or conclusion.

Meaning of “Statistically Significant”

Research reports often state that the results were statistically significant (P-value< 0.05) or make some similar statements.

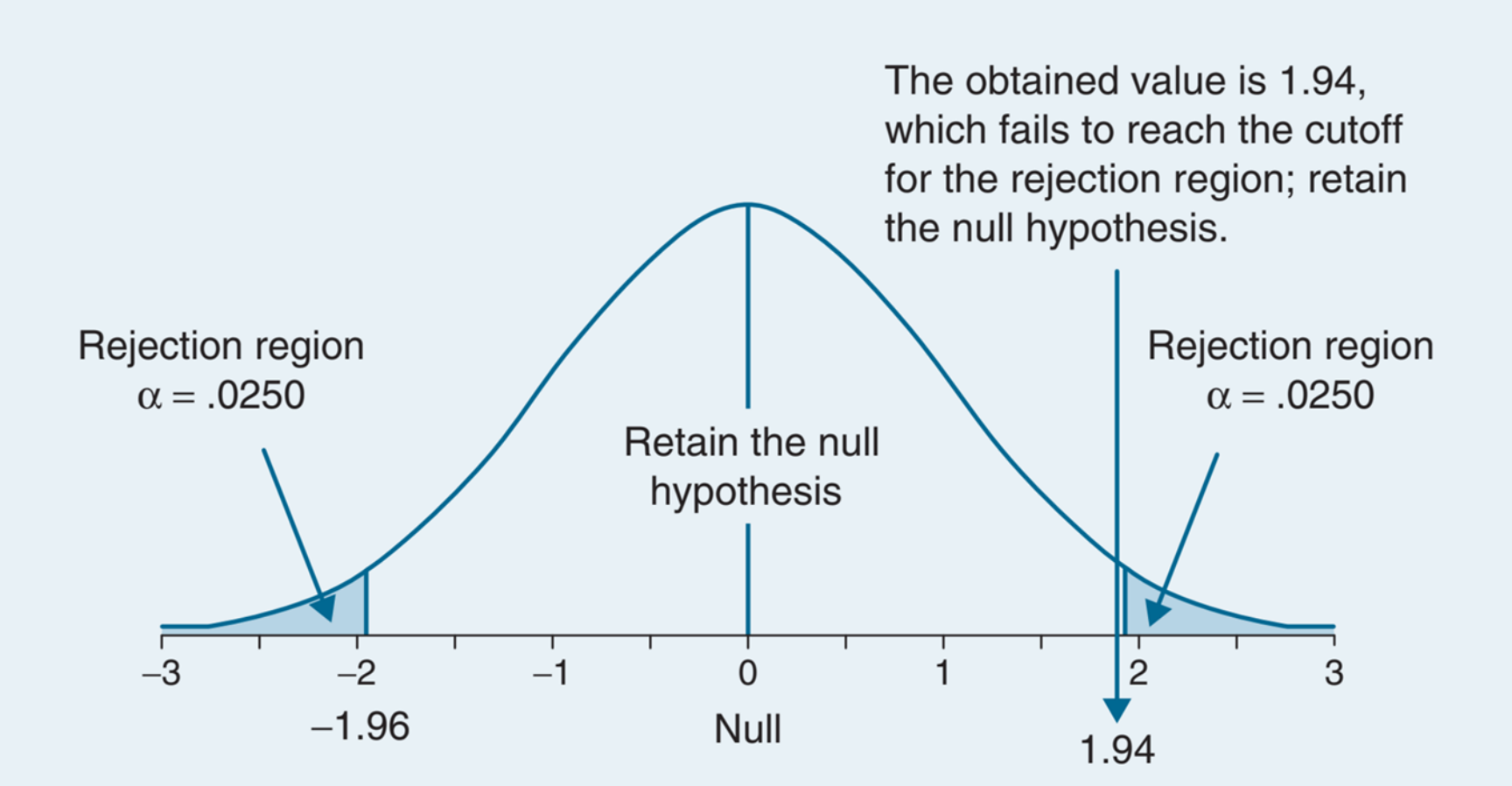
Such a comment means that the observed difference is too significant to be explained by chance alone.

The significant level somewhat arbitrarily selected at such values of α as 0.05, 0.025, 0.01, or 0.001 measures how important a result is.

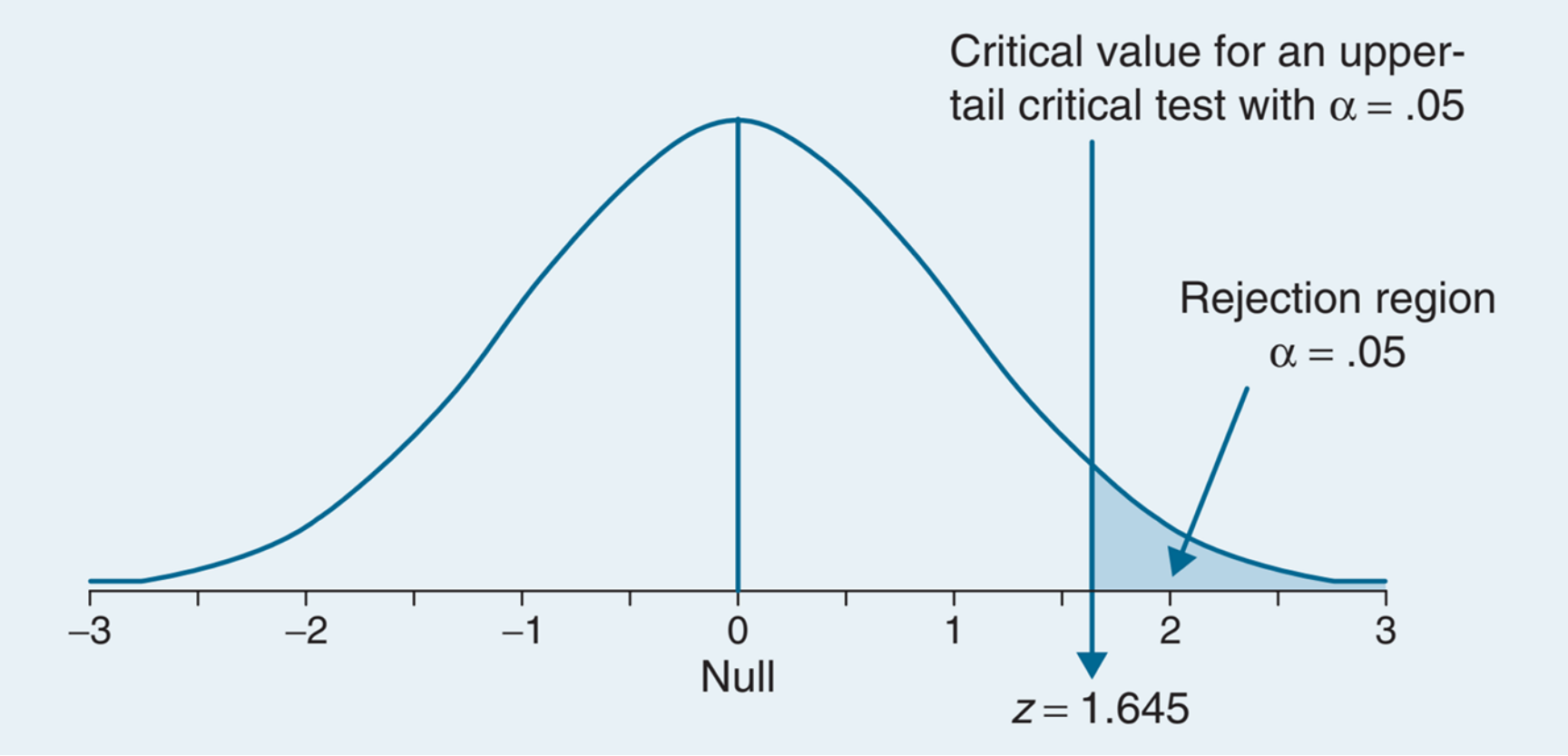
The significant level α also the magnitude of error that one is willing to take in rejecting the null hypothesis.

Statistically significant means that the sample’s evidence is incompatible with the null hypothesis.

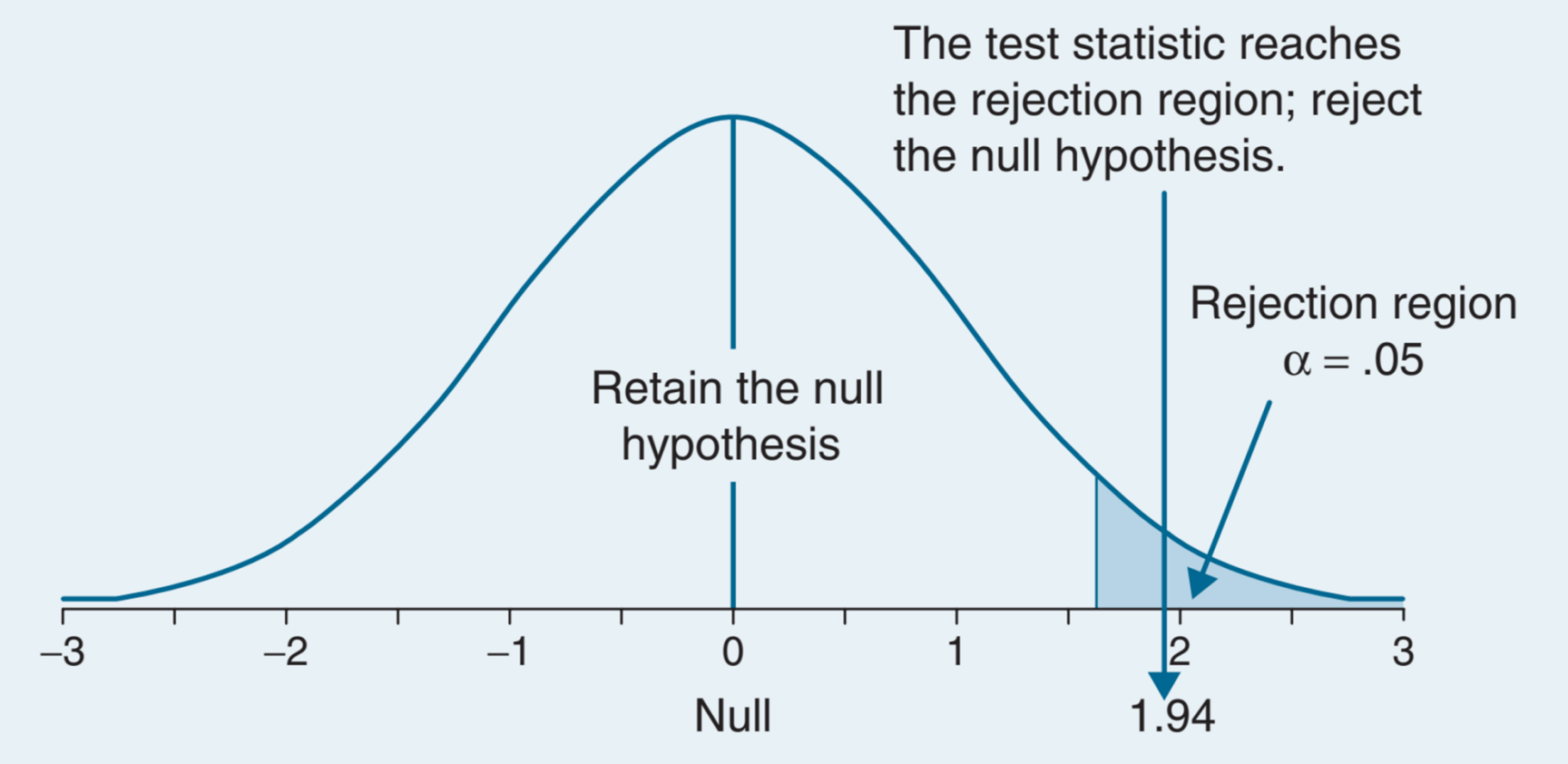
The obtained value fails to reach the rejection region (it is within the critical values of ±1.96); decide to retain the null hypothesis



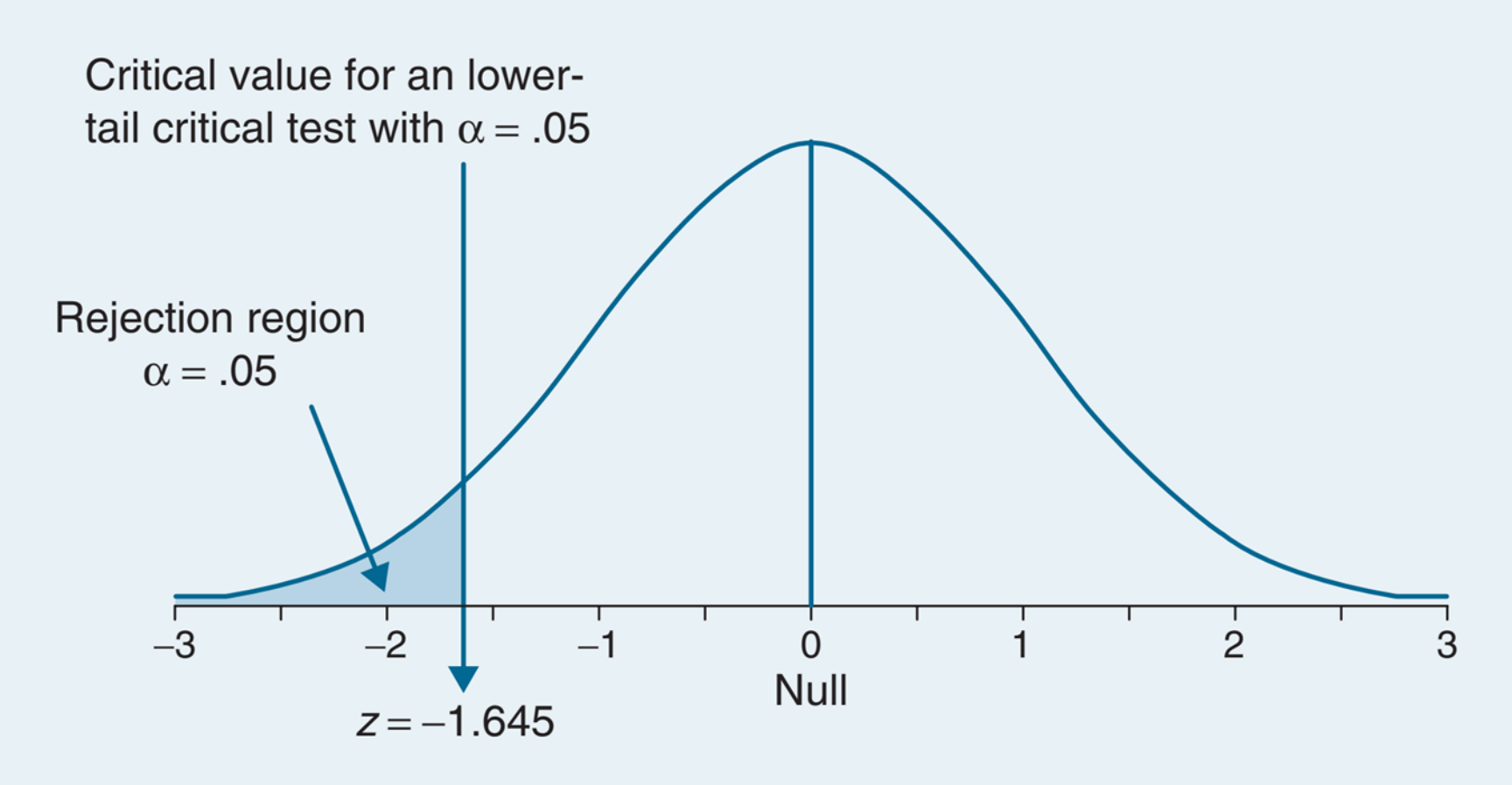
The critical value (1.645) for a directional (upper-tail critical) hypothesis test at α = 0.05 level of significance. When the test statistic exceeds 1.645, we reject the null hypothesis; otherwise, we retain the null hypothesis.



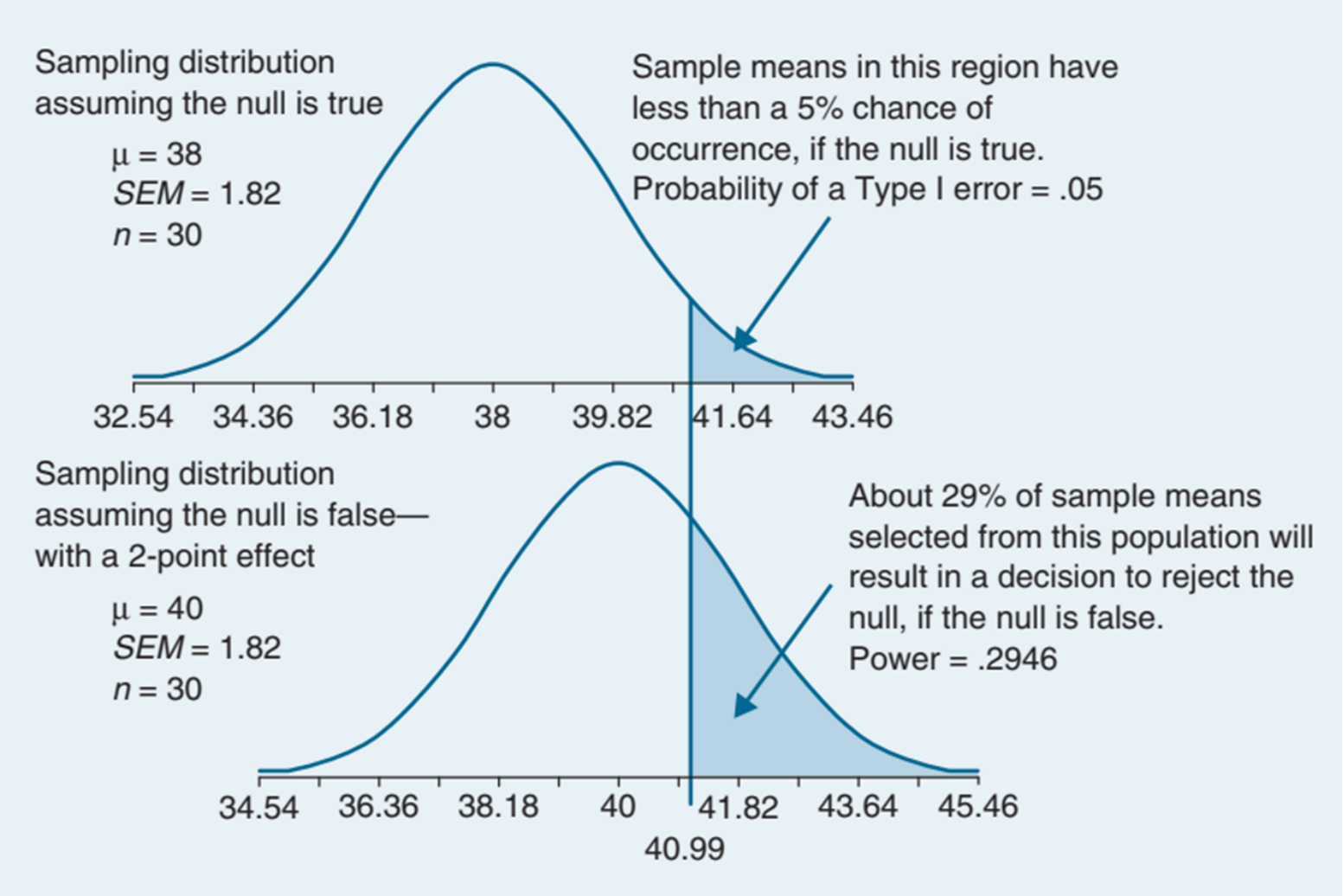
Since the obtained value reaches the rejection region, we decide to reject the null hypothesis



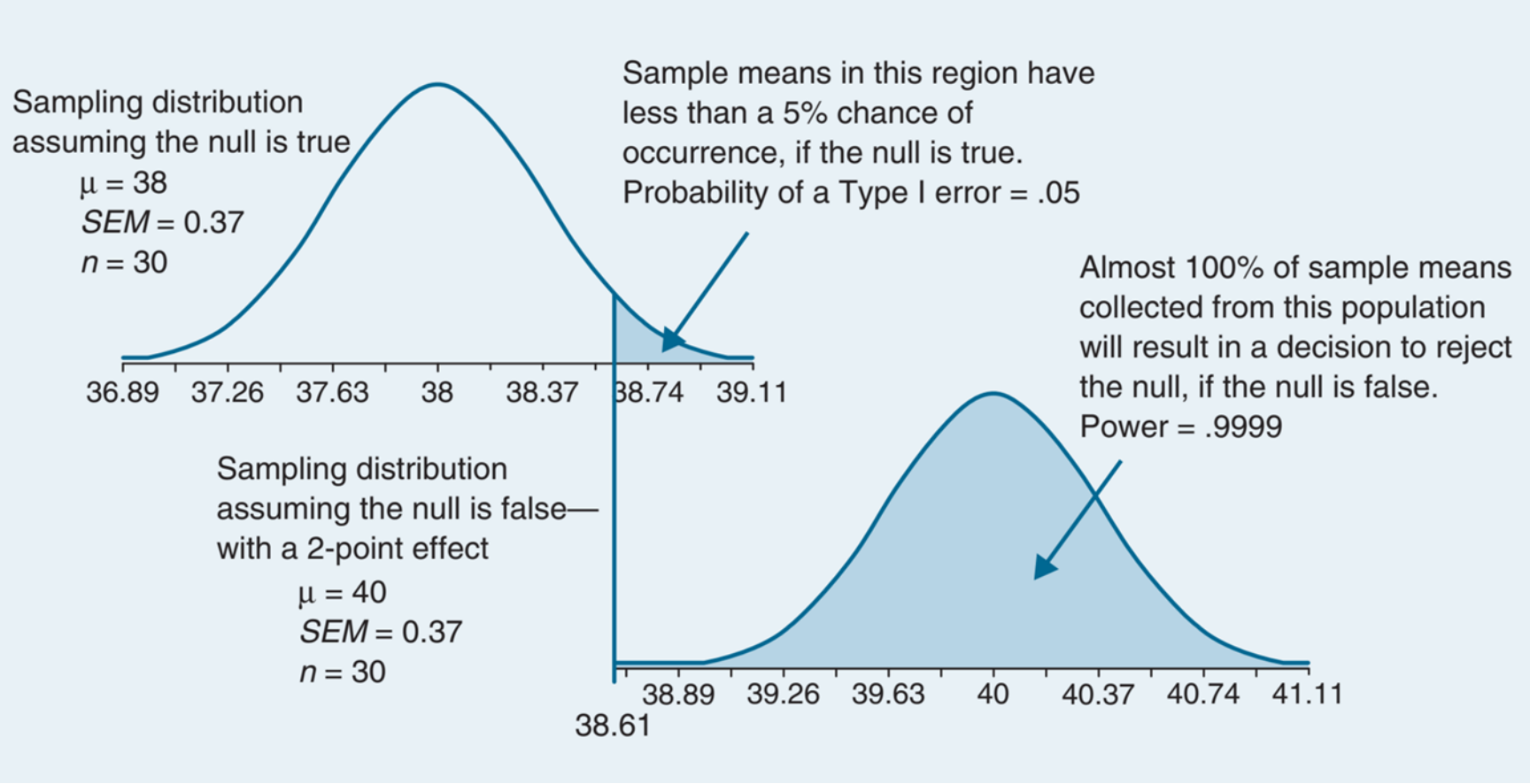
The critical value (-1.645) for a directional (lower-tail critical) test at a .05 level of significance. When the test statistic is less than -1.645, we reject the null hypothesis; otherwise, we retain the null hypothesis.



Small effect size and low power for Class 1



Large effect size and high power for Class 2.



Main Components of the Chapter 3

Title of Chapter 3: Methodology



Quantitative and Qualitative Research Design

- Before we move to the quantitative research design, one more piece of information about the qualitative research design

Qualitative Research Designs

The main objective of qualitative research is:

- To look into the meanings and the ways people understand things.
- To look into patterns of behavior.

Qualitative data:

Qualitative data, depicted in words, images or any other medium, are the product of a process of interpretation.

Qualitative research principles

- It is flexible and elastic.
- It involves mixing various methods in terms of data collection (e.g. interviews and diaries).
- It is focused on understanding a phenomenon or a social setting.
- Tends to be holistic; it involves striving to understand the ‘ whole ’ rather than just understanding a part of the phenomenon being studied.

Qualitative research principles cont.

- Qualitative design requires the researcher to become intensely involved in the study. This can often be over very long periods.
- Qualitative design requires the researcher to become the research instrument.
- Qualitative design requires the ongoing analysis of the collected data (remember, we need some flexibility).

Qualitative research designs – most common

The most common qualitative research designs:

1. Ethnography
2. Grounded theory
3. Phenomenology
4. Historical Studies
5. Case Studies
6. Action Research Studies
7. Participatory action research (PAR)

1- Ethnography

- An ethnographic research study studies people in their natural environment.

- It is a descriptive account of social life and culture within a defined social system and is often thought of as ‘ a portrait of a people ’.
- In effect, it is concerned with a holistic view of culture, including its shared meanings, patterns, and experiences.

Ethnography describes and interprets a cultural or social group or system.

1- Ethnography cont.

-The ethnographic researcher aims to learn from members of a cultural group.

The ethnographic researcher examines the following groups:

- Behavior
- Customs
- Way of life

1. Ethnography (Cont.)

How to collate data?

- Observer (note-taking)
- Interviews (semi-structured, unstructured …etc)

Sample

Relatively small samples are needed for ethnographic research.

The sample is purposive (chosen for the specific purposes of this research study) in that it selects participants.

1. Ethnography (Pros. Vs Cons.)

The conduct of the study and its results is often dependent on the skill and rigor of the researcher. (Cons)

Replication of the study can be difficult. (Cons)

There are issues of researcher objectivity and reactivity (to the members of the group and to situations encountered within the group). (Pros)

1. Ethnography (Pros. Vs Cons.) CONT.

The transferability of findings may be problematic. (Cons)

The links to practice and policy-making may not often be easy to make, or there may not be direct links, making it difficult for implementation. (Cons)

Obtaining funding for ethnographic research can be difficult. (Cons)

2- Grounded Theory

It is a systematic methodology in the social sciences that involves building theories through the methodical gathering and analysis of data.

-The primary purpose of a grounded theory approach is to begin with data collected during the research study and use these data to develop a theory.

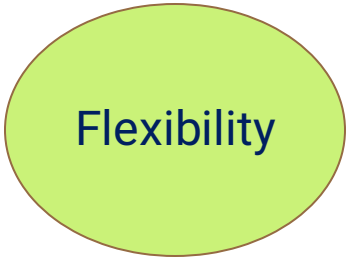
-A grounded theory study uses a prescribed set of procedures for analyzing data and constructing a theoretical model from them.

2- Grounded Theory cont.

The term ‘ grounded ’ in this context refers to the idea that the theory that emerges from the study is derived from data, particularly data collected in the field rather than data found in the research literature.

2- Grounded Theory cont.

How to collate data?



- Observer (note-taking)
- Dairies
- Interviews (semi-structured, unstructured …etc)
- Focus groups

Sample (same as ethnographic)

Relatively small samples are needed for Grounded research.

The sample is purposive (chosen for the specific purposes of this research study) in that it selects participants.

2. Grounded Theory (Pros vs. Cons)

Advantages:

- a systematic and rigorous procedure
- rich data arise from the experiences of individuals participating in the research study.

Disadvantages

- The subjectivity of the data leads to difficulties in establishing the reliability and validity of approaches and information.
- It is difficult to detect or prevent researcher-induced bias.

3- Phenomenology

It is the direct investigation and description of phenomena as they are consciously experienced and without the imposition of any theories that there may be about their causal explanations or their objective reality. In other words, it is the LIVED EXPERIENCE of participants.

-Phenomenology helps us to understand the meaning of people's lived experiences.

3- Phenomenology cont.

A phenomenological study explores what people experience and focuses on their experience of phenomena.

- Phenomenology is the philosophical study of observed unusual people or events that appear without further research or explanation.

- An example of phenomenology is studying the green flash that sometimes happens just after sunset or before sunrise.

3- Phenomenology cont.

How to collate data?

- Observer (note-taking)
- Dairies
- Interviews (semi-structured, unstructured …etc)
- Drawing (non-verbal communication)

Sample (same as ethnographic)

Relatively small samples are needed for ethnographic research.

The sample is purposive (chosen for the specific purposes of this research study) in that it selects participants.

3- Phenomenology cont.

Advantages

- an in-depth understanding of individual phenomena
- the obtaining of rich data based on the experiences of individuals.

Disadvantages

- The subjectivity of the data leads to difficulties in establishing the reliability and validity of approaches and information.
- It is challenging to detect or prevent researcher-induced bias.
- There can be difficulty in ensuring pure bracketing, which can bias the data interpretation.

Disadvantages cont.

- Phenomenology does not produce generalizable data.
- The potential difficulties of participants expressing themselves.
- Participants need to be both interested and articulate. Problems that can cause difficulties in expressing themselves include language, age, brain damage/dysfunction, and embarrassment.

Other qualitative designs (FYI ONLY)

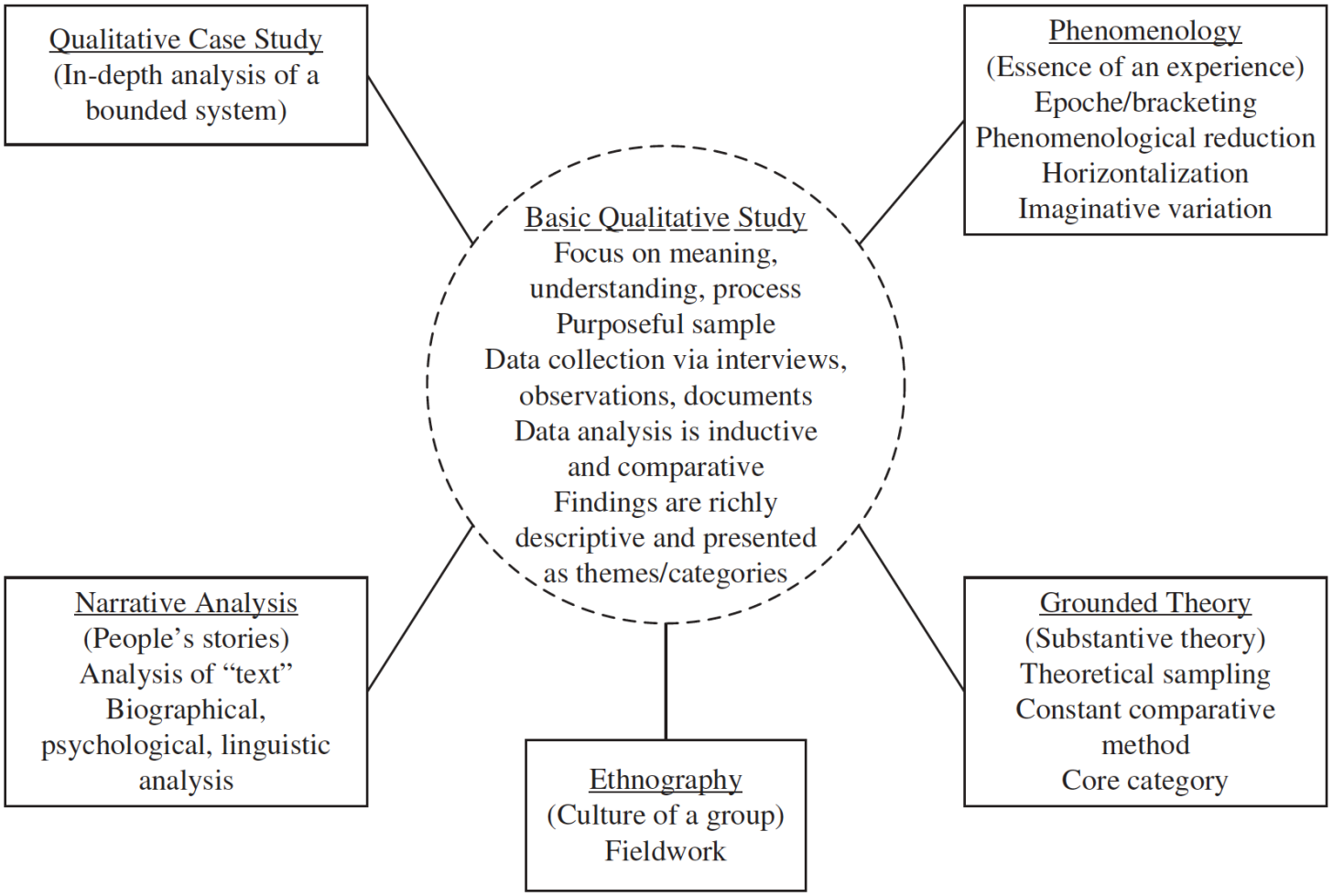


Figure obtained from Merriam, S. B., & Tisdell, E. J. (2015). Qualitative Research: A Guide to Design and Implementation (4 edition). San Francisco, CA: Jossey-Bass.

4. Historical Studies

Historical studies concern the identification, location, evaluation, and synthesis of data from the past. Historical research seeks not only to discover past events but to relate these past happenings to the present and future. Leininger (1985) wrote, “Without a past, there is no meaning to the present, nor can we develop a sense of ourselves as individuals and as members of groups”.

4. Historical Studies cont.

Historical data is frequently referred to as primary and secondary sources. Primary sources provide firsthand information or direct evidence. Secondary sources provide secondhand information (or sometimes third or fourthhand).

4. Historical Studies cont.

Primary sources should be used in historical research when possible. Examples of primary sources include oral histories, written records, diaries, eyewitnesses, pictorial sources, and physical evidence.

4. Historical Studies cont.

The data for historical research should be subjected to two types of evaluation: external criticism and internal criticism. External criticism concerns the authenticity or genuineness of the data and should be considered first. Internal criticism examines the accuracy of the data and is considered after the data are deemed genuine. Whereas external criticism establishes the validity of the data, internal criticism establishes the reliability of the data.

5. Case Studies

Case studies are in-depth examinations of individuals or groups of individuals. They could also examine an institution, such as hospice care for the dying. The case method has its roots in sociology and has also been used extensively in anthropology, law, and medicine.

5. Case Studies cont.

A case study may be considered quantitative or qualitative research depending on the study's purpose and the researcher's chosen design. As is true of other qualitative studies, for a case study to be considered a qualitative study, the researcher must be interested in the meaning of experiences to the subjects themselves rather than in generalizing results to other groups. Case studies are not used to test hypotheses, but hypotheses may be generated from case studies (Younger, 1985).

5. Case Studies cont.

Content analysis is used to evaluate the data from case studies. Content analysis involves examining communication messages. The researcher searches for patterns and themes. After reading the diaries of the individuals who are using insulin pumps, the nurse researcher might come up with themes such as "freedom from the rigid schedule," "more normal life," and "release from self-inflicted pain."

Disadvantages

Case studies are time-consuming and may be quite costly. Additionally, subject dropout may occur during this type of study. The loss of subjects must be considered whenever a study is carried out over an extended period. A person may move from the locality or simply decide to discontinue participation in the study.

6. Action Research Studies

Action research is a type of qualitative research that seeks action to improve practice and study the effects of the action (Streubert & Carpenter, 2002). Solutions are sought to practice problems in one particular hospital or healthcare setting. Unlike quantitative research studies, action research does not aim to generalize the findings. In action research, the implementation of solutions occurs as an actual part of the research process. There is no delay in the implementation of the solutions.

6. Action Research Studies cont.

Action research became popular in the 1940s. Kurt Lewin (1946) was influential in spreading it. He became interested in helping social workers improve their practice. Although many of you may have heard of Lewin and his contribution to change theory, his involvement in action research is not as well known.

7. Participatory action research (PAR)

It is a special kind of community-based action research in which there is a collaboration between the study participants and the researcher in all steps of the study: determining the problem, the research methods to use, the analysis of data, and how the study results will be used. The participants and the researcher are co-researchers throughout the entire research study. According to Kelly (2005), PAR provides an opportunity to involve a community "in the development and assessment of a health program"

Comparison between Qualitative and quantitative Research

Qualitative (relativism)	Quantitative (realism)
Reality is subjective, has multiple voices because of the views of the different participants; variables are complex; insider's point of view. (social studies)	There is an objective reality; outsider ' s point of view. (nature sciences)
Researcher may be closely involved in the study, but explains strategies used to create a distance between the researcher and the study. Empathy is a feature of this type research.	The researcher does not have any personal effect on what is being researched; Detachment and impartiality
Researcher study the topic in its natural setting and use inductive logic and a design that have roots in the way the study has developed.	Cause and effect; generalisations are made, which lead to prediction, explanation and understanding(deductive); the methods are reliable and valid.
The researcher uses words/ phrases provided by the participants of the study.	Begins with a hypothesis and/ or theories
The researcher conducts the investigation in the natural setting with the participants, thus gets a thorough perspective of the participants ' routine. Behaviour is influenced by the setting in which it occurs.	The research may be conducted in a laboratory (not always); formal instruments are used; experimentation may be part of the process. The researcher is independent of what is being studied.
Researchers give a detailed description of the context of the study and its participants; the questions may be revised according to information gleaned from earlier discussions with participants.	All aspects of the study are designed before data are collected. Manipulation and control of variables may be used in the study, components of the study are analysed and deductive analysis is conducted.

Validity & Reliability in Qualitative Research

1) Credibility (parallels internal validity)

Credibility is establishing the study's findings with reality. This demonstrates the believability or truth of the findings. It ascertains whether the study findings are believable based on the participants' original data and participants' views.

√ Prolonged, substantial engagement

√ Persistent observation

√ Peer debriefing

√ Progressive subjectivity

√ Member checks

√ Triangulation

Validity & Reliability in Qualitative Research cont.

2) Transferability (parallels external validity)

This refers to the level at which the study findings can be transferred to other situations. The reader should look at the details of the study, including the methods and rigor applied, and compare this to their own situation to determine if the findings can be applied to their own situation.

√ Thick description

√ Multiple cases

Validity & Reliability in Qualitative Research cont.

3) Dependability (parallels reliability)

Dependability is essential to trustworthiness because it establishes the research study's findings as consistent and repeatable. Researchers aim to verify that their findings are consistent with the collected raw data. They want to make sure that if some other researchers were to look over the data, they would arrive at similar findings, interpretations, and conclusions about the data. This is important to ensure that nothing was missed in the research study or that the researcher was not sloppy or misguided in his or her final report.

Validity & Reliability in Qualitative Research cont.

4) Confirmability (parallels objectivity)

Confirmability questions how the data supports the study findings. It identifies any bias that may have been present. It is the level to which the findings can be confirmed or corroborated. Confirmability is concerned with determining that the researcher's imagination does not makeup data and interpretations of the findings but is derived from the data.

Validity & Reliability in Qualitative Research cont.

5. Member checking: This is a technique in which the data, interpretations, analytical categories, and conclusions are fed back to the participants from whom the data were collected. This process strengthens the data by allowing participants to clarify and correct errors and provide additional information if necessary. It can also highlight how participants and researchers view the data differently.

Validity & Reliability in Qualitative Research cont.

6. Trustworthiness concerns how the researchers establish the study findings as credible, transferable, and confirmable.

Validity & Reliability in Qualitative Research cont.

Reflexivity is the practice of critical self-reflection about oneself as a researcher. It involves looking at one's biases, preferences, and preconceptions. The researcher should reflect on their relationship with the participant and how it affects the participant's answers to questions or decisions in the research. For example, how does being a nurse impact interviewing a patient in your care?

Validity & Reliability in Qualitative Research cont.

Triangulation refers to using different methods of collecting or analyzing data and cross-checking these to see if findings from different methods, analysts, theories, or reliability checks of sources show similar results.

Validity & Reliability (Qualitative Research Vs. Quantitative Research)

Quantitative	Qualitative
Internal Validity	Credibility
External validity	Transferability
Reliability	Dependability
Objectivity	Confirmability

Important points to remember about qualitative research

√Direct Observations – The researcher has to record everything (recall bias?).

√Detailed Data

√Holistic

√Validity and actor's perceptions

√Hawthorne effect

Important points to remember about qualitative research cont.

√ Time requirement

√ Presentation of results

√ Reliability – Working alone, therefore verifying might be a bit problematic.

√ Interviewer's effect.

√ Invasion of privacy

√ Gaining access to data

Guidelines for Critiquing Qualitative Designs

1. Does the phenomenon lend itself to study by qualitative methods, or would a quantitative approach have been more appropriate?
2. Does the study focus on the subjective nature of human experience?
3. Is the specific qualitative approach named and described?
4. Will the study findings have significance for the profession of physiotherapy?
5. Does the researcher clearly describe how participants were selected?
6. How was the sample size determined?

Guidelines for Critiquing Qualitative Designs cont.

7. Is the data collection and recording process fully presented?
8. Is it clear how researcher bias in data collection was avoided?
9. Is the data analysis method consistent with the purpose and approach of the study?
10. Are the study findings presented and study limitations acknowledged?
11. Based on the study findings, are suggestions made for further research?

Main Components of the Proposal

Chapter 3: Methodology	
Section	Description
3.1	Study design
3.2	Sample and sampling
3.3	Ethical consideration
3.4	Study instruments (Study Tools)

- Sampling method
- Questionnaire
- Physical measurements
- Biochemical measures (If any)

Main Components of the Proposal

Chapter 3: Methodology	
Section	Description
3.5	Quality Control
3.6	Data collection • Data handling
3.7	Statistical analysis
3.8	Research flow chart • (Optional)

10

Epidemiological Methods

Descriptive studies

A. Individual
   i. Case report
   ii. Case series
   iii. Cross-sectional

B. Population
   iv. Ecological (Correlational)
   v. Comparative cross-sectional

Analytic studies

A. Observational studies
   i. Cross–sectional
   ii. Case–control
   iii. Cohort

B. Experimental studies
   iv. Therapeutic trials
   v. Preventive trial

Epidemiological Studies

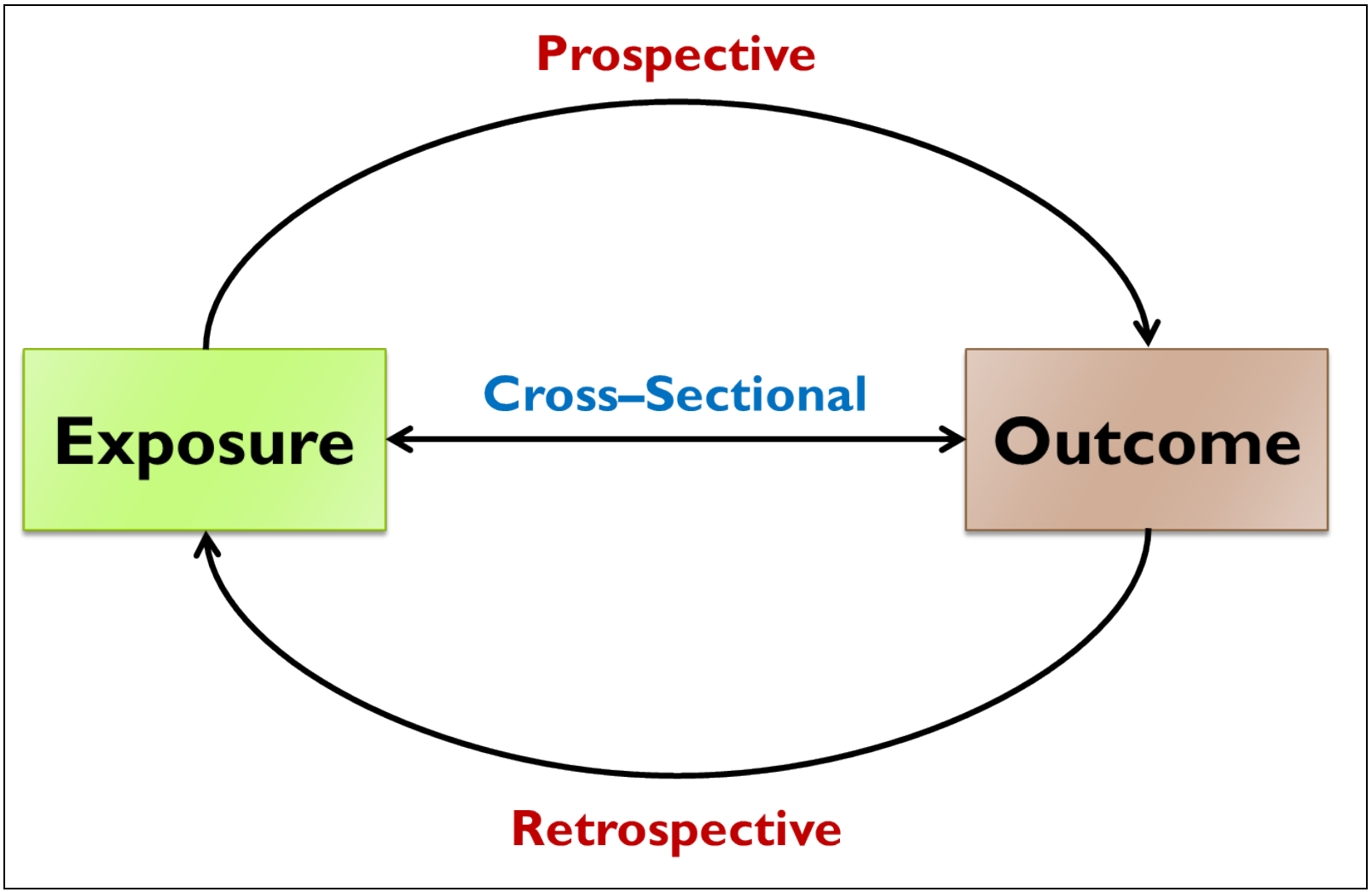
1.. Descriptive Studies: These studies organize and summarize health-related data according to time, place, and person.

2.Analytic Studies:

a) An experimental trial, also known as a clinical trial, is one in which subjects are exposed to N-exposure randomized groups for some time and the results are compared.

b) Observational (Non-intervention): Changes or differences in one characteristic are studied about changes or differences in others without the investigator's intervention.

The major types of the observational studies



Exploratory studies

An exploratory study is a small-scale, relatively short-duration study carried out when little is known about a situation or a problem.

It may include a description as well as a comparison.

If the problem and its contributing factors are poorly defined, it is always advisable to do an exploratory study before embarking on a large-scale descriptive or comparative study.

10.1.

Descriptive Studies

Descriptive studies

Descriptive studies may be defined as studies that describe the patterns of disease occurrence and other health-related conditions by person, place, and time.

Personal variables include basic demographic factors, such as age, sex, marital status or occupation, and the consumption of various types of food or medication use.

Characteristics of place refer to the geographic distribution of disease, including variation among or within countries, such as between urban and rural areas.

Concerning time, descriptive studies may examine seasonal patterns in disease onset, etc.

Uses of descriptive studies

1. They can be done quickly and easily.
2. Allow planners and administrators to allocate resources
3. Provide the first important clues about possible determinants of disease (useful for formulating hypotheses).

Types of descriptive studies:

1. Case reports and case series
2. Ecological studies
3. Cross-sectional studies (Remember !!)

1. Case reports and Case series

Case report: a careful, detailed report by one or more clinicians of the profile of a single patient.

The individual case report can be expanded to a case series describing the characteristics of several patients with a given disease.

Uses

- Important link between clinical medicine and epidemiology
- One of the first steps in outbreak investigation
- Often useful for hypothesis generating and examining new diseases, but conclusions about etiology cannot be made.

2. Ecological studies

Data from entire populations are used to compare disease frequencies between different groups during the same period or in the same population at various points in time.

Example: Countries with low cigarette consumption have lower lung cancer rates than those countries with high cigarette consumption.

Ecological studies are usually quick and easy to do and can be done with already available information.

Since ecological studies refer to whole populations rather than individuals, it is impossible to link exposure to disease occurrence in the same person.

3. Cross-sectional studies – 1

A cross-sectional (prevalence) study provides information concerning the situation at a given time.

In this type of study, the status of an individual concerning the presence or absence of exposure and disease is assessed simultaneously.

- Usually involves the collection of new data.
- In general, measure prevalence rather than incidence.
- It is not suitable for studying rare diseases or diseases with short durations; it is also not ideal for studying rare exposures.

For factors that remain unaltered over time, such as sex, blood group, etc., the cross-sectional survey can provide evidence of a valid statistical association.

3. Cross-sectional studies – 2

As can be noted from the above explanation, a cross-sectional study can be either analytical or descriptive according to its purpose.

If data are collected on exposures and outcomes of interest, and if the data are analyzed to demonstrate differences between exposed and non-exposed groups concerning the outcome or between those with the outcome and those without the outcome concerning the exposure, then this is an analytical cross-sectional study.

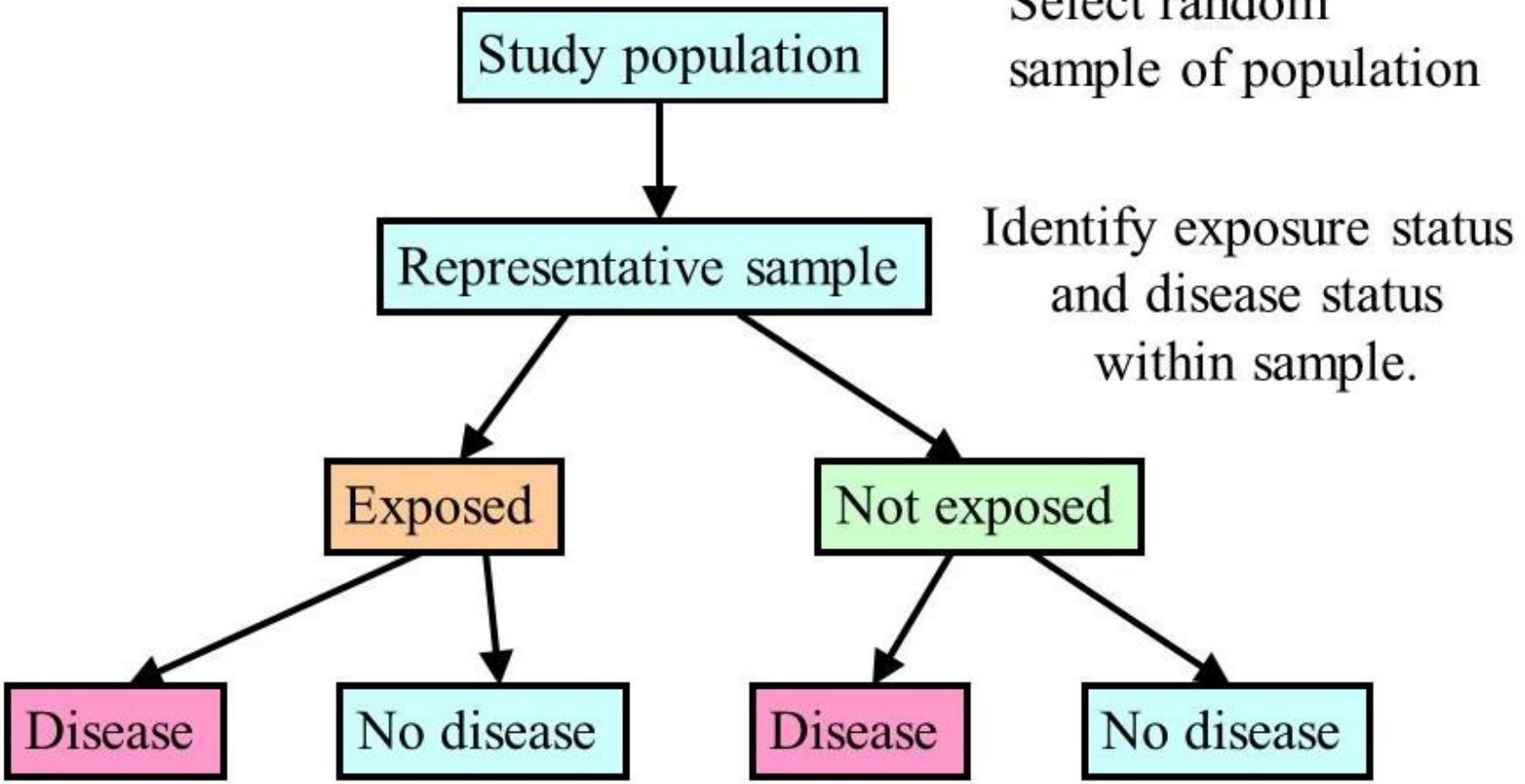
3. Cross-sectional studies – 3

If the information collected is purely of a descriptive nature, not involving the comparison of groups formed based on exposure or outcome status, then this is a descriptive cross-sectional study.

Often, a cross-sectional study may have both descriptive and analytical components.

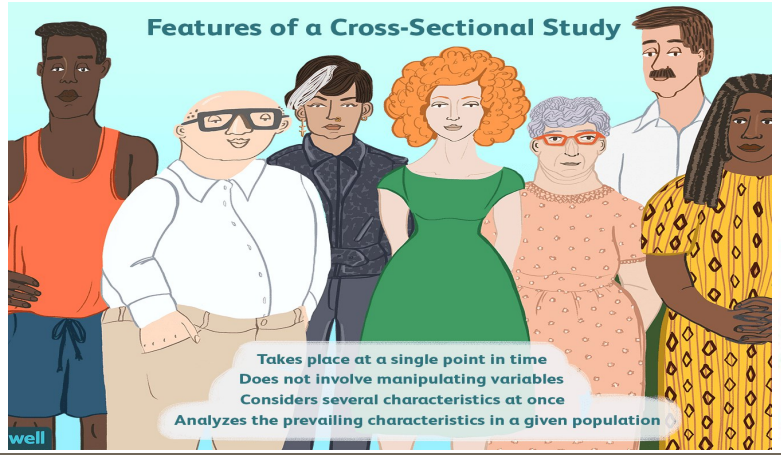
Nowadays, there is an increasing emphasis on the value of longitudinal studies in which observations are repeated in the same community over a prolonged period (i.e., longitudinal studies provide the required data at more than one point in time unlike cross–sectional surveys).

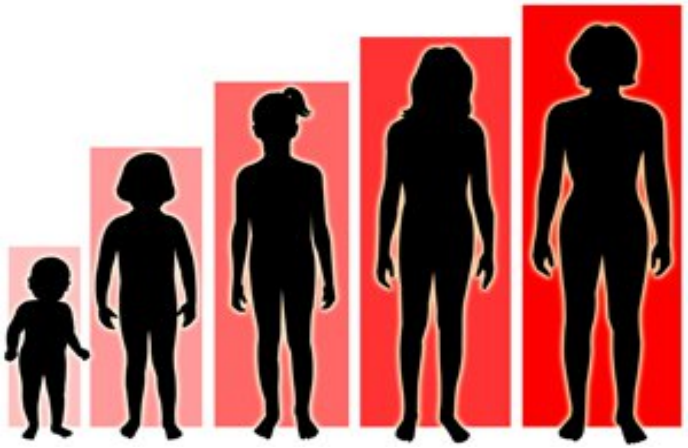
Cross-Sectional Design



Differences between cross-sectional and longitudinal studies

Feature	Cross-sectional	Longitudinal studies
Timeline	One in point in time	Multiple points in time
Sample	Different	Same
Results	Delivers snapshot in given point of time	Provides details of changes over time





Examples of cross-sectional – 1

Master Thesis (1999)

A Researcher from Gaza completed a cross-sectional study to determine the prevalence of low back pain (LBP) among health professionals at the health department, UNRWA, Gaza

The main results revealed that 71% of the interviewed staff reported a history of LBP at some time in life and 37.1% point prevalence at the time of applying the questionnaire.

The same study indicates that psychosocial factors, sitting posture, and bending activities are risk factors for LBP.

Examples of cross-sectional – 2

https://doi.org/10.3329/jhpn.v31i4.20031

Ihab Naser (2013) examined nutritional outcomes of household food insecurity among mothers in rural Malaysia.

A cross-sectional survey of low-income households was conducted, and 223 households with mothers aged 18–55 years, who were non-lactating, non-pregnant, and had at least one child aged 2–12 years, were purposively selected.

Of the households, 16.1% were food-secure, whereas 83.9% experienced some kind of food insecurity: 29.6% of households were food-insecure, 19.3% contained food-insecure individuals, and 35.0% fell into the 'child hunger' category.

Examples of cross-sectional – 3

https://doi.org/10.1007/s10389-012-0540-9

Marwan Jalambo determined the prevalence of anemia among female secondary students in the Gaza Strip and assessed the important risk factors relevant to the development of anemia.

Self-administered questionnaires collected data in a sample of 316 students aged 15–18.

The prevalence of anemia among students was 33.5%. Anemia was found to be more prevalent among adolescents with low socioeconomic status. Anemia was found to be statistically significant regarding the father's job status, monthly expenditure average, and individual monthly expensing. Skipping breakfast was reported by 60.8 % students and 71.2 % ate only 1–2 meals/day.

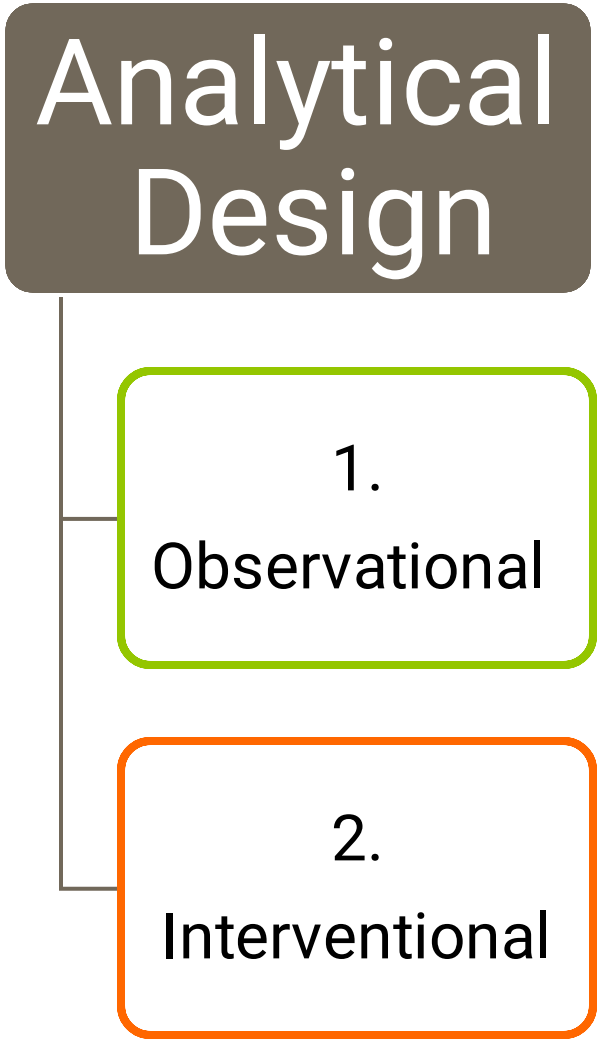
10.2.

Analytical Studies

Analytical Studies

Analytic studies may be used to test hypotheses concerning the relationship between a suspected risk factor and an outcome and to measure the magnitude of the association and its statistical significance.

Analytic study designs can be divided into broad design strategies: Observational and Interventional.



1. Observational studies

No human intervention is involved in assigning study groups; observe the relationship between exposure and disease.

Subject to many potential biases, but by careful design and analysis, many can be minimized.

Examples of observational studies:

A. Comparative cross-sectional (Review the previous section)
B. Cohort (Prospective),
C. Case-control studies (Retrospective).

Observational Studies – 1

Observational studies draw inferences about the effect of an "exposure" or intervention on subjects, where the investigator observes the assignment of subjects to groups rather than manipulates (e.g., through randomization).

Observational research involves the direct observation of individuals in their natural setting.

As such, who does or does not receive an intervention is determined by individual preferences, practice patterns, or policy decisions.

It is, therefore, important for readers of observational research to consider if alternative explanations for study results exist.

This issue (known as "confounding") is a primary challenge of observational research.

Observational Studies – 2

Data for observational research is either collected by the investigator for the purpose of the study (primary data) or has already been collected for another purpose but is used by the investigator to examine a novel research question (secondary data).

The primary trade-offs between using primary and secondary data relate to time, resources, and control of the collection and measurement of study variables.

A common source of secondary data used for observational research is administrative data.

For example, data from Medicare claims allow researchers to study the healthcare utilization of large groups of individuals.

Observational Studies – 3

Studies using Medicare claims data are observational because the investigator observes the subjects' health care utilization without any contact or involvement with them.

It is secondary data because the data were collected by the Centers for Medicare and Medicaid Services for purposes other than the investigators' study.

Other data sources typically used for observational research include hospital administrative databases, data obtained from medical chart reviews, or data obtained from previously conducted research studies.

A. Comparative cross-sectional

A cross-sectional study is observational in which exposure and outcome are determined simultaneously for each subject. It is often described as taking a “snapshot” of a group of individuals.

The cross-sectional study design has been used to understand the prevalence of various conditions, treatments, services or other outcomes and the factors associated with such outcomes.

The cross-sectional study design was efficient for evaluating a large sample of patients receiving hospice, understanding the prevalence of specific services, and generating hypotheses.

Similarly, we used a cross-sectional design to estimate the association between hospice ownership and the provision of specific types of hospice services.

Issues to consider in evaluating a cross-sectional design – 1

First, the primary limitation of the cross-sectional study design is that because the exposure and outcome are simultaneously assessed, there is generally no evidence of a temporal relationship between exposure and outcome.

That is, although the investigator may determine that there is an association between an exposure and an outcome, there is generally no evidence that the exposure caused the outcome.

If the exposure is a characteristic such as gender or race and the outcome developed over time, the temporal nature of the exposure-outcome association is more plausible; however, for studies in which the exposure is not an inherent trait but one that developed over time, causality is often unclear.

Issues to consider in evaluating a cross-sectional design – 2

Second, a cross-sectional study evaluates prevalent rather than incident outcomes and thus excludes people who develop the outcome but die before the study.

In a cross-sectional study, the measured association is between exposure and having the outcome instead of exposure and developing the outcome.

As such, there is a bias toward including individuals with more favourable survivorship in the study.

For example, early cross-sectional studies that observed the beneficial effects of postmenopausal estrogen use on cardiovascular disease in older women failed to account for the increase in cardiovascular-related mortality that, as a result of randomized studies that followed women over time, we now know occurs within the first several years of hormone replacement therapy.

Issues to consider in evaluating a cross-sectional design – 3

Third, the reader needs to assess if alternative explanations for study results have been appropriately ruled out.

Issues to Consider: Cross-sectional Study

1. If causality is asserted, is it appropriate?
2. Is the distinction between prevalent and incident cases a factor?
3. Have alternative explanations been ruled out?

Bias in cross-sectional study

Nonresponse Errors

arises from the fact that there are usually differences between the ideal sample pool of respondents and the sample that responds to a survey.

Nonresponse bias usually cannot be avoided.

Affects negatively by creating errors in a statistical measurement.

Researchers must, therefore, account for nonresponse either when planning their survey or when analyzing their survey results.

Neyman Bias

Prevalence-incidence bias occurs when individuals with severe or mild disease are excluded, resulting in an error in the estimated association between an exposure and an outcome.

Prevalent cases of chronic diseases may not be representative of all cases.

- Early fatalities subjects die before they are diagnosed, OR
- Silent cases when the evidence of exposure disappears when the disease begins
- Occurs when there is a time gap between exposure and selection of study subjects and, in worst cases, have died
- Existing cases are those who survive.
- Are different from patients who have died due to severity and staging.

2 x 2 Table: cross-sectional design

	Outcome			
Exposure		Diseased	Not diseased	
	Yes	A	B	A+B
	No	C	D	C+D
		A+C	B+D	N

- Calculate the statistical values according to the formula:

$$\textbf{Prevalence Value (for diseased)} = \frac{(A+C)}{N}$$

$$\textbf{Prevalence of non-exposed among diseased} = \frac{C}{(A+C)}$$

B. Cohort studies

Enrollment into the study is based on exposure characteristics or membership in a group.

The outcomes such as Disease, Death, or other health-related outcomes are then ascertained and compared for those who were exposed and those who were not.

A cohort is a well-defined group of people who have had a common experience or exposure and are then followed up for the incidence of new diseases or events, as in a cohort or prospective study.

- A birth cohort is a group of people born during a particular period or year (cohort means a group of people).

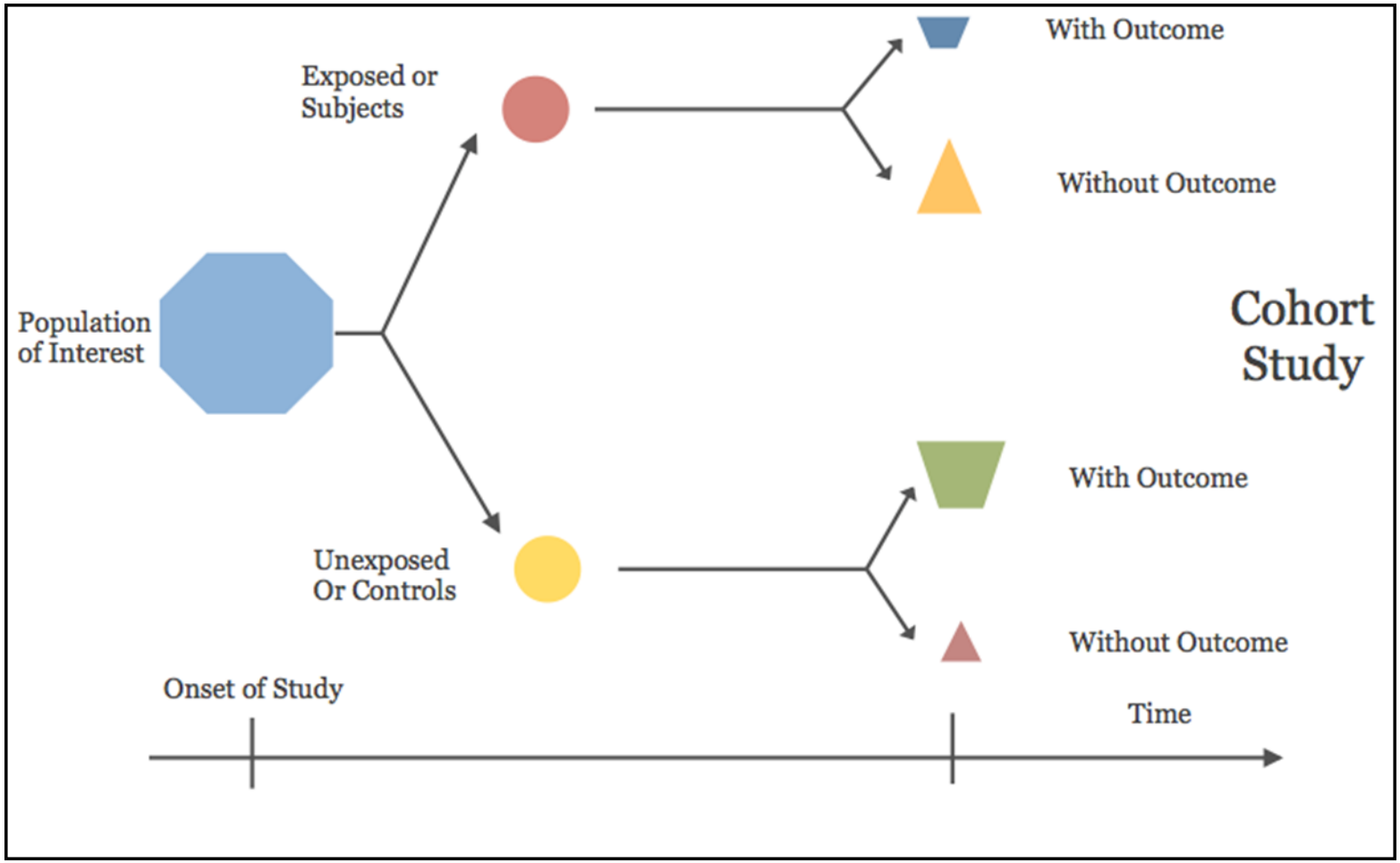
Cohort in Perspective

Cohort studies are a type of medical research used to investigate disease causes and establish links between risk factors and health outcomes.

These long-term studies are sometimes called longitudinal studies. An example of an epidemiological question that can be answered using a cohort study is:

- Whether exposure to X (e.g., smoking) associated with outcome Y (e.g., lung cancer)?
- The groups could be matched in terms of many other variables, such as economic status and other health status, so that the variable being assessed, the independent variable (e.g., smoking), can be isolated as the cause of the dependent variable (e.g., lung cancer).

Distribution of cohort studies subjects'

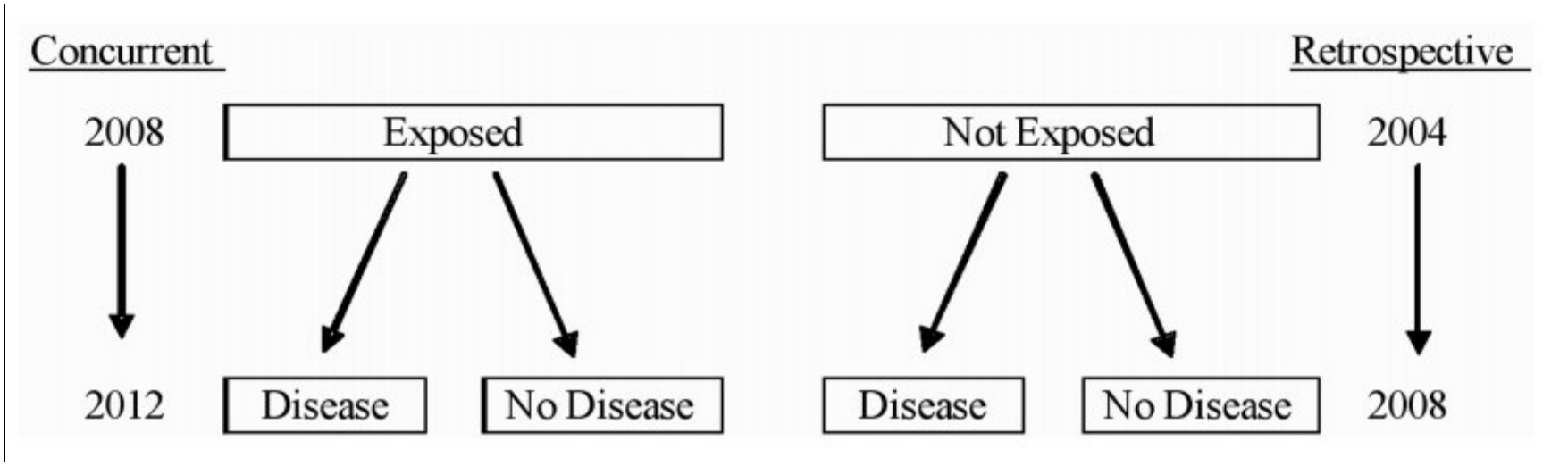


Facts about Cohort Studies – 1

The identifying feature of a cohort study design is that the subjects are followed over time.

Cohort studies begin with individuals who are exposed and not exposed to a factor and then evaluate the subsequent development of an outcome.

Cohort studies may be concurrent or retrospective, the distinction being when, relative to the current time, the subjects are identified.



Facts about Cohort Studies – 2

Cohort studies are an appropriate study design when:

1. There is good evidence to suggest an association between an exposure and an outcome;
2. The interval between exposure and development of the outcome is relatively short to minimize loss to follow-up;
3. The outcome is not too rare (so that the size of the cohort is reasonable).

The advantages of the cohort study design are that because the investigator identifies new or “incident” cases of the outcome, one can look at disease progression, staging, and natural history.

Facts about Cohort Studies – 3

Cohort designs can yield incidence rates and relative risks, and cohort studies may be able to assess causality due to the temporal nature of the study design.

Issues to Consider: Cohort Study

1. Was there a significant loss to follow-up?
2. Was the loss to follow-up differential?
3. Was there potential bias in outcome assessment?
4. Have alternative explanations been ruled out?

Strengths

- Is of particular value when the exposure is rare
- Can examine multiple effects of a single exposure
- Allows direct measurement of incidence of disease in the exposed and non-exposed groups.

Limitations

- Is inefficient for the evaluation of rare diseases
- Expensive and time consuming
- Validity of the results can be seriously affected by losses to follow-up.

2 x 2 Table: cohort design

		Developing Disease		
Exposure		Yes	No	
	Yes	A	B	A+B
	No	C	D	C+D
		A+C	B+D	N

- **Incidence among Exposed** = $\frac{A}{A+B}$
- **Incidence among non Exposed** = $\frac{C}{C+D}$
- **Relative Risk** = $\frac{Incidence\ among\ Exposed}{Incidence\ among\ non\ Exposed}$

Examples of Cohort studies

1.. In 1951, the British Doctors Study began a cohort that included both smokers (the exposed group) and non-smokers (the unexposed group). The study continued through 2001. By 1956, it provided convincing proof of the association of smoking with the incidence of lung cancer.

2. The Framingham Heart Study is a long-term, ongoing cardiovascular cohort study of residents of the city of Framingham, Massachusetts. The study began in 1948 with 5,209 adult subjects from Framingham, and is now on its third generation of participants. Prior to the study almost nothing was known about the epidemiology of hypertensive or arteriosclerotic cardiovascular disease. Much of the now-common knowledge concerning heart disease, such as the effects of diet, exercise, and common medications such as aspirin, is based on this longitudinal study.

C. Case–Control Study

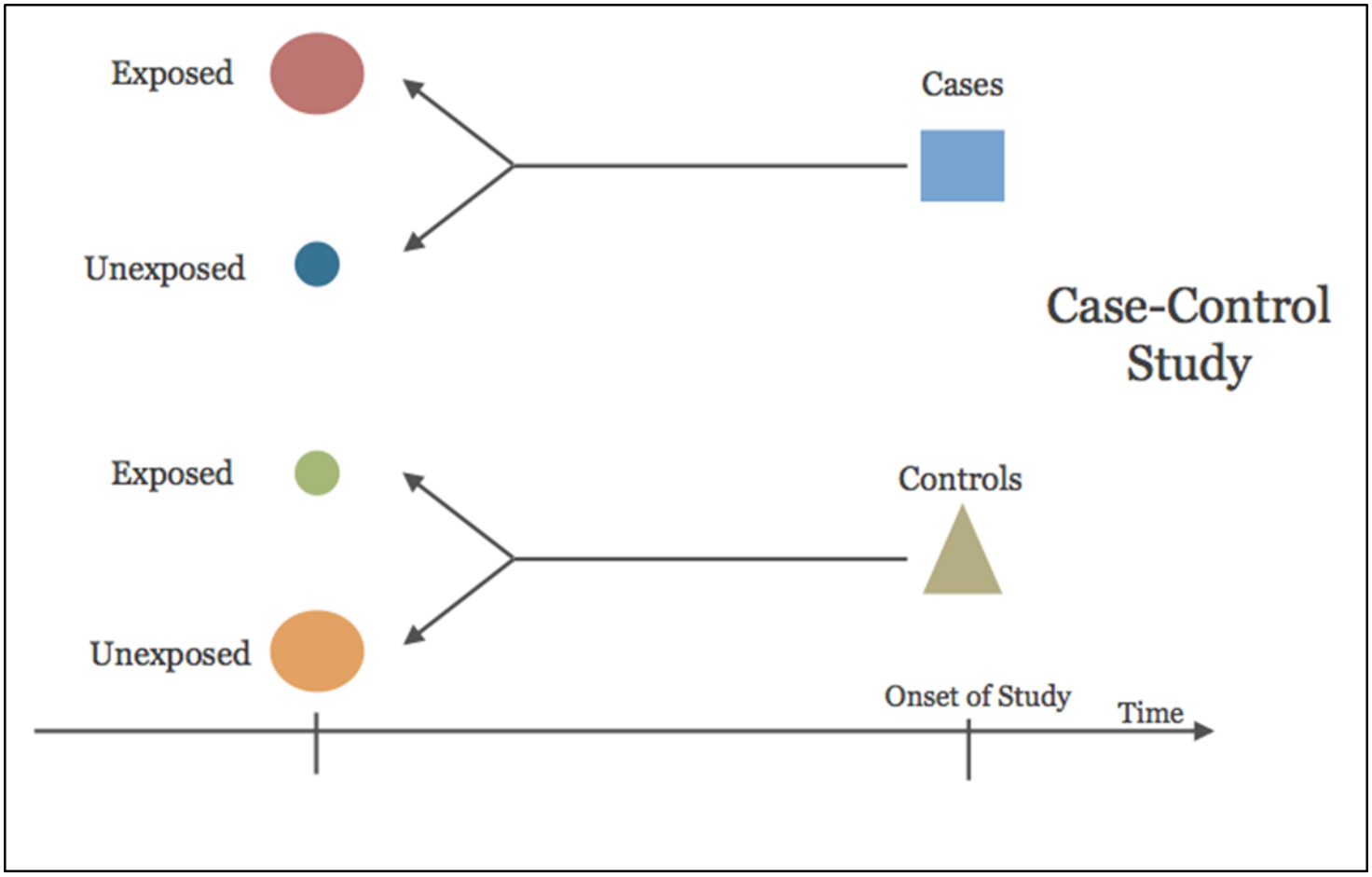
Enrollment in the study is based on the presence of an outcome (case) or the absence of the same outcome as a disease (control).

In a Case-Control Study, the investigator starts with a group of subjects who have already experienced a problem of concern (cases) and then selects a second group of people who have not (controls).

Both groups are compared for history of exposures.

Using the retrospective design to describe the act referring to that case and control in terms of the antecedent of factors believed to cause the group difference.

Distribution of case-control studies subjects'



Facts about case-control studies – 1

Case-control studies begin with individuals who have the outcome (“cases”) and compare them to individuals who do not have the outcome (“controls”) according to their history of exposure to a factor.

Case-control studies are appropriate when:

1.. The outcome is rare

2. there is reliable evidence of past exposure

One issue to consider in ascertaining past exposure is recall bias.

Past exposure is generally ascertained by interviewing subjects or analyzing historical records or charts.

If cases and controls differentially recall past exposures or there is more or less thorough documentation on cases compared to controls, study results may be biased.

Facts about case-control studies – 2

For case-control studies, the general concern is that cases will be more likely than controls to recall past exposures because they have already considered the potential causes of their disease.

Similarly, interviewer bias occurs when study investigators interview cases more thoroughly regarding past exposures than controls because they know the subject is a case.

Readers of case-control studies should consider the potential extent of recall or interviewer bias and whether study investigators attempted to mitigate these issues.

Facts about case-control studies – 3

Sample selection in a case-control study is complex.

Cases may be selected from various sources, including hospital patients, patients in a physician's practice, clinic patients, and cancer registries.

It is desirable to Selecting cases from multiple institutions (e.g., multiple hospitals in the community instead of one hospital) is desirable to obtain more generalizable results.

Criteria for case eligibility should be carefully specified in the Methods section.

It is preferable to use incident ("newly diagnosed") cases so that the risk factors identified are not related to survival with the outcome as opposed to the development of the outcome.

Strengths

- Is relatively quick and inexpensive
- Is optimal for the evaluation of rare diseases.
- Can examine multiple etiologic factors for a single disease.

Limitations

- Is inefficient for the evaluation of rare exposures
- Cannot directly compute disease incidence rates in exposed and non-exposed individuals.
- Is prone to bias compared with other analytic designs, particularly selection and recall bias.

Selection of the cases and controls – 1

The most critical issue to consider in critically evaluating a case-control study is the process by which controls were selected and the resulting comparability of cases and controls.

The selection of controls is the most complex and controversial aspect of conducting a case-control study.

Controls should be similar to cases in all respects other than having the disease or identical to the general population from which the cases arose.

Common sources of controls include the case's spouse, friend, or neighbour, an individual hospitalized at the same time as the case but for a different reason, or an individual chosen randomly from the general population.

Selection of the cases and controls – 2

Controls were chosen from the general population to be often ascertained from random digit dialling.

However, the increased use of answering machines, "Do Not Call" lists, and cell phones has rendered random digit dialling a less effective option as individuals reached by landline phone may no longer represent the general population.

Some case-control studies use matching, which refers to selecting controls similar to cases in specific characteristics (e.g., race, age, gender, socioeconomic status).

Cases that cannot be matched are often excluded from the analyses.

Readers should determine the proportion of cases excluded from the analyses because a high proportion could limit

Selection of the cases and controls – 3

A. Cases:

1. Diagnostic criteria
2. Sources: hospitals - general population.
3. Incident and prevalent cases

B. Control:

4. Comparability to the cases except having the Disease
5. Representative of all non-diseased population
6. Practical: feasibility – Fund
7. Sources of control:
   a) Population of defined area.
   b) Hospital patients
   c) Neighbors: walk phone letter
   d) Friends or associates of cases
   e) Siblings, spouses, and relatives

Methodological issues of Cases and controls

1. Matching:
   1. Group matching.
   2. Individual matching.

   Problems:

   3. Matching for many variables: make it difficult to find appropriate control.
   4. Can not explore possible association of disease with any variable on which cases and controls have been matched.
2. Multiple controls:

   Similar or different types.

Examples of case-control studies – 1

https://doi.org/10.1016/j.clnesp.2021.03.007

Objective: To evaluate the level of vitamin C deficiency and its association with metabolic complications in adults with and without chronic respiratory diseases (Asthma and COPD) in the Gaza Strip.

Methods: A matched case-control study, including 52 cases of CRD patients and 52 controls of apparently healthy participants, was matched with cases by age, gender, waist circumferences, and body mass index, aged 18-60 years, conducted on three centres of the secondary health care system of the ministry of health (MOH) in Gaza Strip.

Examples of case-control studies – 2

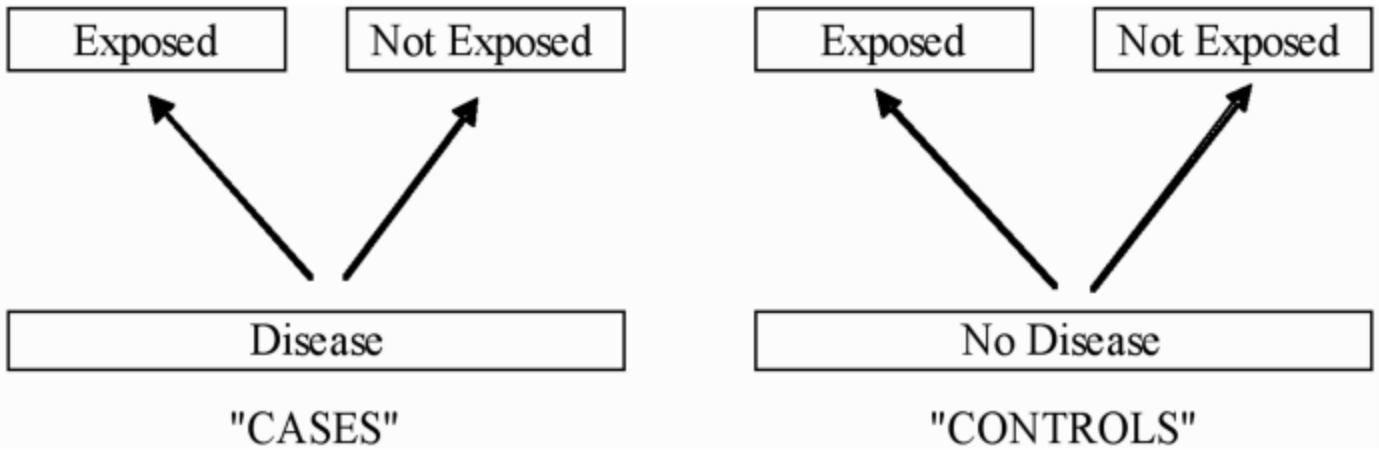
https://doi.org/10.1016/j.clnu.2018.06.961

Background & aims: Nutrient pattern analysis is an easy way to compare nutrient intakes across different nations due to nutrients' universality. This study aimed to investigate the relationship between patterns of nutrient intake and Glioma in a case-control study of Iranian adults.

Methods: Hospital-based case-control study, enrolled 128 pathologically confirmed new cases of Glioma and 256 age and sex-matched controls. The dietary intakes of study participants were assessed using the validated Block-format 123-item semi-quantitative FFQ. Data on potential confounders were also collected through the use of the pre-tested questionnaire.

Glioma is a type of tumor in the brain and spinal cord.

Design of case-control



Issues to Consider: Case-Control Study

1. Do controls come from the same hypothetical population as cases?
2. Is recall or interviewer bias an issue?
3. Have alternative explanations been ruled out?

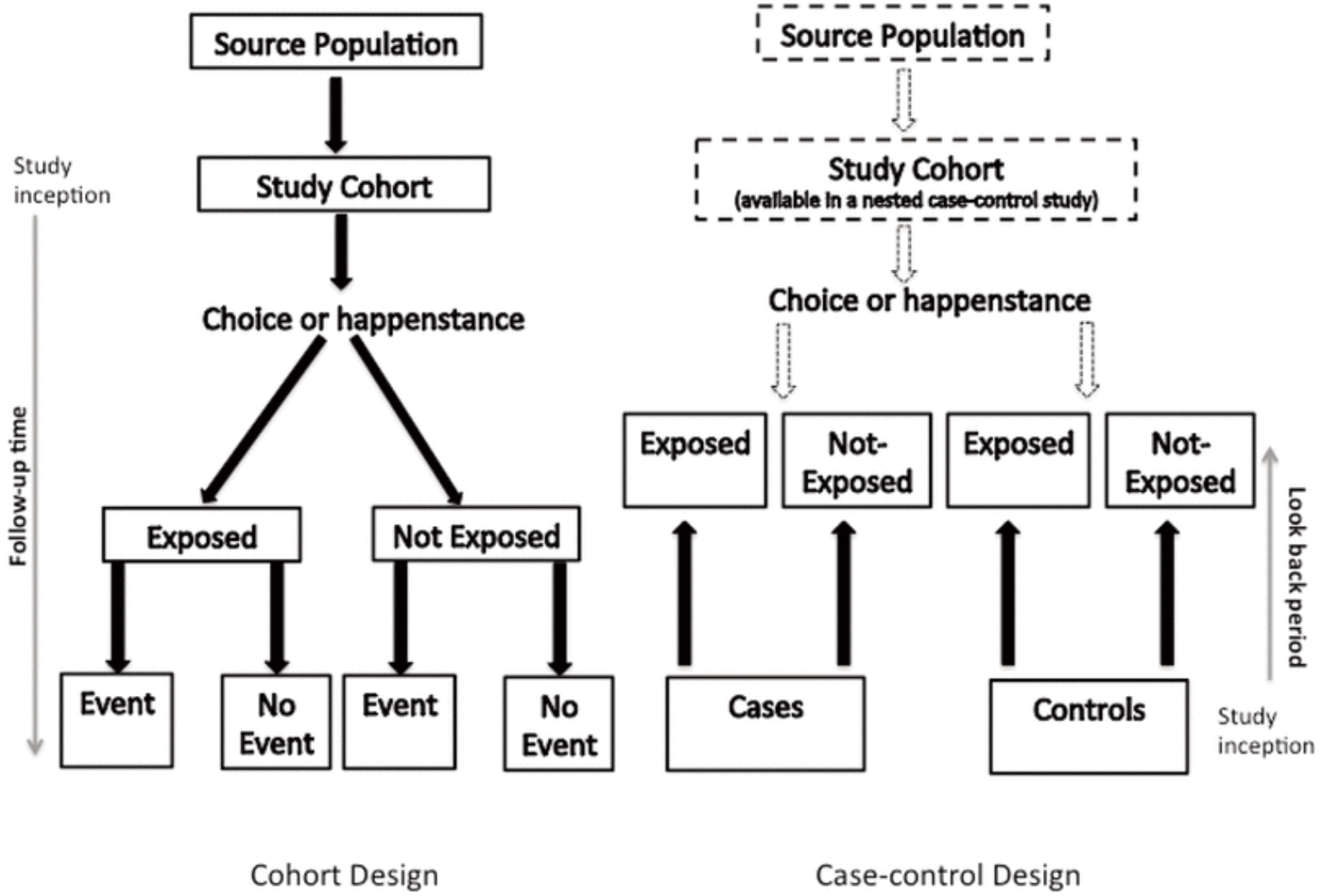
2 x 2 Table: case-control design

		Population		
Exposure		Case	Control	
	Yes	A	B	A+B
	No	C	D	C+D
		A+C	B+D	N

- Calculate the Odd's Ratio according to the formula:

$$\textbf{Odd's Ratio} = \frac{A \times D}{B \times C}$$

Observational Analytical Studies



2. Intervention studies (Clinical Trials)

- In intervention studies, the researcher manipulates a situation and measures the effects of this manipulation.
- Usually (but not always), two groups are compared: one group in which the intervention takes place (e.g. treatment with a certain drug) and another group that remains 'untouched' (e.g. treatment with a placebo).

- The three categories of intervention studies are:
  A. Experimental studies
  B. Quasi-experimental studies
  C. One group intervention

Ethical Considerations in a Clinical Trial

1. Is the proposed treatment safe (unlikely to bring harm) to the Patient?
2. For the sake of a controlled trial, can a treatment ethically be withheld from any patient in the doctor's care?
3. What patients may be brought into a controlled trial and allocated to any of the different treatments?
4. Is it ethical to use a placebo or dummy treatment?
5. Is it proper for the trial to be in any way blind?

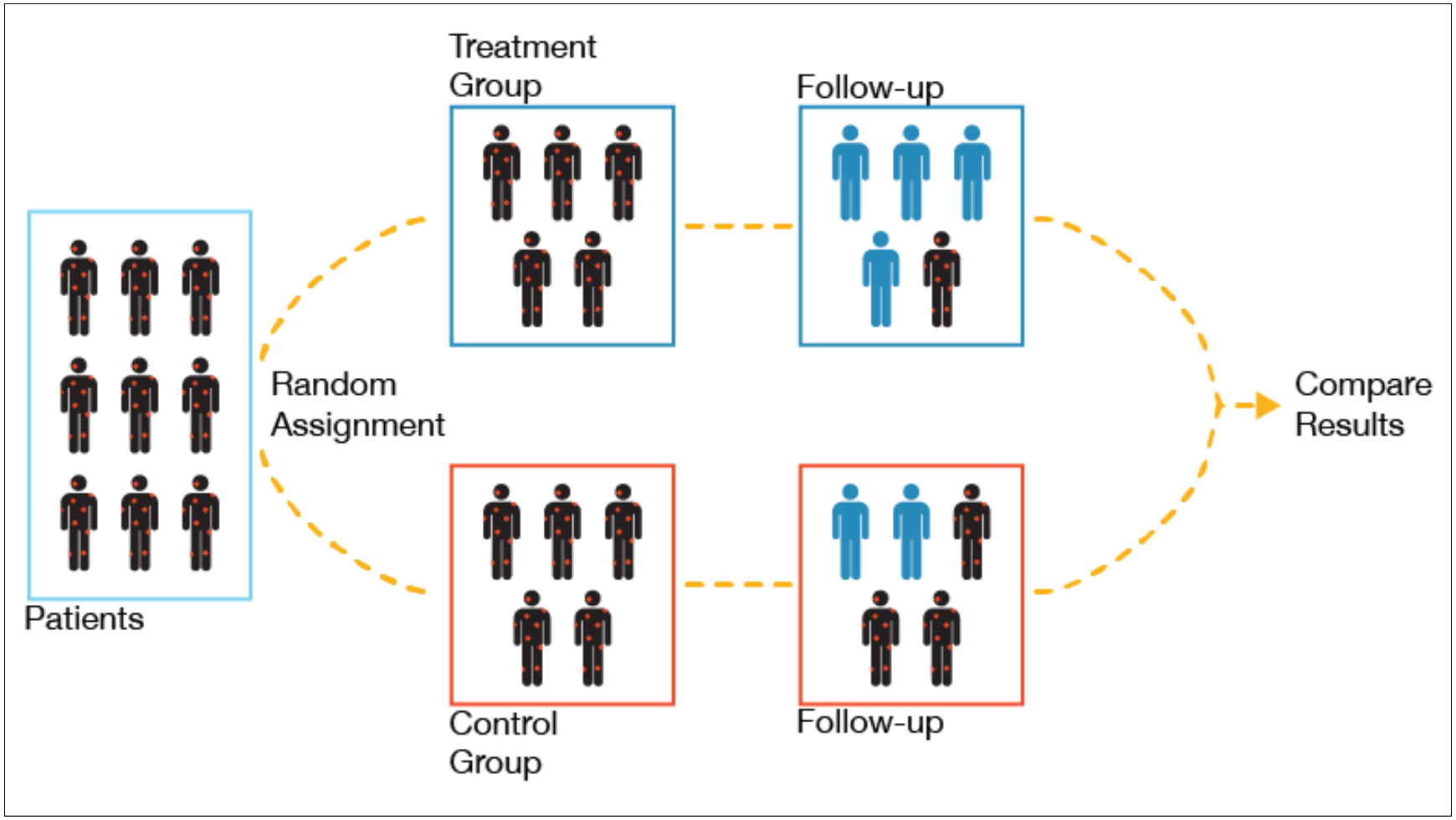
A. Experimental studies – 1

- An experimental design is a study design that gives the most reliable proof for causation.
- In an experimental study, individuals are randomly allocated to at least two groups.
- One group is subject to an intervention or experiment, while the other group(s) is not.
- The outcome of the intervention (effect of the intervention on the dependent variable/problem) is obtained by comparing the two groups.
- A number of experimental study designs have been developed. These are widely used in laboratory settings and clinical settings.

A. Experimental studies – 2

- For ethical reasons, the opportunities for experiments involving human subjects are restricted. However, randomized control trials of new drugs are common.
- At the community level, where health research is frequently undertaken, we experience ethical and practical problems in carrying out experimental studies.
- In real-life settings, assigning persons randomly to two groups or maintaining a control group is often impossible.
- Therefore, experimental research designs may have to be replaced by quasi-experimental designs.

Experimental studies



Example of RCT

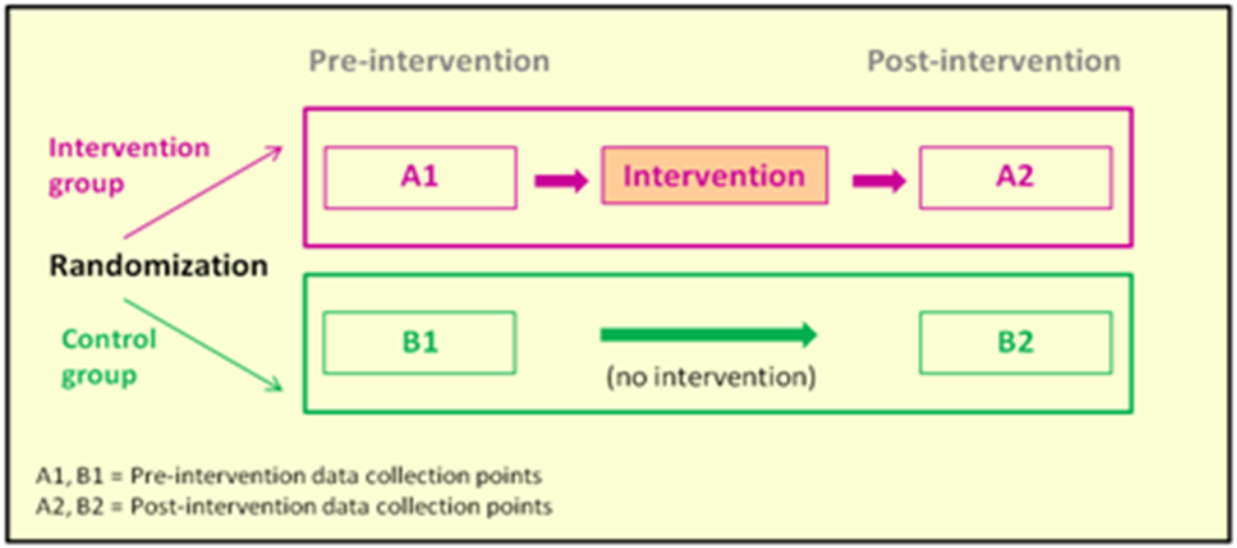
https://doi.org/10.3402/fnr.v60.29268

- Objective: To identify the effects of LC ω-3 PUFAs on reducing inflammatory markers in hypertensive and/or diabetic obese adults.
- Materials and methods: Sixty-four patients who were hypertensive and/or diabetic obese with high levels of inflammatory markers from primary healthcare centers of Gaza City, Palestine, enrolled in two groups of an open-label, parallel, randomized, controlled trial for 8 weeks. Thirty-three patients were in the control group, and 31 were in the experimental group. The experimental group was treated with a daily dose of 300 mg of eicosapentaenoic acid and 200 mg of docosahexaenoic acid.

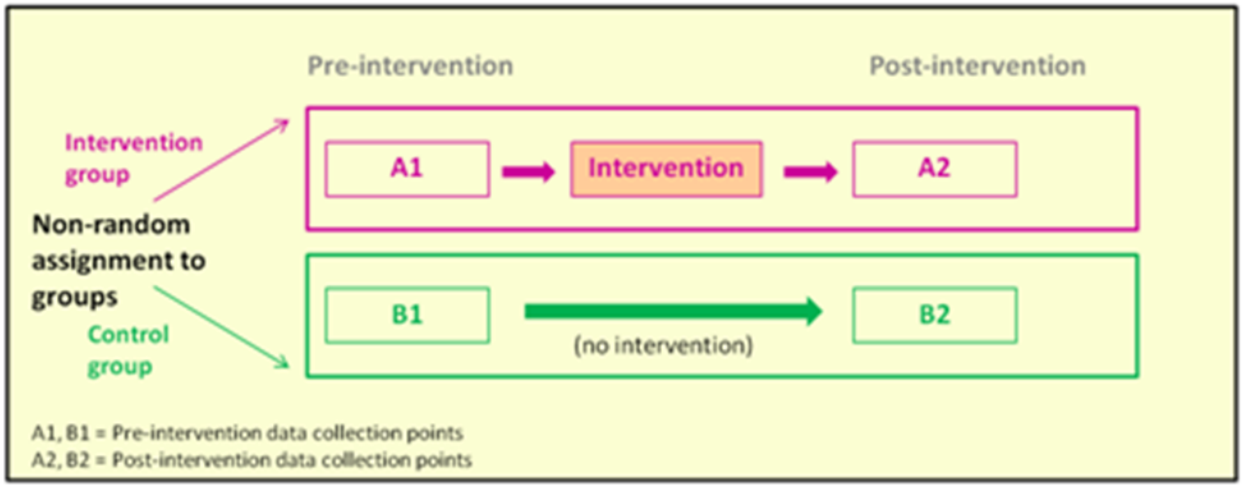
B. Quasi–experimental studies

- In a quasi-experimental study, one characteristic of a true experiment is missing, either randomization or the use of a separate control group.
- A quasi-experimental study, however, always includes the manipulation of an independent variable which is the intervention.
- One of the most common quasi-experimental designs uses two (or more) groups, one of which serves as a control group in which no intervention takes place.
- Both groups are observed before as well as after the intervention, to test if the intervention has made any difference.
- (This quasi-experimental design is called the 'non-equivalent control group design' because the subjects in the two groups (study and control groups) have not been randomly assigned.)

Classical Design of Randomized Experiments



Classical Quasi-Experimental Design



Example for quasi-experimental study

https://doi.org/10.1371/journal.pone.0213982

- A quasi-experimental study was conducted among 115 mild to moderately anaemic pregnant women attending antenatal clinics to assess the effect of nutrition education on nutritional knowledge, haemoglobin level and dietary intake of anaemic pregnant women.
- Pregnant women were consecutively enrolled and assigned to receive nutrition education and diet plans in the intervention group (n = 58) and general education only in the control group (n = 57).
- Nutrition education was given to pregnant women on an individual basis at the time of enrollment, and follow-ups were done through biweekly phone calls and every four weeks during ANC visits. Baseline data were collected using a semi-structured questionnaire for interviews, and haemoglobin level was also measured. Data were collected after 10 weeks of nutrition education intervention.

Similarities Between Experimental and Quasi-Experimental Research

- Similarities between experimental and quasi-experimental research can be seen in the following aspects:
  1. Cause-and-effect relationship is hypothesized
  2. Participants are randomly assigned (experimental) or non-randomly assigned (quasi-experimental)
  3. Application of an experimental treatment by the researcher
  4. Following the treatment, all participants are measured on the dependent variable
  5. Data are usually quantitative and analyzed by looking for significant differences in the dependent variable

C. One group intervention

- Another type of design often chosen because it is quite easy to set up uses only one group in which an intervention is carried out.
- The situation is analyzed before and after the intervention to test if there is any difference in the observed problem.
- This is called a 'BEFORE-AFTER' study.
- This design is considered a 'pre-experimental' rather than a 'quasi-experimental' design because it involves neither randomization nor using a control group.

About experimental studies

- Intervention (experimental) studies can also be considered therapeutic or preventive.

1. Therapeutic trials are conducted among patients with a particular disease to determine the ability of an agent or procedure to diminish symptoms, prevent recurrence, or decrease the risk of death from that disease.
2. A preventive trial (community trial) evaluates whether an agent or procedure reduces the risk of developing disease among people free from that condition at enrolment.
   - Thus, preventive trials can be conducted among individuals at usual risk (e.g. vaccine trials)

Types of clinical trials

Type	Examples
Therapeutic	1. Iron supplementation for the treatment
	2. Vitamin C to alleviate inflammatory
Intervention	1. Omega-3 fatty acids to reduce the risk of
	2. Physical exercise for decreasing the risk
Preventive	1. BCG vaccination for tuberculosis
	2. Salt fortification to prevent hypothyroidism

Internal and External Validity of Experimental Research

- When to use experimental research design :
  - If you strongly suspect a cause-and-effect relationship exists between two conditions and
  - The independent variable can be introduced to participants and can be manipulated, and
  - The resulting dependent variable can be measured for all participants
- Validity of research refers to the degree to which the conclusions are accurate and generalizable
- Both experimental and quasi-experimental research are subject to threats to validity
- If threats are not controlled for, they may introduce errors into the study, leading to misleading conclusions.

Threats to External Validity – 1

External validity: the extent to which the results can be generalized to other groups or settings

1. Population validity: degree of similarity among sample used, the population from which it came, and target population
2. Ecological validity: physical or emotional situation or setting that may have been unique to the experiment
   - If the treatment effects can be obtained only under a limited set of conditions or by the original researcher, the findings have low ecological validity.

Threats to External Validity – 2

3. Selection bias
   - If the sample is biased, you cannot generalize to the population.
4. Reactive effects
   - Experimental setting: Differs from natural setting.
   - Testing: Pretest influences how subjects respond to the treatment.
5. Multiple-treatment inference
   - If the subjects are exposed to more than one treatment, then the findings could only be generalized to individuals exposed to the same treatments in the same order of presentation.

Threats to Internal validity – 1

Internal validity: the extent to which differences in the dependent variable directly result from the manipulation of the independent variable.

1. History: when factors other than treatment can exert influence over the results; problematic over time
2. Maturation: when changes occur in the dependent variable that may be due to natural developmental changes; problematic over time
3. Testing: pretest may give clues to treatment or post-test and may result in improved post-test scores
4. Instrumentation: The nature of the outcome measure has changed.

Threats to Internal validity – 2

5. Regression: Tendency of extreme scores to be nearer to the mean at retest
6. Differential selection of participants: participants are not selected/assigned randomly
7. Attrition (mortality): loss of participants
8. Experimental treatment diffusion: Control conditions receive experimental treatment.

COMPARISON OF EXPERIMENTAL AND OBSERVATIONAL STUDY DESIGNS

	Experimental	*Observational*		
Study design	Randomized Control Trial	Cross-sectional	Cohort	Case-control
Study population	Highly selected population; highly controlled environment	Diverse population observed in a range of settings	Diverse population observed in a range of settings	Diverse population observed in a range of settings
Directionality	Exposure is assigned before outcome is ascertained	Exposure and outcome ascertained simultaneously	Exposure is ascertained before outcome is ascertained	Outcome is ascertained before exposure is ascertained
Primary Use	Demonstrating efficacy of an intervention	Screening hypotheses; prevalence studies	Assessing association between multiple exposures and outcomes over time	Assessing associations between exposures and rare outcomes
Analysis	Straight-forward	Sophisticated multivariate techniques may be required to account for confounding	Sophisticated multivariate techniques may be required to account for confounding	Sophisticated multivariate techniques may be required to account for confounding
Internal validity	High	Low	Low	Low
External validity	Low-Moderate	High	High	High

11

Methodological Issues of Clinical Trials

RCTs: Randomized Controlled Trials

General Principles of RCTs – 1

- The RCT is one of the most straightforward but powerful research tools.
- In essence, the RCT is a study in which people are randomly allocated to receive one of several clinical interventions.
- On most occasions, the term "intervention" refers to treatment, but it should be used in a much wider sense to include any clinical maneuver offered to study participants that may have an effect on their health status.
- Such clinical maneuvers include prevention strategies, screening programs, diagnostic tests, interventional procedures, the setting in which health care is provided, and educational models.
- RCTs in nutrition can play a major role in the assessment of screening programs, diagnostic tests, and procedures in intervention.

General principles of RCTs – 2

- RCTS are used to examine the effect of interventions on particular outcomes such as death or the recurrence of disease.
- Some consider RCTs to be the best of all research designs or “the most powerful tool in modern clinical research”, mainly because the act of randomizing patients to receive or not receive the intervention ensures that, on average, all other possible causes are equal between the two groups.
- Thus, any significant differences between groups in the outcome event can be attributed to the intervention and not to some other unidentified factor.

General principles of RCTs – 3

- However, RCTs are not a panacea to answer all clinical questions; for example, the effect of a risk factor such as smoking cannot ethically be addressed with RCTs.
- Furthermore, in many situations, RCTs are not feasible, necessary, appropriate, or even sufficient to help solve important problems.
- RCTs are not appropriate for cancer screening, a situation in which the outcome is rare and frequently occurs only after a long delay.
- Thus, although the test for appraising the ultimate value of a diagnostic test may be a large, well-designed RCT with patient outcomes as the end point, the trial should presumably be performed after other smaller studies have examined the predictive value of the test against some accepted standard.

General principles of RCTs – 4

- An overview of the publications on mammography screening is an excellent example of the controversies that can arise with RCTs.
- The most important references concern the article by Miettinen et al. [Lancet 2002;359:404–405] linking screening for breast cancer with mammography and an apparently substantial reduction in fatalities and the responses that it elicited.
- RCTs may not be appropriate for assessing interventions with rare outcomes or effects that take a long time to develop.
- In such instances, other study designs, such as case-control studies or cohort studies, are more appropriate.

General principles of RCTs – 5

- In other cases, RCTs may not be feasible because of financial constraints or because of the expectation of low compliance or high drop-out rates.
- Many RCTs involve large sample sizes because many treatments have relatively small effects.
- The size of the expected effect of the intervention is the primary determinant of the sample size necessary to conduct a successful RCT.
- Obtaining statistically significant differences between two samples is easy if large differences are expected.
- However, the smaller the expected effect of the intervention, the larger the sample size needed to be able to conclude, with enough power, that the differences are unlikely to be due to chance.

General principles of RCTs – 6

- For example, let us assume that we wish to study two groups of patients undergoing different interventions, one of which is a new procedure.
- We expect a 10% decrease in the morbidity rate with the new procedure.
- We need 80 patients in each treatment arm to detect this difference with a probability (power) of 80%.
- If the expected difference in effect between the two groups increases to 20%, the number of patients required per arm decreases to 40.
- Conversely, if the difference between the groups is expected to be only 1%, the study population must increase to 8,000 per treatment arm.
- The sample size required to achieve power in a study is inversely proportional to the treatment effect squared.

Randomization: The Strength of the RCT – 1

- The randomization procedure gives the RCT its strength.
- Random allocation means that all participants have the same chance of being assigned to each study group.
- The allocation, therefore, is not determined by the investigators, the clinicians, or the study participants.
- The purpose of the random allocation of participants is to ensure that the characteristics of the participants are as likely to be similar as possible across groups at the start of the comparison (also called the baseline).

Randomization: The Strength of the RCT – 2

- Proper randomization reduces the risk of a serious imbalance in known and unknown factors that could influence the clinical course of the participants.
- No other study design allows investigators to balance these factors.
- The investigators should follow two rules to ensure the success of the randomization procedure.
- They must first define the rules governing the allocation and then follow those rules strictly throughout the study.
- The crucial issue is that after the procedure for randomization is determined, it should not be modified at any point during the study.

Randomization: The Strength of the RCT – 3

- There are many adequate methods of randomization, but their common element is that no one should be able to determine ahead of time which group a given patient will be assigned.
- Numerous methods are also available to ensure that the patient sample is balanced whenever a small predetermined number of patients have been enrolled.
- Unfortunately, the allocation methods in randomized studies are poorly and infrequently reported.
- As a result, it is not possible to determine, on most occasions, whether the investigators used proper methods to generate random allocation sequences.

Bias in Randomized Controlled Trials – 1

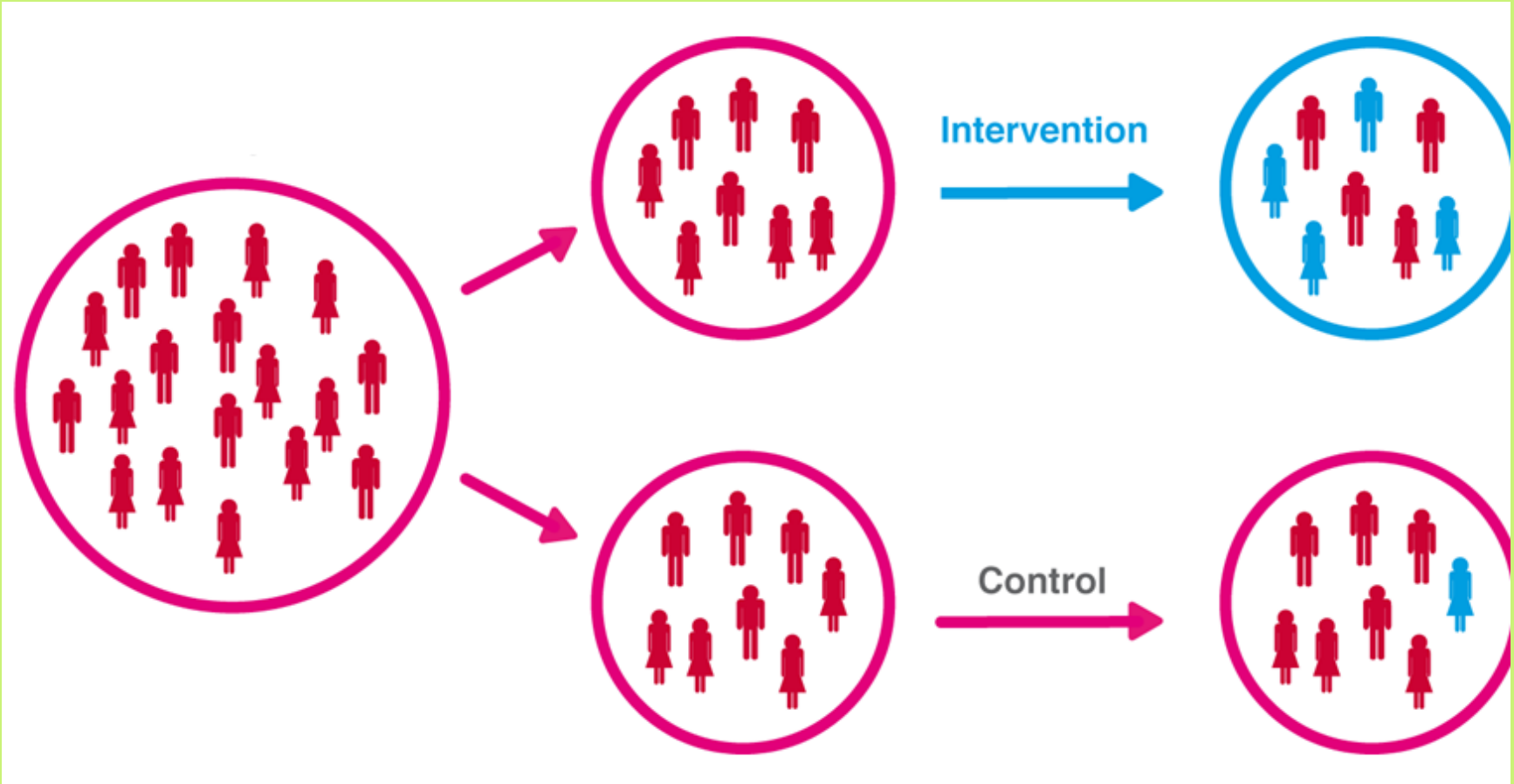
- The main appeal of the RCT in health care derives from its potential for reducing allocation bias.
- No other study design allows researchers to balance unknown prognostic factors at baseline.
- Random allocation does not protect randomized controlled trials against other types of bias.
- Recently, RCTs have been the subject rather than the tool of important, albeit isolated, research efforts usually designed to generate empiric evidence to improve the design, reporting, dissemination, and use of RCTs in health care.
- Such studies have shown that RCTs are vulnerable to multiple types of bias at all stages of their work span.

Bias in Randomized Controlled Trials – 2

- In summary, RCTs are quantitative, comparative, controlled experiments in which a group of investigators studies two or more interventions by administering them to groups of individuals who have been randomly assigned to receive each intervention.
- Alternatively, each individual might receive a series of interventions in random order (crossover design) if the outcome can be uniquely associated with each intervention through, for example, the use of a "washout" period.
- This step ensures that the effects from one test are not carried over to the next one and subsequently affect the independent evaluation of the second test administered.
- Apart from random allocation to comparison groups, the elements of a randomized controlled trial are no different from those of any other type of prospective, comparative, quantitative study.

Types of Randomized Controlled Trials

Use the following Book: Jadad AR. Randomised controlled trials: a user's guide. London, England: BMJ Books, 1998



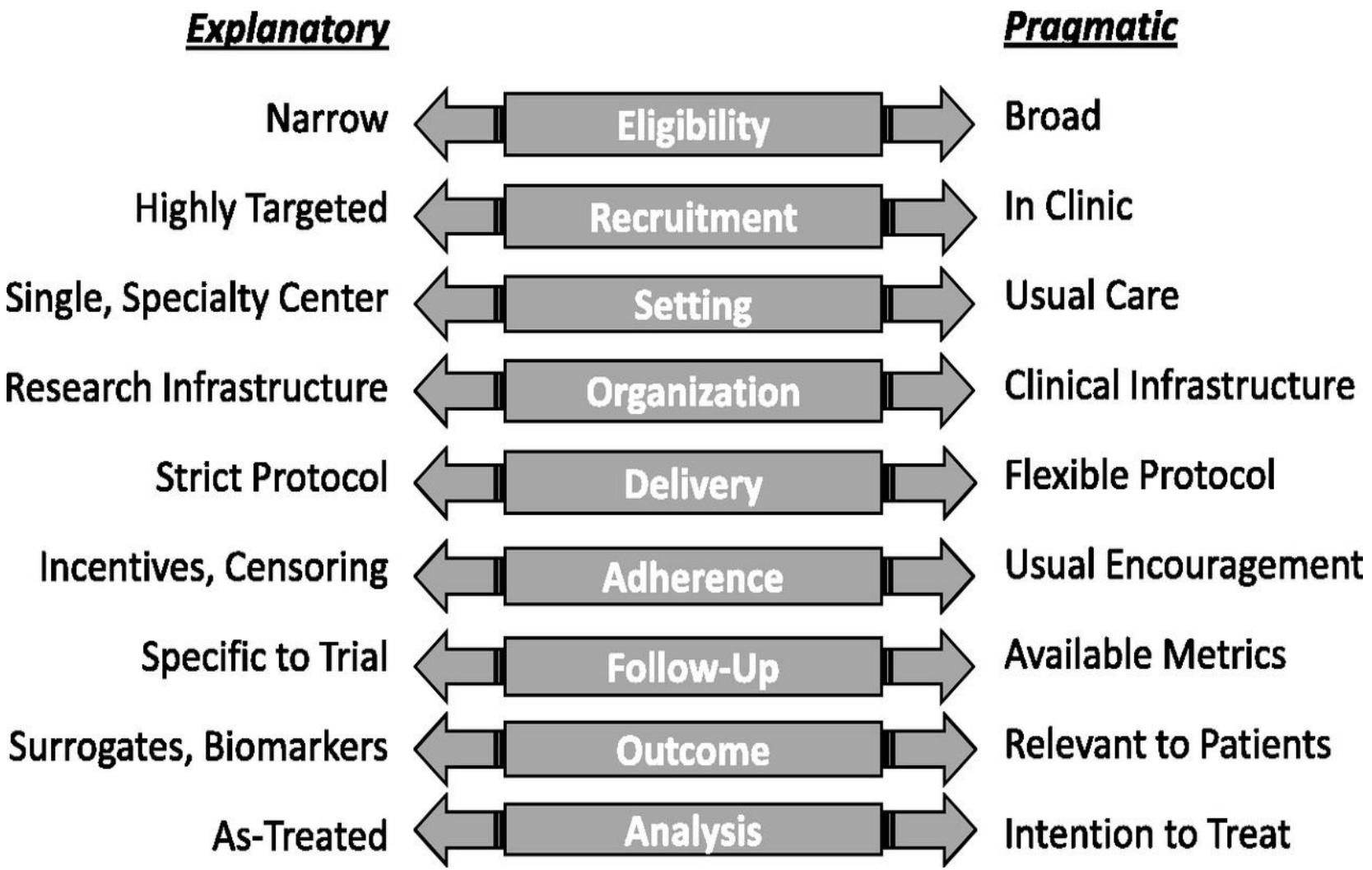
Types of Randomized Controlled Trials

Classification	Types
Classified according to the different aspects of interventions evaluated	Explanatory or pragmatic trials
	Efficacy or effectiveness trials
	Phase 1, 2, 3, and 4 trials
Classified according to participants' exposure and response to the intervention	Parallel design
	Crossover design
	Factorial design
Classified according to the number of participants	N-of-one trials
	Sequential trials
	Fixed trials
Classified according to the level of blinding	
Classified according to nonrandomized participant preference	

Explanatory or Pragmatic trials

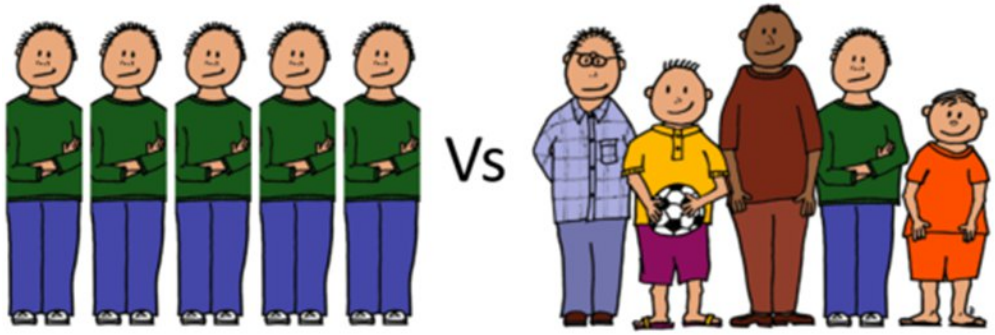
- Explanatory trials are designed to answer a simple question: Does the intervention work?
- If it does, the trial attempts to establish how it works.
- Pragmatic trials, on the other hand, are designed not only to determine whether the intervention works but also to describe all the consequences of the intervention and its use under circumstances corresponding to daily practice.
- Although both explanatory and pragmatic approaches are reasonable and even complementary, it is important to understand that they represent extremes of a spectrum, and most RCTs combine elements of both.

Explanatory or pragmatic trials



Efficacy or effectiveness trials

- RCTs are also often described in terms of whether they evaluate the efficacy or effectiveness of an intervention.
- Efficacy refers to interventions carried out under ideal circumstances, whereas effectiveness evaluates the effects of an intervention under circumstances similar to those found in daily practice



Efficacy vs. Effectiveness

Phase 1, 2, 3, and 4 trials – 1

- These terms describe the types of trials used to introduce a new intervention, traditionally a new drug. Still, they could also encompass trials used to evaluate a new embolization material or type of prosthesis, for example.
- Phase 1 studies are usually conducted after the safety of the new intervention has been documented in animal research, and their purpose is to document the safety of the intervention in humans.
- Phase 1 studies are usually performed on healthy volunteers. Once the intervention passes phase 1, phase 2 begins.

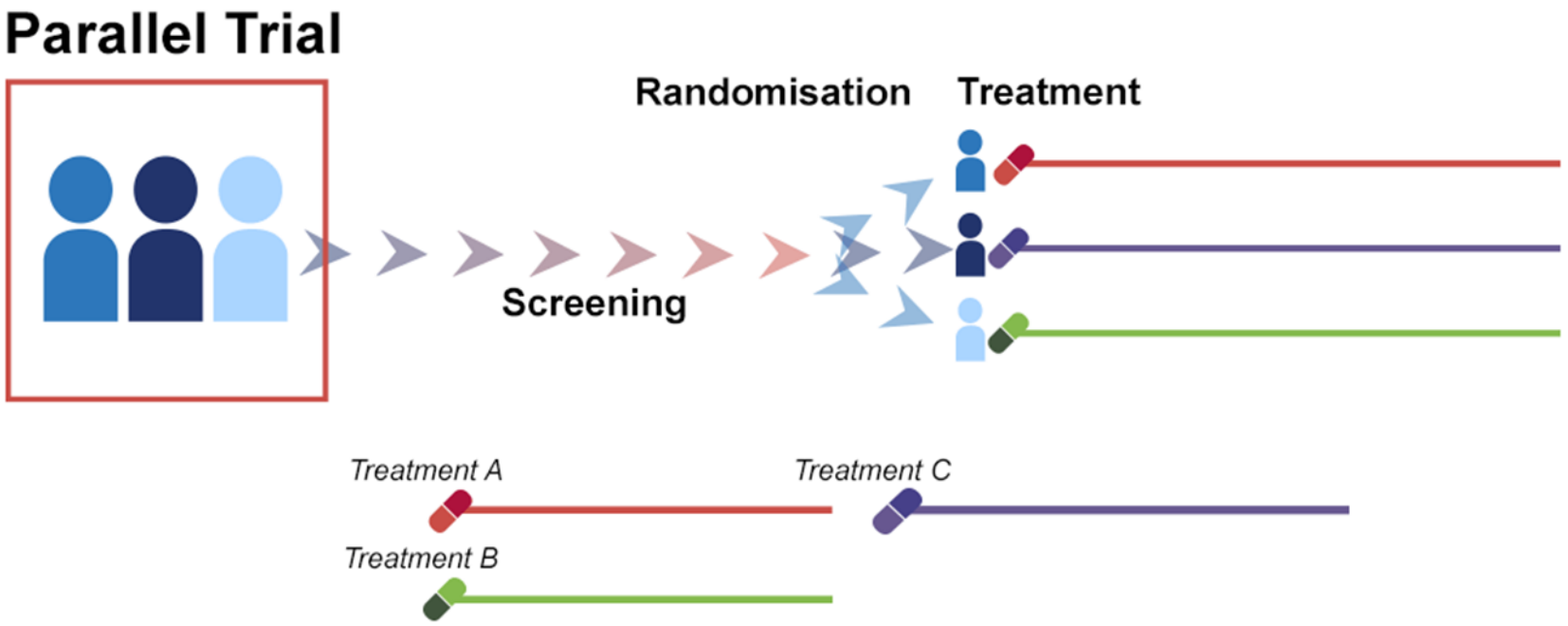
Phase 1, 2, 3, and 4 trials – 2

- Typically, the intervention is given to a small group of actual patients, and this study aims to evaluate the efficacy of different modes of administration of the intervention to patients.
- Phase 2 studies focus on efficacy while still providing safety information.
- Phase 3 studies are typically effectiveness trials performed after a given procedure has been shown to be safe with a reasonable chance of improving patients' conditions.
- Most phase 3 trials are randomized controlled trials.
- Phase 4 studies are equivalent to postmarketing studies of the intervention; they are performed to identify and monitor possible adverse events not yet documented

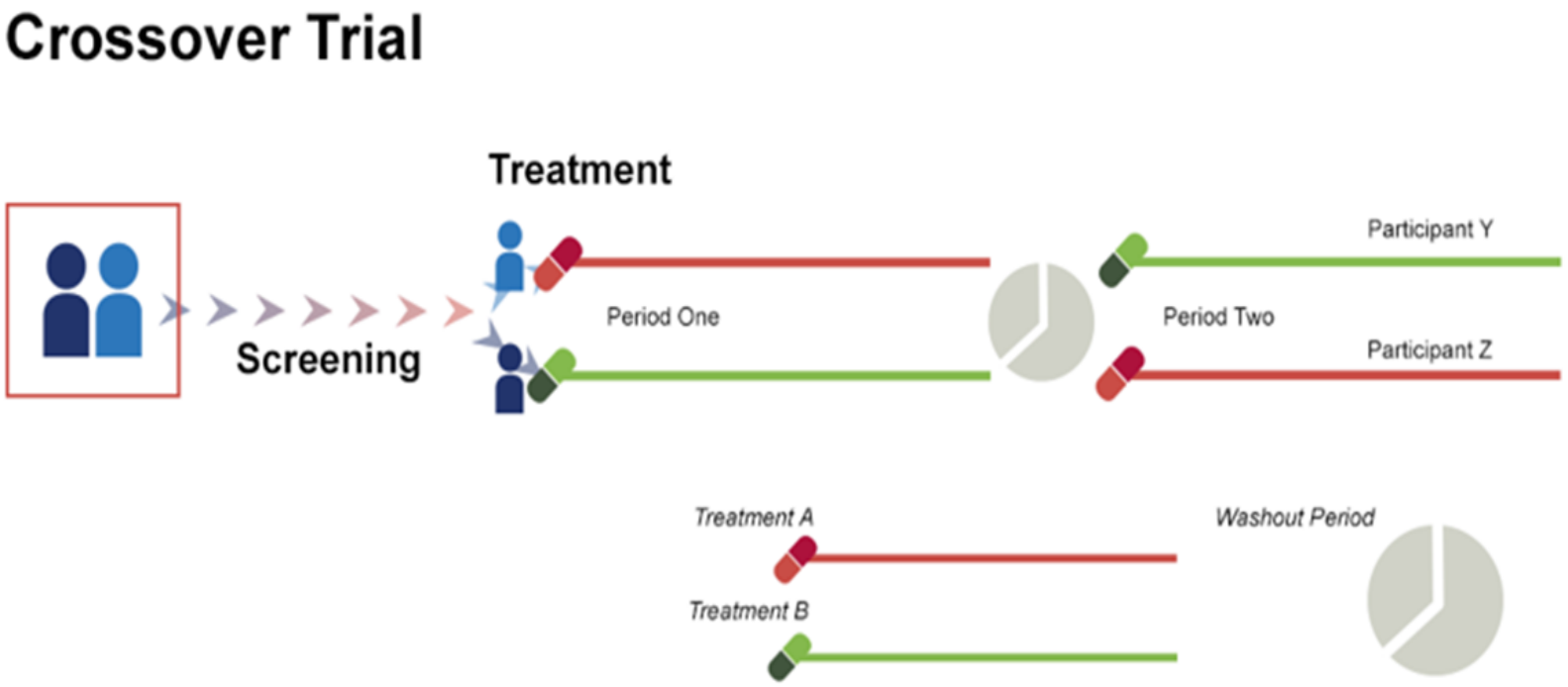
Parallel design vs. Crossover design

- Parallel design:
  - Most RCTs have parallel designs in which each group of participants is exposed to only one of the study interventions.
- Crossover design:
  - A crossover design refers to a study in which each participant receives all of the study interventions in successive periods.
  - The order in which the participants receive each study intervention is determined randomly.
  - This design is appropriate only for chronic conditions that are fairly stable over time and for interventions that last a short time within the patient and do not interfere with one another.
  - Otherwise, false conclusions about the effectiveness of an intervention could be drawn.

Parallel RCT design



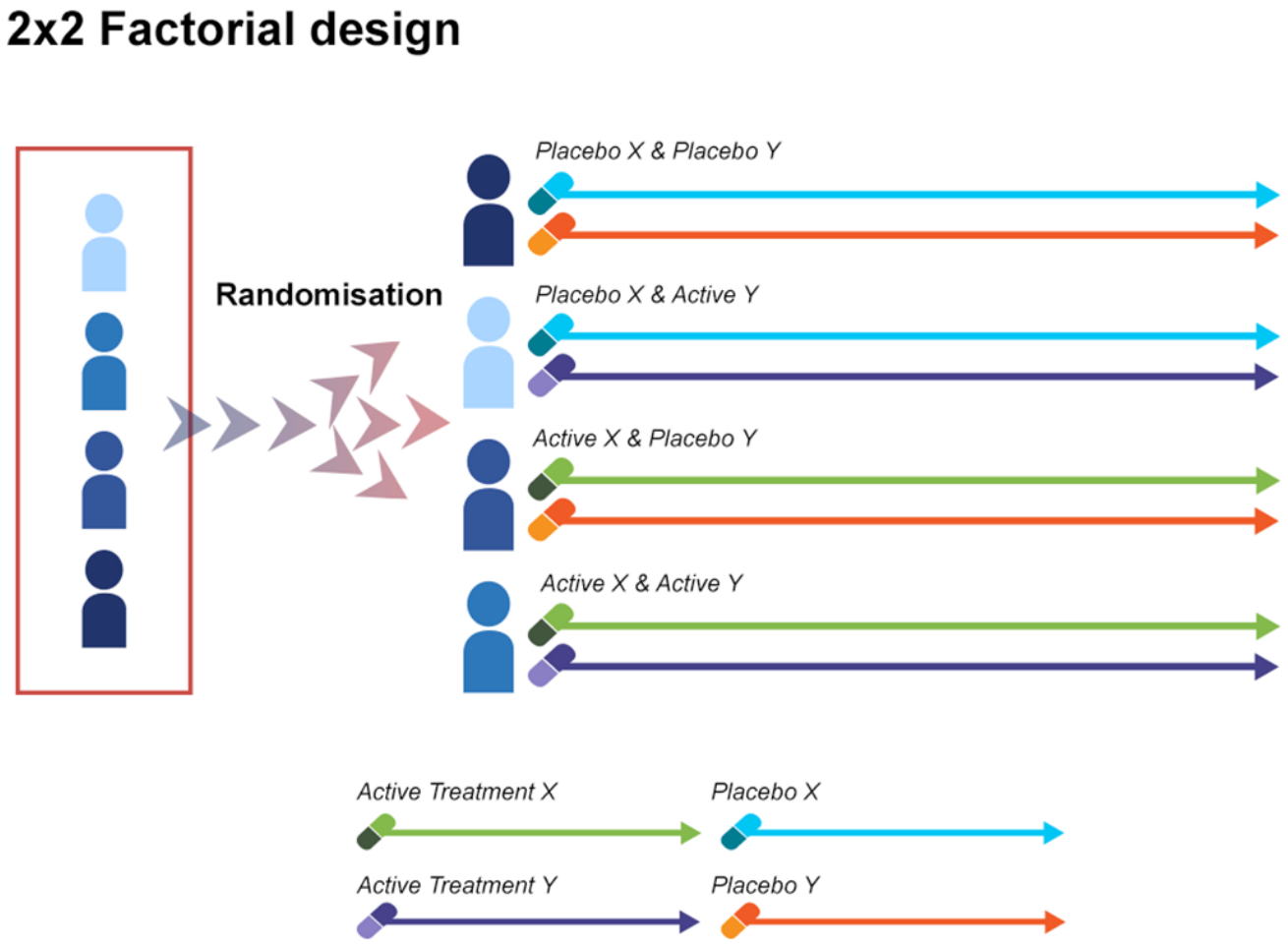
Crossover RCT design



Factorial design

- An RCT has a factorial design when two or more experimental interventions are evaluated separately, in combination, and against a control.
- For example, a 2 × 2 factorial design generates four sets of data to analyze: data on patients who received none of the interventions, patients who received treatment A, patients who received treatment B, and patients who received both A and B.
- More complex factorial designs involving multiple factors are occasionally used.
- The strength of this design is that it provides more information than parallel designs.
- In addition to the effects of each treatment, factorial design allows evaluation of the interaction between two treatments.
- Because RCTs are generally expensive to conduct, the more answers that can be obtained, the better

Factorial RCTs design

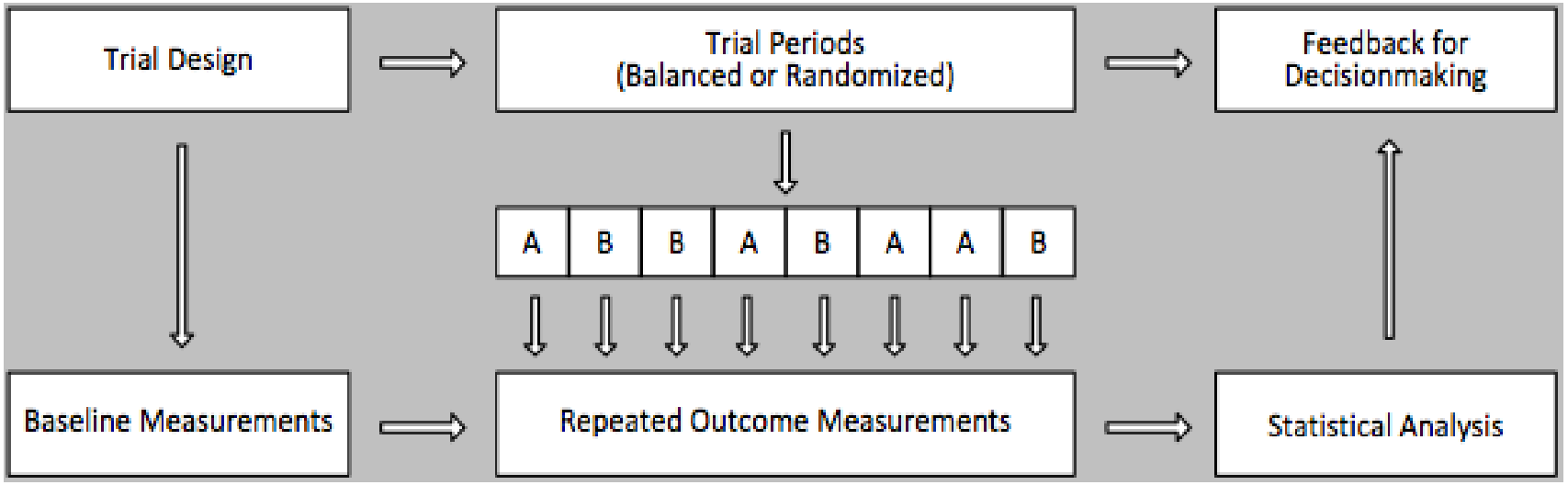


N-of-one trials – 1

- RCTs with only one participant are called "N-of-one trials" or "individual patient trials."
- RCTs with a simple design that involves thousands of patients and limited data collection are called megatrials. Usually, megatrials require the participation of many investigators from multiple centres and different countries.
- An N-of-one trial is a clinical trial in which a single patient is an entire trial, a single case study.
- A trial in which random allocation can be used to determine the order in which an experimental and a control intervention are given to a patient is an N-of-one trial randomized controlled trial.
- The researcher can also fix the order of experimental and control interventions.

N-of-one trials – 2

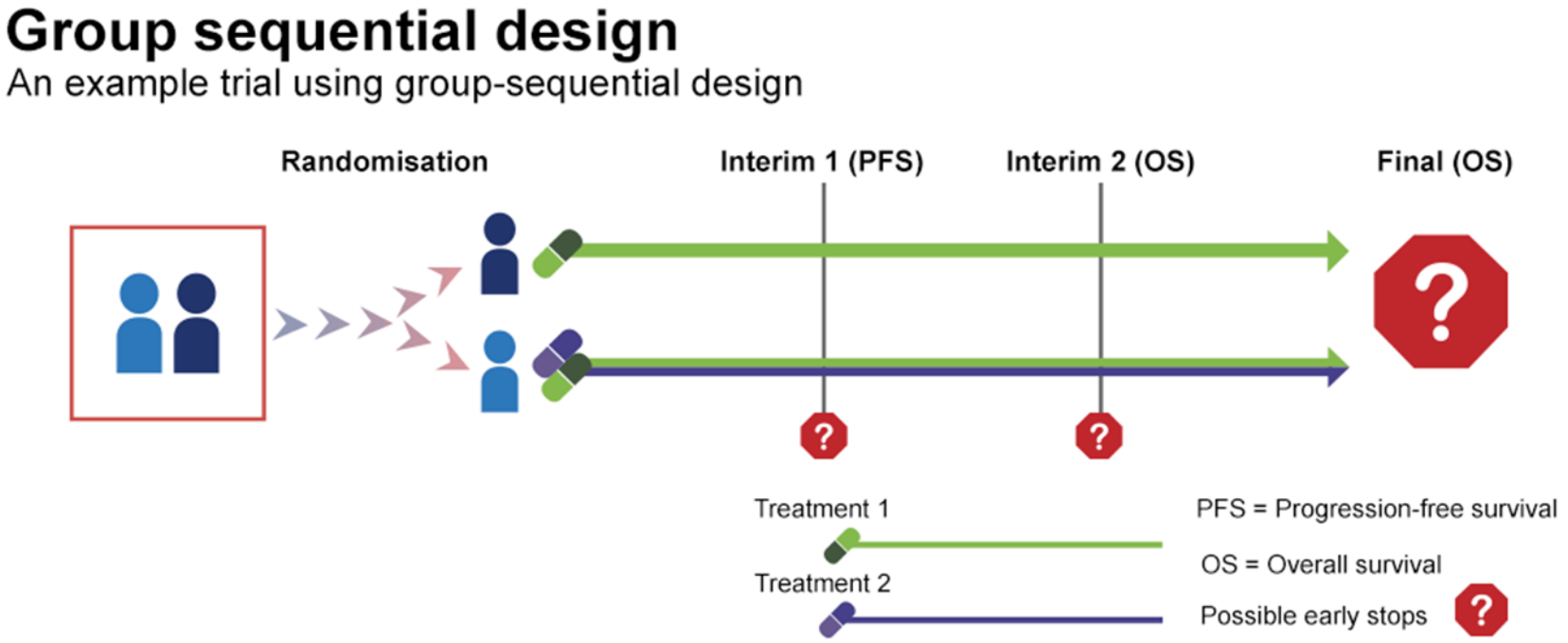
- This type of study has enabled practitioners to achieve experimental progress without the overwhelming work of designing a group comparison study.
- It can be very effective in confirming causality. This can be achieved in many ways; full experiment or quasi-experiment.



Sequential trials

- A sequential trial is a study with a parallel design in which the investigators do not specify the number of participants beforehand.
- Instead, the investigators continue recruiting participants until they observe a clear benefit of one of the interventions or until they become convinced that there are no important differences between the interventions.
- This element applies to comparing some diagnostic interventions and some procedures in interventional radiology.
- Strict rules govern when trials can be stopped based on cumulative results, and important statistical considerations come into play.

Sequential RCT design



Fixed trials

- Alternatively, in a fixed trial, the investigators establish deductively the number of participants (sample size) that will be studied.
- This number can be decided arbitrarily or calculated using statistical methods.
- The latter is a more commonly used method.
- Even in a fixed trial, the trial's design usually specifies whether there will be one or more interim data analyses.
- If a clear benefit of one intervention over the other can be shown with statistical significance before all participants are recruited, it may not be ethical to pursue the trial, and it may be prematurely terminated.

RCTs classified according to the level of blinding – 1

- In addition, Besides randomization, the investigators can incorporate other methodologic strategies to reduce the risk of biases.
- These strategies are known as "blinding."
- The purpose of blinding is to reduce the risk of ascertainment and observation bias.
- An open RCT is one in which everybody involved in the trial knows which intervention is given to each participant.
- Many studies are open RCT because blinding is not feasible or ethical.
- For example, one cannot perform an interventional procedure with its associated risks without revealing to the patient and the treating physician which group the patient has been randomized to.

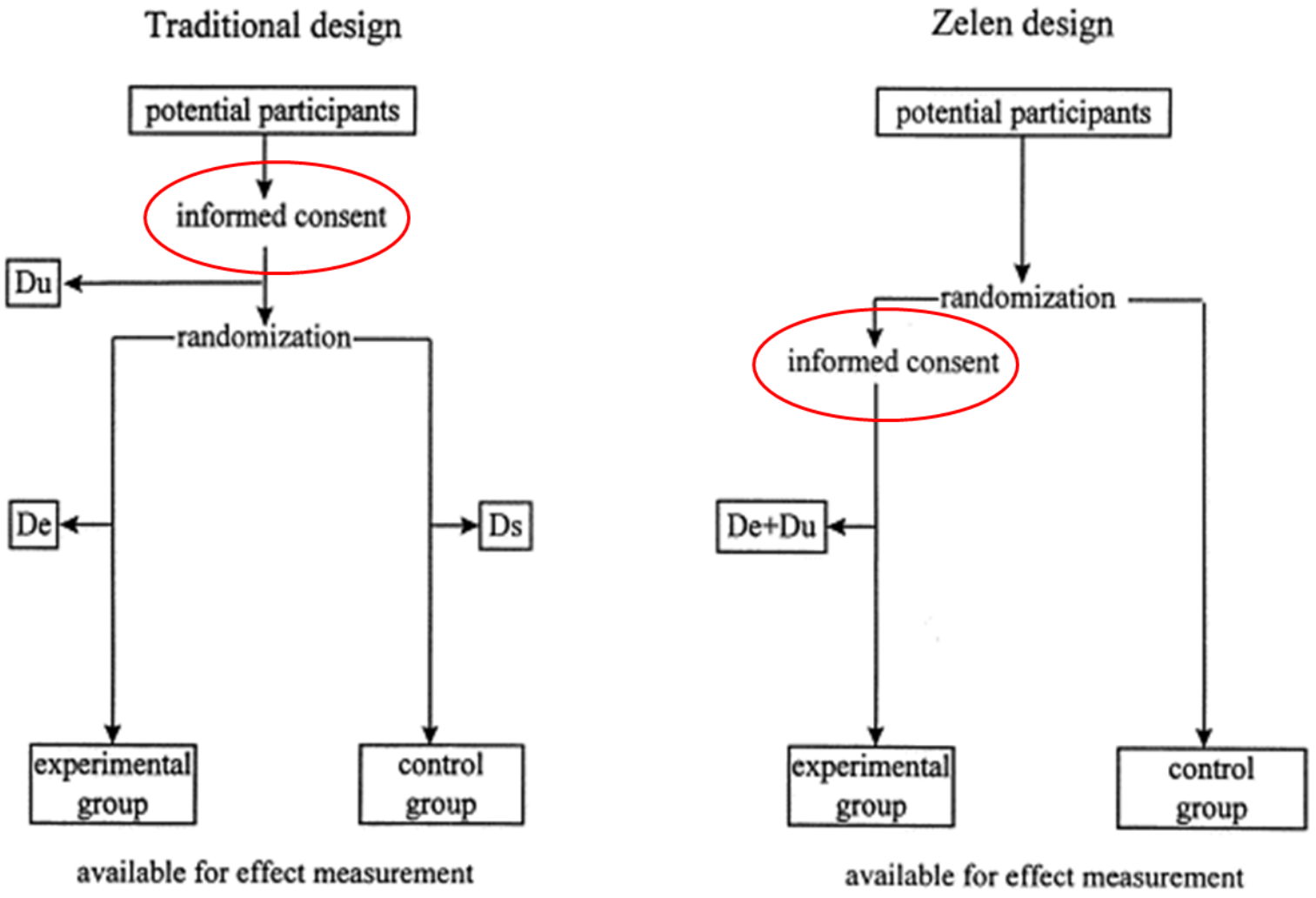
RCTs classified according to the level of blinding – 2

- A single-blinded RCT is one in which a group of individuals involved in the trial (usually patients) does not know which intervention is given to each participant.
- A double-blinded RCT, on the other hand, is one in which two groups of individuals involved in the trial (usually patients and treating physicians) do not know which intervention is given to each participant.
- Beyond this, triple-blinded (blinding of patients, treating physicians, and study investigators) and quadruple-blinded RCT (blinding of patients, treating physicians, study investigators, and statisticians) have been described but are rarely used.

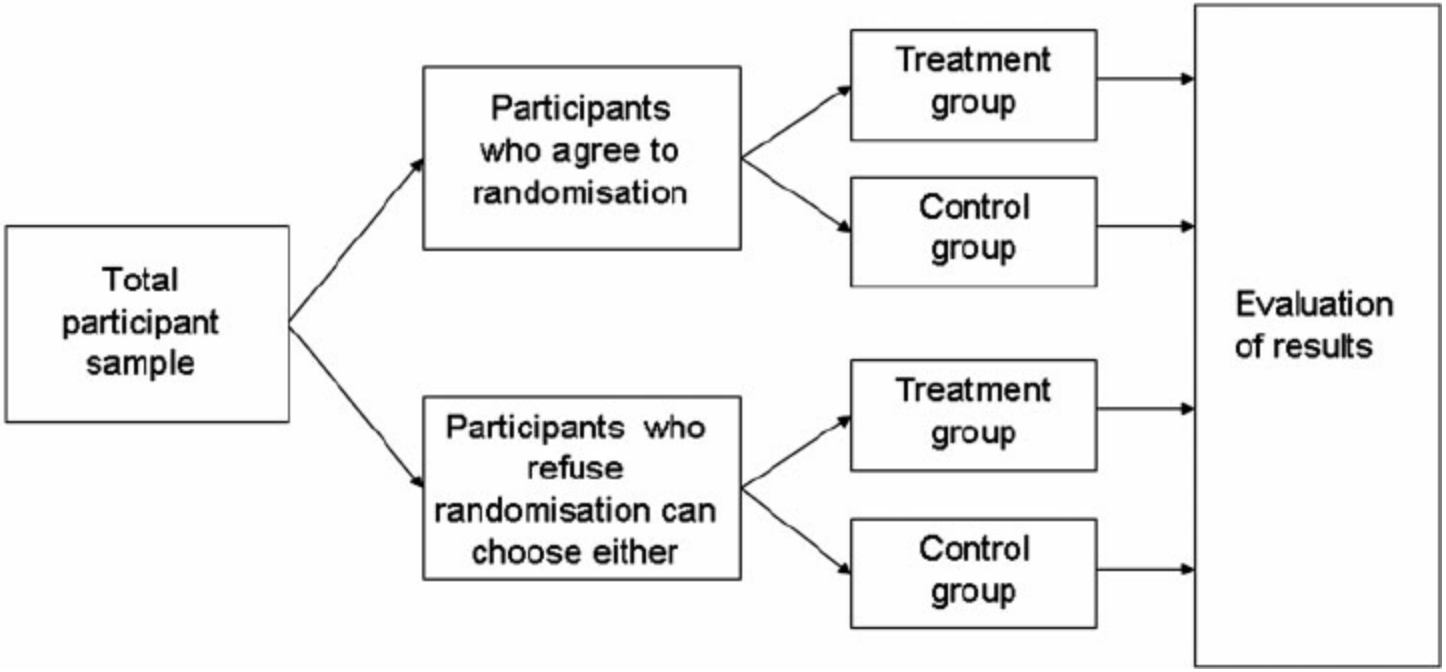
RCTs classified according to Nonrandomized Participant Preferences

- Eligible individuals may refuse to participate in an RCT.
- Other eligible individuals may decide to participate in an RCT but clearly prefer one of the study interventions.
- At least three types of RCTs consider the preferences of eligible individuals as to whether or not they take part in the trial.
- These are called preference trials because they include at least one group in which the participants can choose their preferred treatment from several options offered.
- Such trials can have a Zelen design, comprehensive cohort design, or Wennberg's design.

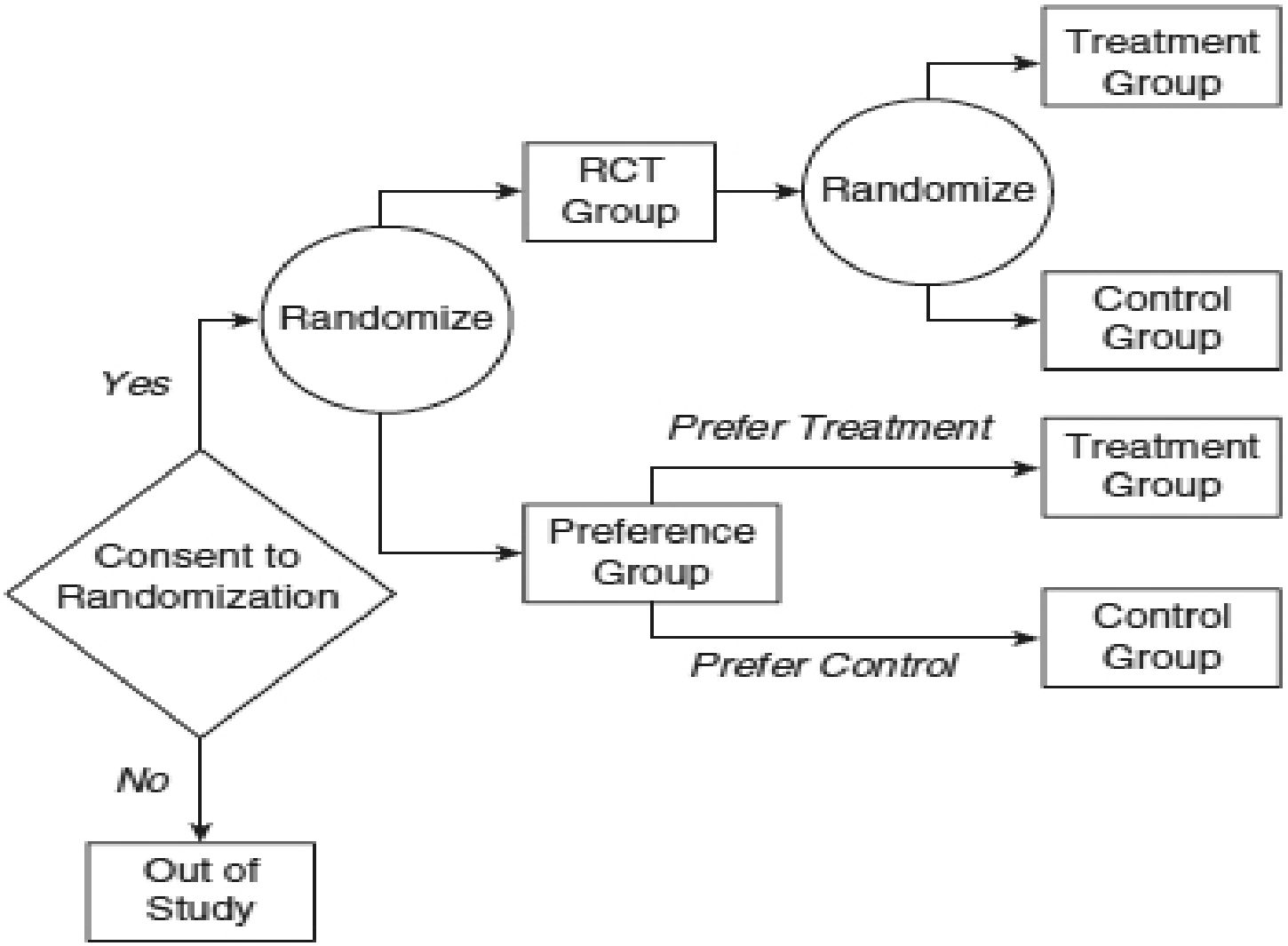
Zelen design



Comprehensive cohort design



Wennberg's design



How to do random allocation (Randomization for RCTs) – 1

- RCTs are known as the best method to prove causality despite various limitations.
- Random allocation is a technique that chooses individuals for treatment groups and control groups entirely by chance, disregarding the will of researchers or patients' condition and preference.
- This allows researchers to control all known and unknown factors that may affect results in treatment and control groups.
- Allocation concealment is a technique used to prevent selection bias by concealing the allocation sequence from those assigning participants to intervention groups, until the moment of assignment.
- Allocation concealment prevents researchers from influencing which participants are assigned to a given intervention group.

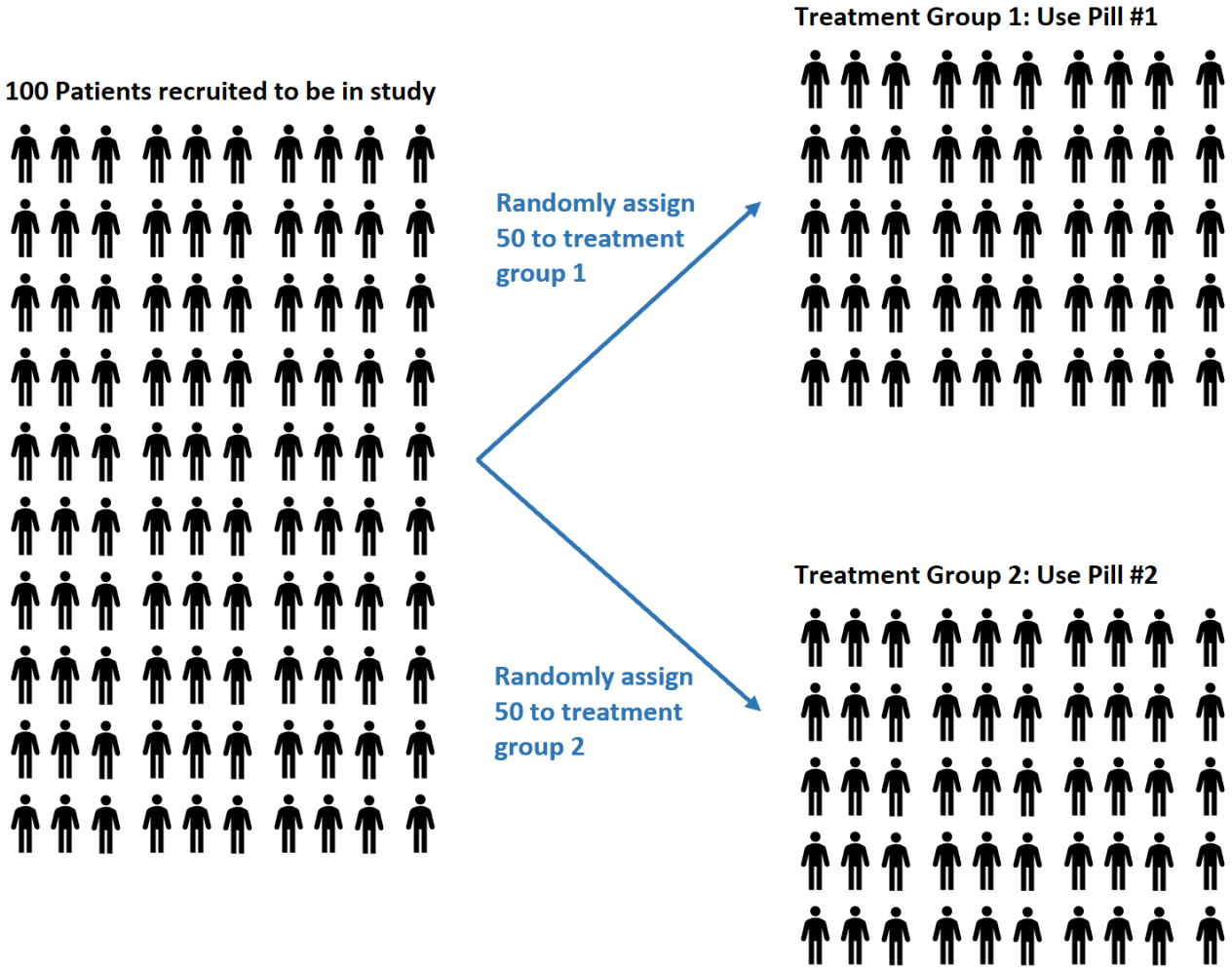
How to do random allocation (Randomization for RCTs) – 2

- This process must be included in the experiment for the success of any RCT.
- Blinding refers to keeping trial participants, health care providers, assessors or data collectors unaware of the assigned intervention, so that they will not be influenced by that knowledge.
- This process is conducted to minimize possible bias in dropouts, implementation, measurements, etc.
- Blinding is not always feasible for RCT but should be implemented if possible.
- Randomization, allocation concealment and blinding should be well implemented and should be described in the paper.

Categories of Randomization for RCTs: Simple Randomization

- The easiest method is simple randomization.
- If you assign subjects to groups A and B, you randomly assign subjects to each group for every assignment.
- Even though this is the most basic way, if the total number of samples is small, sample numbers are likely to be assigned unequally.
- For this reason, we recommend using this method when the total number of samples is more than 100.

Simple Randomization



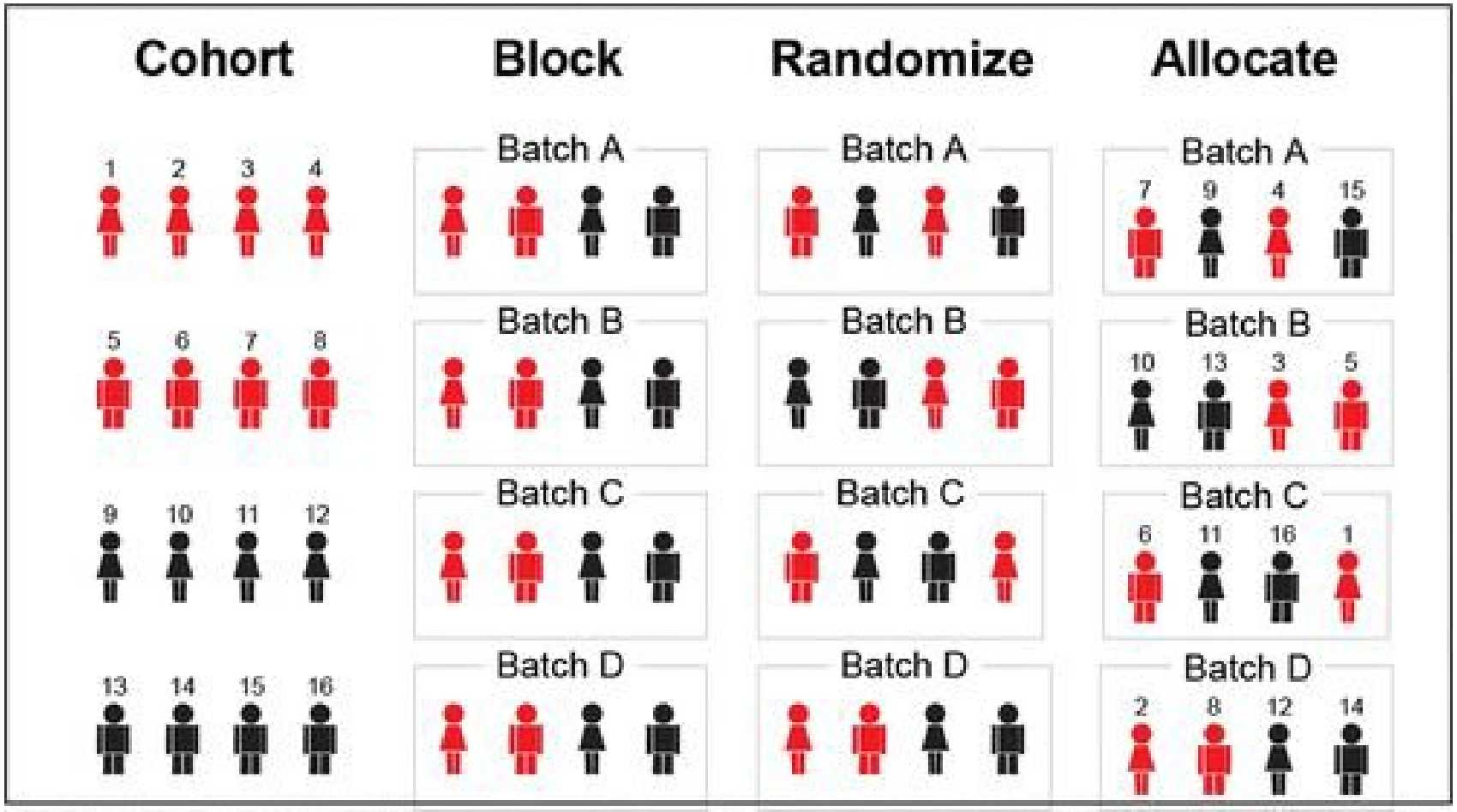
Categories of Randomization for RCTs: Block Randomization – 1

- We can create a block to assign sample numbers equally to each group and assign the block.
- If we specify two in one block (the so-called block size is two), we can make two possible sequences of AB and BA.
- When we randomize them, the same sample numbers can be assigned to each group.
- If the block size is four, we can make six possible sequences: AABB, ABAB, ABBA, BAAB, BABA, and BBAA, and we randomize them.

Categories of Randomization for RCTs: Block Randomization – 2

- However, there is a disadvantage in that the executor can predict the next assignment.
- If the block size is two, we can quickly know that B comes after A, and if the block size is four, we can predict what every fourth sample is.
- This is discordant with the principle of randomization.
- To solve this problem, the allocator must hide the block size from the executor and use randomly mixed block sizes.
- For example, the block size can be two, four, and six.

Block Randomization



Categories of Randomization for RCTs: Stratified Randomization – 1

- Randomization is important because it is almost the only way to assign all the other variables equally except for the factors (A and B) we are interested in.
- However, some important confounding variables can often be assigned unequally to the two groups.
- This possibility increases when the number of samples is smaller, and we can stratify the variables and assign the two groups equally in this case.
- For example, if the smoking status is very important, what will you do? First, we have two methods of randomization that we learned previously.
- There are two randomly assigned separate sequences for smokers and non-smokers.

Categories of Randomization for RCTs: Stratified Randomization – 2

- Smokers are assigned to the smoker's sequences, and non-smokers are assigned to the non-smoker's sequences.
- Therefore, smokers and non-smokers groups will be placed equally and have the same numbers.
- So, we can use:
  1. Simple randomization with/without stratification, or
  2. Block randomization with/without stratification.
- However, if multiple stratified variables exist, it isn't easy to place samples in both groups equally with the same numbers.
- Usually, two or fewer stratified variables are recommended.

12

Study Population

Study Population

- This could be a group of units: Persons, families, schools, medical records, death certificates or specimens.
- Define your study population:
  - Who they are?
  - Where do they live?
  - Characters to be examined; are they cases or controls?
  - Describe the method of selection of the study population.
- Decide the sampling method you will use (if needed) and your sample size.
- Study Population = Sampled population = Parent population
- Reference population is used if the study population is considered typical of a broader population.

Selection of the study population

- Appropriateness and accessibility are two major factors to consider in the selection process of the study population.

  1. Appropriateness: The suitability for the attainment of the study objectives.
  2. Accessibility: When the investigator decides to select a specific group, he has to know:
     - Can the investigator reach them?
     - Are they ready to participate?
     - Are they mobile?

Special study population

- The study population could be selected from the general population or special groups as:
  1. Volunteer population
  2. Hospital or clinic population
  3. Healthy population
  4. Population appears frequently
- In case-control studies, the study population includes a group of people with the condition of interest (Cases) and those who are free from the same condition (Controls).
- Accordingly, the selection of the study population necessitates the selection of cases and controls.

A. Selection of cases – 1

- During the selection of cases, investigators have to consider these important aspects.

1. Diagnostic criteria:
   - There should be specific criteria for diagnosis of the case.
   - These criteria are based on the operational definition of the disease or the condition under investigation.
   - It could be as simple as defining anaemic children as those with haemoglobin values less than 11-gm/ 100-ml blood.
   - Complex criteria are used depending on clinical and/ or laboratory findings as the criteria used to define meningitis or rheumatic fever.

A. Selection of cases – 2

2.Sources:

- Cases could be selected from hospitals or from the general population.
- Sever cases are hospital cases, while simple or chronic diseases are more common in the community.
- Records could be used as a sampling frame for case selection.

3.Incidents and prevalent cases:

- Prevalent cases appear more than once during the research process.
- A diabetic patient could appear more than once in the study, affecting the pattern of the disease among the population.

B. Selection of controls – 1

- Apparently, the selection of controls is more straightforward than in cases.
- Actually, control selection is a complex process and may be more complicated than case selection.
- These points have to be considered in control selection:

1. Comparability to the cases except having the disease: Controls have to be as similar to cases as possible as the investigator can accept having the disease or the condition under investigation.
2. Representative of all non-diseased population: The selection of controls should not be from a sector that is known to be healthier than other sectors.
   - Controls must represent all non-diseased populations regardless of their status if they are comparable to the cases.

B. Selection of controls – 2

3. Practical: The selection process should be practical; otherwise, the investigator will be blocked by difficulties such as feasibility, availability of controls, and shortage of funds.
   - Time is an important factor for a study to be practical.
4. Sources of Control: The investigators choose controls from one or more of these sources:
   a) Population of defined area.
   b) Hospital patients either directly or from records.
   c) Neighbors: different approaches are used to select controls from neighbours, such as a walk. A phone or a letter.
   d) Friends or associates of cases are easy to participate.
   e) Siblings, spouses and relatives are willing more than other groups to participate as controls.

B. Selection of controls – 3

5. Methodological issues: consider issues in control selection.

- Matching: Group matching and individual matching is used.
  - Group matching is used when a group of people as a class in a school is exposed to an event under investigation.
  - In this case, a similar class could be used as a control, and the investigator could compare the cases with the control group.
  - Individual matching is applied by selecting one or more controls for each case.
- Matching is faced with problems such as:

a) Matching for many variables makes it difficult to find appropriate control

B. Selection of controls – 4

b) We can not explore the possible association of disease with any variable on which cases and controls have been matched and where cases and controls will be similar for this variable.

- Multiple controls: More than one control could be selected.
  - They are either similar or different types.
  - In case of similar controls, two, three or even four controls could be chosen for each case.
  - We can select two different types of controls.
  - For example, in the study of risk factors for persistent diarrhea

Advantages and disadvantages of using primary and secondary data

Primary Data	Secondary Data
Advantages	Advantages
● Investigator controls all aspects	● Relatively fast and inexpensive
Disadvantages	Disadvantages
● Time consuming	● Data may not include all variables of interest
● Expensive	● Maybe difficult to understand how and why data elements were collected
● All variables of interest can be measured	● Unobtrusive to subjects

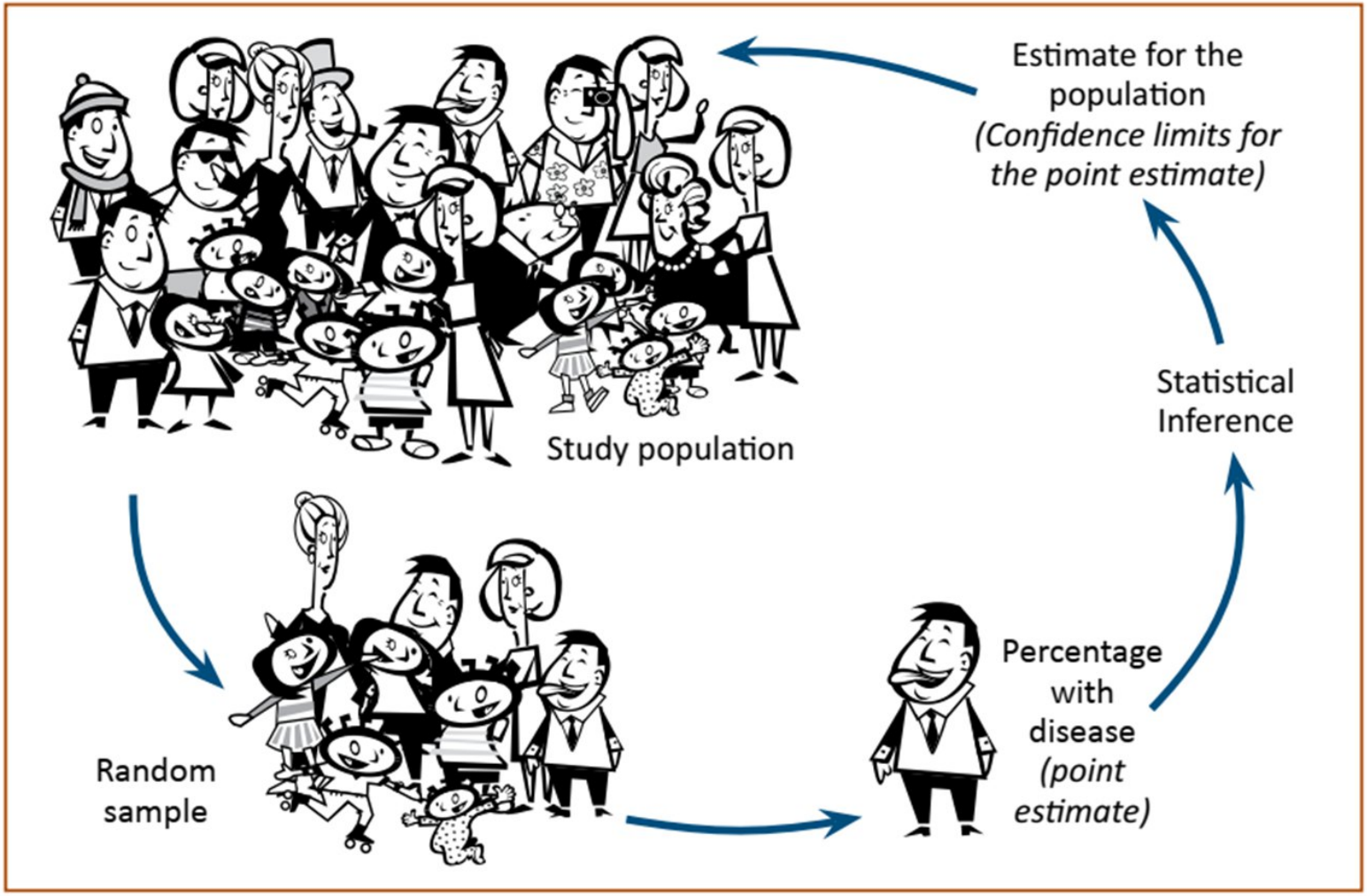
collection and follow-up methods

geographical area and thus provide ability to assess national trends

13

Sampling

Sampling and statistical inference



Inclusion and Exclusion Criteria

Inclusion and exclusion criteria are used to ensure precision in a study.

We clearly determine who can be considered a potential sample and who cannot.

These criteria are based on such factors as age, geographical location, disease severity or stage, previous treatment, presence of other medical conditions, etc.

1. Inclusion criteria: are the criteria for including a patient in the study.
2. Exclusion criteria: are the criteria for excluding patients from the study.

It is important that these criteria be clearly defined objectively so that everyone involved in the study is consistent in selecting the sample.

The selection criteria also determine to whom the study results can apply.

Why sampling?

The cost of research work is generally high.

The cost includes personal time, equipment, material, and required tests and examinations.

In addition to the cost, hard work is required to complete the research activities, including study design, fieldwork, required tests, examinations, data management, and reporting.

If a research question could be answered by testing 100 individuals, do not try to examine 1000.

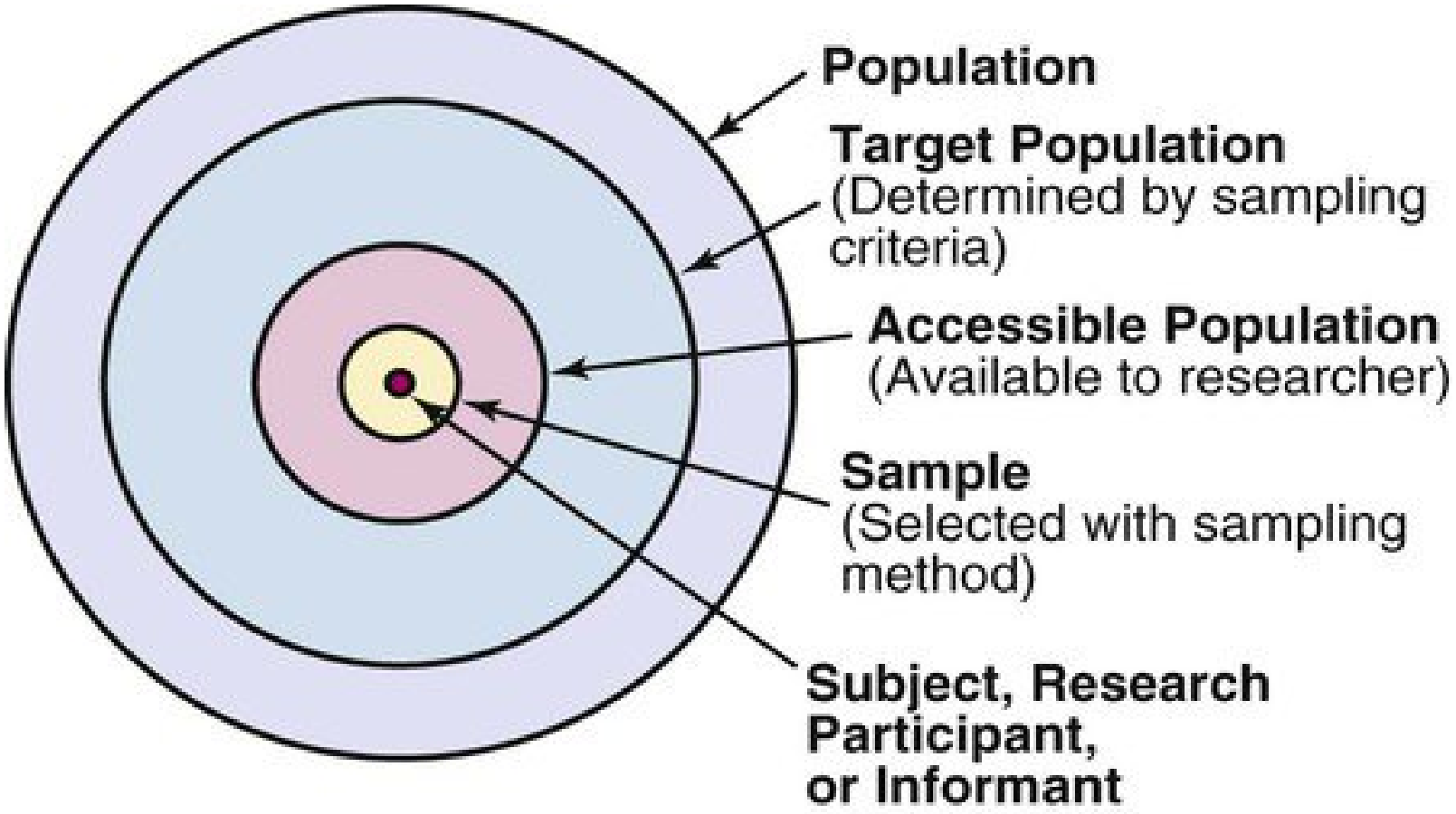
Accordingly, a representative sample could answer a research question with less cost, time and effort.

Sampling terms

Before going into details of sampling, the investigator should be familiar with these relevant terms:

1. Population: The total number of inhabitants living in a defined geographical area.
2. Study population: Units from which the sample is drawn. It is not essential to include all the population.
3. Survey population: the group of people involved in the survey, who are also part of the study population.
4. Sampling unit: The number of subjects selected for the study.
5. Sampling frame: This is the site where you list all the subjects included in the survey population. These could be lists, records, files, or blocks of buildings.

Sampling


Population
Target Population
(Determined by sampling criteria)
Accessible Population
(Available to researcher)
Sample
(Selected with sampling method)
Subject, Research Participant, or Informant

Sampling methods

A. Non-probability (Non-random) sampling:

Nonrandom sampling or convenient sampling involves selecting a group of individuals according to the investigator's desire and judgment.

There is no objective way of judgment.

Non-random sampling is a sample of convenience or accidental or incidental samples

Non-probability sampling method does not provide representative data for the total population.

B. Probability (Random) sample:

Each individual in the sample frame has the same probability of appearing in the sample.

13.1.

Non-probability Sampling

This is the sampling procedure which does not afford any basis for estimating the probability that each item in the population has of being included in the sample .

Types of Non-probability Sampling – 1

1.Convenience Sampling

It is also known as accidental or haphazard sampling. In this method, the researcher studies all the persons most conveniently available or accidentally encountered.

2.Purposive Sampling/ Judgmental Sampling

It is a sampling technique that allows a researcher to use cases that have the required information regarding the objectives of his study.

Cases of subjects are therefore handpicked because they are informative or pose the required characteristics.

Types of Non-probability Sampling – 2

3.Quota Sampling

The researcher purposely selects subjects to fit the quotas identified.

The selection of actual participants is not random since subjects are picked as they fit into identified quotas.

4.Snowball Sampling

In this method subjects with the desired characteristics are picked using purposive sampling.

The few identified subjects name others that they know have the required characteristics until the researcher gets number of cases he requires.

Types of Non-probability Sampling – 3

5. Census
   - This means using the entire population as the sample.
   - It is very attractive for small populations but very costly for large populations.
   - A census eliminates sampling error and provides data on all the individuals in the population.
6. Consecutive sampling
   - Also known as total enumerative sampling, consecutive sampling is the process of conducting research, including all the people who meet the inclusion criteria and are conveniently available as part of the sample.
7. Using a Sample of a Similar Study
   - Another approach is to use the same sample size as those of studies similar to the one you plan to do.
   - There is a risk, though, of repeating errors that were made in determining the sample size for the other research.

13.2.

Probability Sampling

Sampling of Probability Sample – 1

1. Simple random sample (SRS) or random sample
   - Prepare a sampling frame
   - Decide on the size of the sample
   - Select the required number by drawing lots or using the random table.
2. Systematic sample:
   - Prepare a sampling frame
   - Decide on the size of the sample
   - Select every K^{th} number
   - K = Division of frame individuals by sample size
   - First number could be selected randomly

Sampling of Probability Sample – 2

3. Cluster samples:

   Simple random sample not for individual subjects but for a group (cluster) of individuals

4. Stratified sample:

   Divide the sample frame into groups (stratum) according to one or more characteristics such as age and sex.

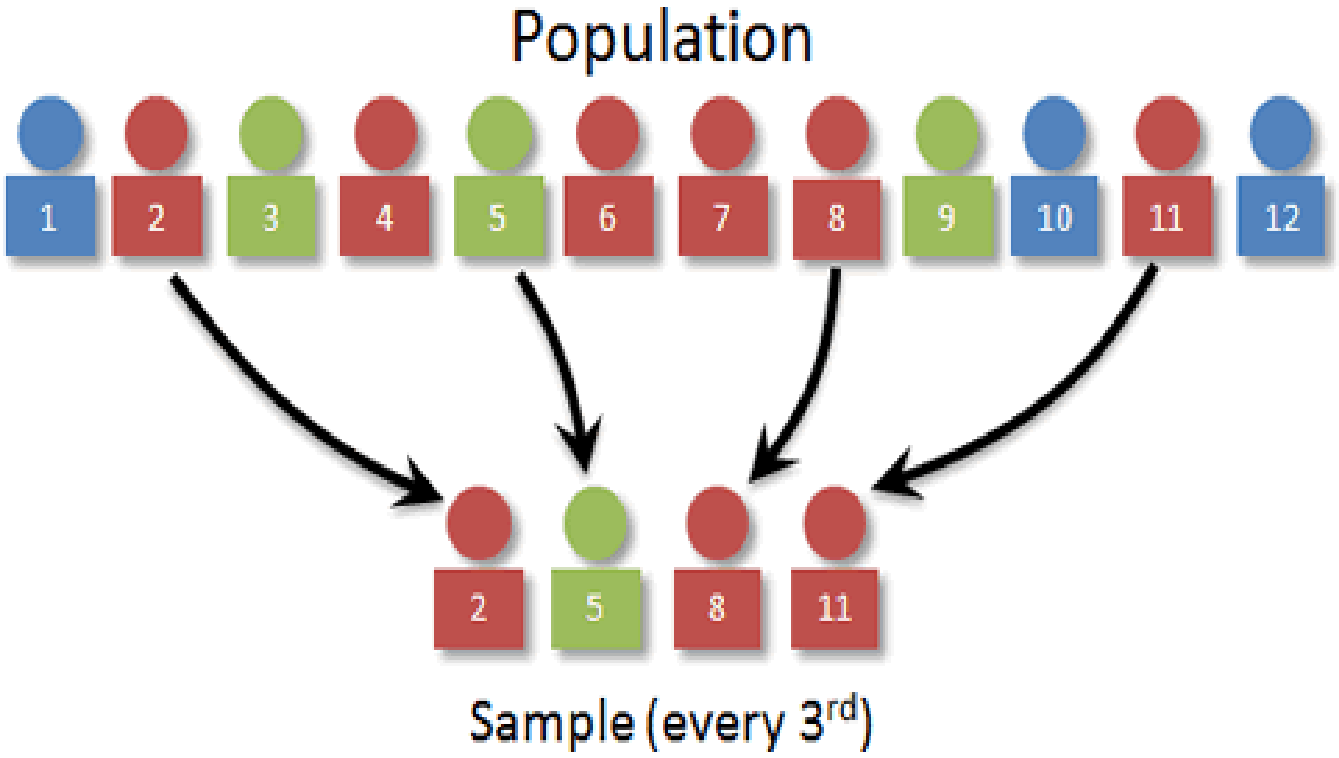
   From each stratum select the required number either randomly or systematically

5. Multistage sample:

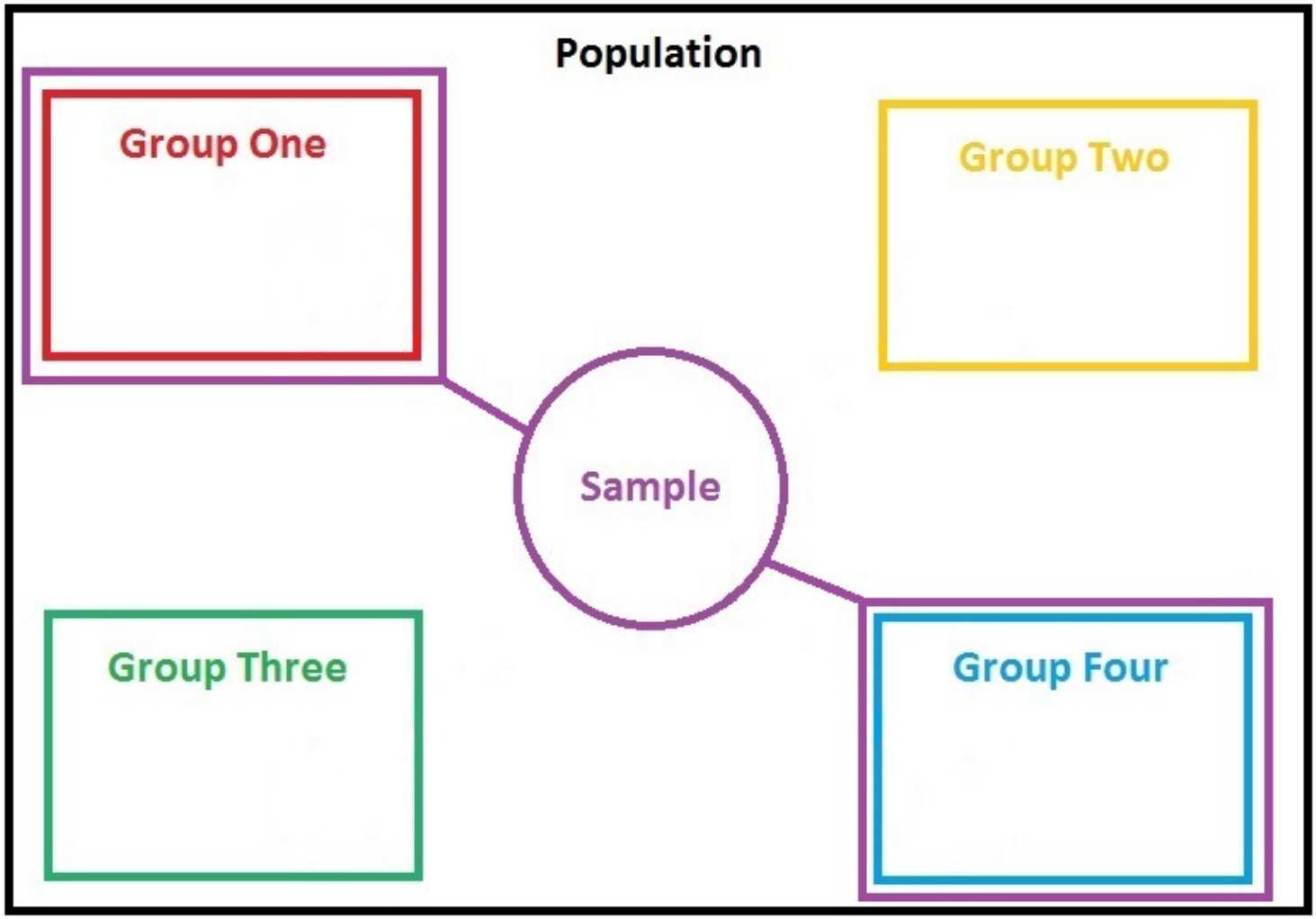
   Combination of more than one method.

   Appropriate when the population is large and widely scattered.

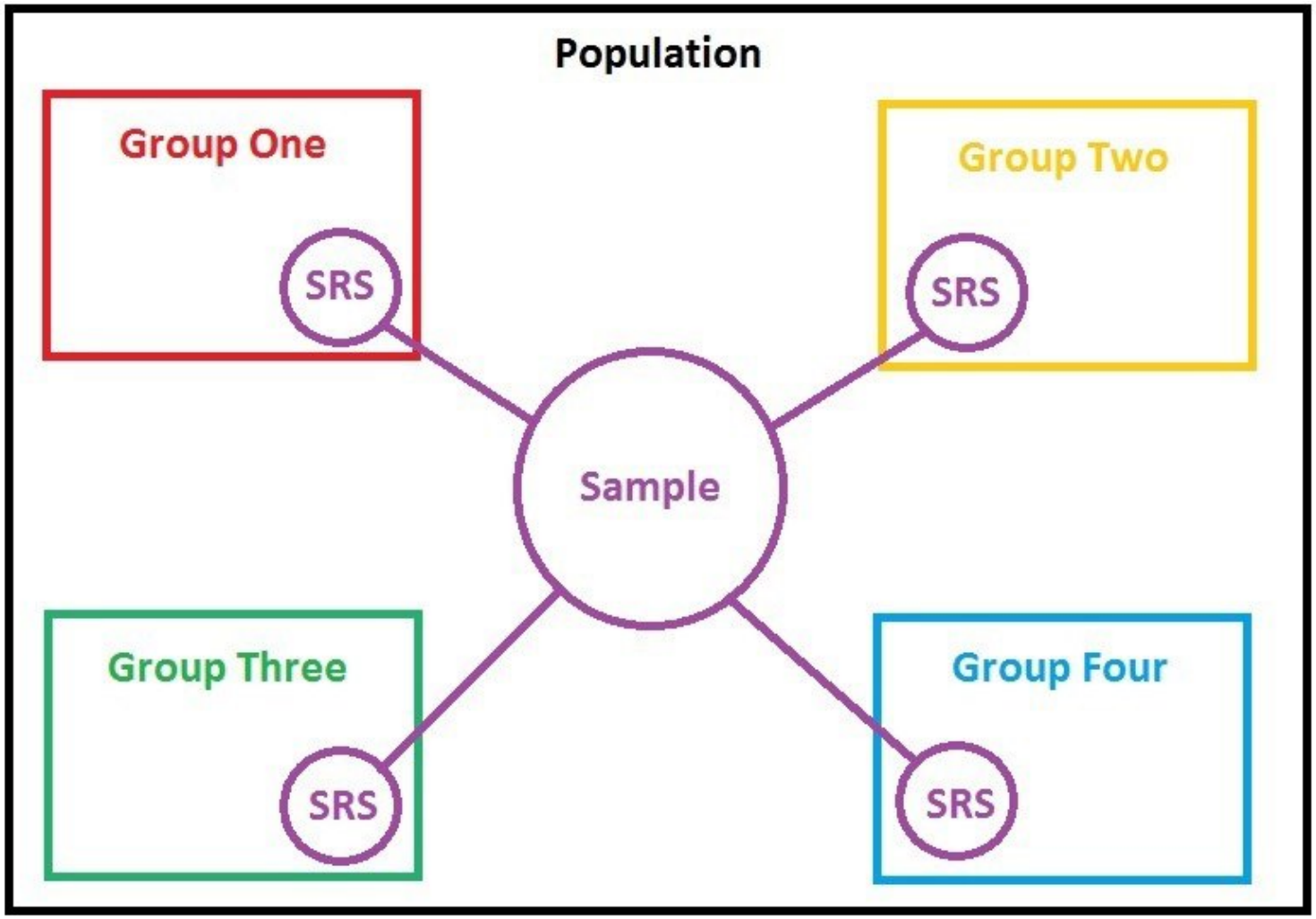
Systematic Random Sampling



Cluster Random Sampling



Stratified Random Sampling



13.3.

Sampling Error

Sampling error (i.e., Random Error) – 1

Random error, the opposite of reliability (i.e., Precision or repeatability), consists of random deviations from the true value, which can occur in any direction.

Sampling error (random error) can be minimized by increasing the sample size.

Reliability (or precision):

- This refers to the repeatability of a measure, i.e., the degree of closeness between repeated measurements of the same value.
- Reliability addresses the question of how close the measurements are to each other if the same thing is measured several times.

Sampling error (i.e., Random Error) – 2

The sources of variation resulting in poor reliability include:

1. Variation in the characteristic of the subject being measured. Example: blood pressure
2. The measuring instruments, e.g. questionnaires
3. The persons collecting the information (observer variation)

Inter-observer variation: differences between observers in measuring the same observation

Intra-observer variation: differences in measuring the same observation by the same observer on different occasions.

Non-Sampling error (i.e., Bias) – 1

Bias, the opposite of validity, consists of systematic deviations from the true value, always in the same direction.

It is possible to eliminate or reduce the non-sampling error (bias) by carefully designing the sampling procedure.

Validity refers to the degree of closeness between a measurement and the true value of what is being measured.

Validity addresses the question: how close is the measured value to the true value?

Non-Sampling error (i.e., Bias) – 2

To be accurate, a measuring device must be valid and reliable.

However, if one cannot have both, validity is more important when we are interested in the absolute value of what is being measured.

Reliability, on the other hand, is more critical when it is not essential to know the absolute value, but instead, we are interested in finding out if there is a trend or to rank values.

Examples of types of bias in sampling include:

Bias resulting from the incompleteness of the sampling frame: accessibility bias, seasonability bias, self-reporting bias, volunteer bias, non-response bias, etc.

Non-response bias – 1

Non-response bias refers to the failure to obtain information on some of the subjects in the study sample.

It results in significant bias when the following two situations are fulfilled.

1. When non-respondents constitute a significant proportion of the sample.
2. When non-respondents differ significantly from respondents.

The issue of non-response should be considered during the planning stage of the study:

A. Non-response should be kept to a minimum, e.g. below 15%.

Methods that may help in maintaining non-response at a low level could be:

Non-response bias – 2

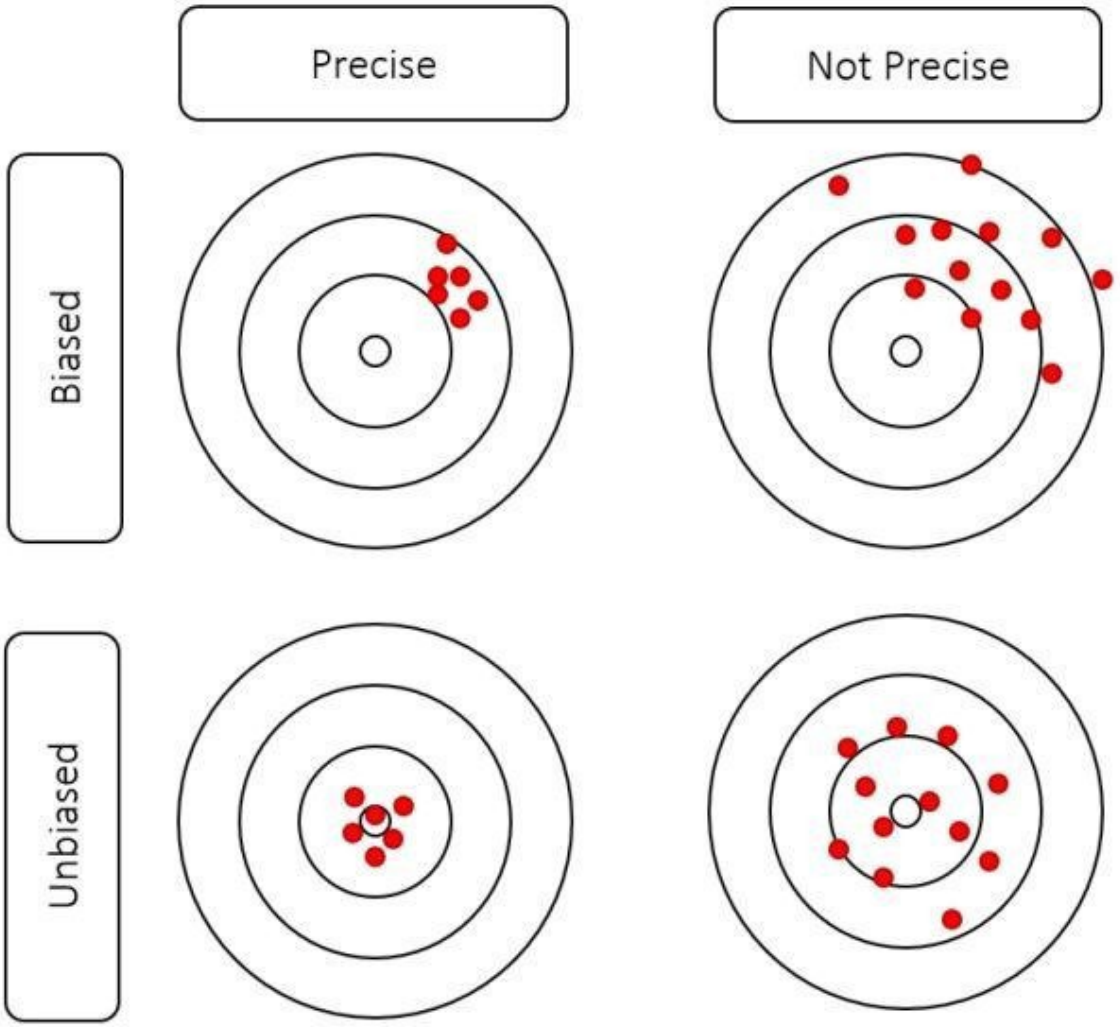
1. Training data collectors to initiate contact with study subjects respectfully and convince them about the importance of the given study (this minimizes the refusal type of non-response)
2. Offer incentives to encourage participation (this should be done by considering the potential problems that may arise in conducting future research).
3. By making repeated attempts (at least 3 times) to contact study subjects absent at the initial visit.

B. The number of non-responses should be documented according to type to facilitate an assessment of the extent of bias introduced by non-response.

C.. As much information as possible should be collected from non-respondents to see how they may differ from respondents.

Selection bias cannot be corrected by increasing the sample size.

Bias & Precision: from Sample to Population



13.4.

Validity in Quantitative Research

Types of validity in quantitative research

Validity tells you how accurately a method measures something.

If a method measures what it claims to measure, and the results closely correspond to real-world values, then it can be considered valid.

<u>There are four main types of validity:</u>

1. Construct validity: Does the test measure the concept that it's intended to measure?
2. Content validity: Is the test fully representative of what it aims to measure?
3. Face validity: Does the content of the test appear to be suitable to its aims?
4. Criterion validity: Do the results correspond to a different test of the same thing?

1. Construct validity

Construct validity evaluates whether a measurement tool represents what we are interested in measuring. It's central to establishing a method's overall validity.

What is a construct?

A construct refers to a concept or characteristic that can't be directly observed but can be measured by observing other associated indicators.

Constructs can be characteristics of individuals, such as intelligence, obesity, job satisfaction, or depression; they can also be broader concepts applied to organizations or social groups, such as gender equality, corporate social responsibility, or freedom of speech.

What is a construct?

Example:

There is no objective, observable entity called “depression” that we can measure directly.

But based on existing psychological research and theory, we can measure depression based on a collection of symptoms and indicators, such as low self-confidence and low energy levels.

What is construct validity?

Construct validity is ensuring that the measurement method matches the construct you want to measure.

If you develop a questionnaire to diagnose depression, you need to know: does the questionnaire measure the construct of depression? Or is it measuring the respondent’s mood, self-esteem, or other construct?

What is construct validity?

To achieve construct validity, you must ensure that your indicators and measurements are carefully developed based on relevant knowledge.

The questionnaire must include only relevant questions that measure known indicators of depression.

The other types of validity described below can all be considered as evidence for construct validity.

2. Content validity

Content validity assesses whether a test is representative of all aspects of the construct.

To produce valid results, the content of a test, survey or measurement method must cover all relevant parts of the subject it aims to measure.

If some aspects are missing from the measurement (or irrelevant aspects are included), the validity is threatened.

Example

A mathematics teacher develops an end-of-semester algebra test for her class. The test should cover every form of algebra taught in the class. If some types of algebra are left out, then the results may not accurately indicate students' understanding of the subject. Similarly, if she includes algebra-related questions, the results are no longer a valid measure of algebra knowledge.

3. Face validity

Face validity considers how suitable the content of a test seems to be on the surface. It's similar to content validity, but face validity is a more informal and subjective assessment.

As face validity is a subjective measure, it's often considered the weakest form of validity. However, it can be useful in the initial stages of developing a method.

Example

You create a survey to measure the regularity of people's dietary habits. You review the survey items, which ask questions about every meal of the day and snacks eaten in between for every day of the week. On its surface, the survey seems like a good representation of what you want to test, so you consider it to have high face validity.

4. Criterion validity

Criterion validity evaluates how closely the results of your test correspond to the results of a different test.

What is a criterion?

The criterion is an external measurement of the same thing. It is usually an established or widely-used test that is already considered valid.

What is criterion validity?

To evaluate criterion validity, you calculate the correlation between the results of your measurement and the results of the criterion measurement.

If there is a high correlation, this gives a good indication that your test is measuring what it intends to measure.

Example of a criterion

A university professor creates a new test to measure applicants' English writing ability.

To assess how well the test does measure students' writing ability, she finds an existing test that is considered a valid measurement of English writing ability. She compares the results when the same group of students take both tests.

If the outcomes are very similar, the new test has a high criterion validity.

13.5.

Reliability in Quantitative Research

Types of reliability in quantitative research

Reliability tells you how consistently a method measures something. Applying the same method to the sample under the same conditions should get the same results.

If not, the method of measurement may be unreliable.

There are four main types of reliability:

1. Test-retest: The same test over time.
2. Inter-rater: The same test is conducted by different people.
3. Parallel forms: Different versions of a test designed to be equivalent.
4. Internal consistency: The individual items of a test.

1. Test-retest reliability

Test-retest reliability measures the consistency of results when you repeat the same test on the same sample at a different point in time. You use it when measuring something that you expect to stay constant in your sample.

A test of colour blindness for trainee pilot applicants should have high test-retest reliability because colour blindness is a trait that does not change over time.

Why it's important?

Many factors can influence your results at different times: for example, respondents might experience different moods, or external conditions might affect their ability to respond accurately.

Test-retest reliability can be used to assess how well a method resists these factors over time. The smaller the difference between the two sets of results, the higher the test-retest reliability.

How to measure test-retest reliability

To measure test-retest reliability, you conduct the same test on the same group of people at two different points in time. Then, you calculate the correlation between the two sets of results.

Test-retest reliability example

You devise a questionnaire to measure a group of participants' IQ (a property unlikely to change significantly over time). You administer the test two months apart to the same group of people, but the results are significantly different, so the test-retest reliability of the IQ questionnaire is low.

Improving test-retest reliability

1. . When designing tests or questionnaires, try to formulate questions, statements, and tasks that are uninfluenced by participants' moods or concentrations. When planning your data collection methods, try to minimize the influence of external factors and ensure all samples are tested under the same conditions.
2. Remember that changes can be expected to occur in the participants over time, and take these into account.

2. Inter-rater reliability

Inter-rater reliability (also called inter-observer reliability) measures the degree of agreement between different people observing or assessing the same thing.

You use it when researchers collect data by assigning ratings, scores, or categories to one or more variables.

For example, in an observational study in which a team of researchers collects data on classroom behaviour, inter-rater reliability is important. All researchers should agree on categorizing or rating different types of behaviour.

Why it’s important?

People are subjective, so different observers’ perceptions of situations and phenomena naturally differ. Reliable research aims to minimize subjectivity as much as possible so that a different researcher can replicate the same results.

When designing the scale and criteria for data collection, it’s essential to ensure that different people will rate the same variable consistently with minimal bias. This is especially important when multiple researchers collect or analyse data.

How to measure inter-rater reliability

To measure inter-rater reliability, different researchers conduct the exact measurement or observation on the same sample. Then, you calculate the correlation between their various sets of results. If all the researchers give similar ratings, the test has high inter-rater reliability.

Inter-rater reliability example

A team of researchers observe the progress of wound healing in patients. Rating scales are used to record the stages of healing, with criteria to assess various aspects of wounds.

The results of different researchers assessing the same set of patients are compared, and there is a strong correlation between all sets of results, so the test has high inter-rater reliability.

Improving inter-rater reliability

1. Clearly define your variables and the methods that will be used to measure them.
2. Develop detailed, objective criteria for how the variables will be rated, counted or categorized.
3. If multiple researchers are involved, ensure they all have the same information and training.

3. Parallel forms reliability

Parallel forms reliability measures the correlation between two equivalent test versions.

You use it when you have two different assessment tools or questions designed to measure the same thing.

Why it's important

If you want to use multiple versions of a test (for example, to avoid respondents repeating the same answers from memory), you first need to ensure that all the sets of questions or measurements give reliable results.

Parallel forms reliability

e.g.: In educational assessment, it is often necessary to create different versions of tests to ensure that students don't have access to the questions in advance. Parallel form reliability means that, if the same students take two different versions of a reading comprehension test, they should get similar results in both tests.

How to measure parallel form reliability?

The most common way to measure parallel form reliability is to produce a large set of questions to evaluate the same thing, then divide these randomly into two question sets.

The same group of respondent answers both sets, and you calculate the correlation between the results. High correlation between the two indicates high parallel form reliability.

Parallel forms reliability example

A set of questions is formulated to measure financial risk aversion in a group of respondents. The questions are randomly divided into two sets, and the respondents are randomly divided into two groups. Both groups take both tests: group A takes test A first, and group B takes test B first. The results of the two tests are compared, and the results are almost identical, indicating high parallel form reliability.

Improving parallel form reliability

Ensure that all questions or test items are based on the same theory and formulated to measure the same thing.

4. Internal consistency

Internal consistency assesses the correlation between multiple items in a test that are intended to measure the same construct.

You can calculate internal consistency without repeating the test or involving other researchers, so it's a good way of assessing reliability when you only have one data set.

Why it's important

When you devise a set of questions or ratings that will be combined into an overall score, you have to make sure that all of the items really do reflect the same thing. If responses to different items contradict one another, the test might be unreliable.

How to measure internal consistency

e.g.: To measure customer satisfaction with an online store, you could create a questionnaire with a set of statements that respondents must agree or disagree with. Internal consistency tells you whether the statements are all reliable indicators of customer satisfaction.

Two common methods are used to measure internal consistency.

1. Average inter-item correlation: For a set of measures designed to assess the same construct, you calculate the correlation between the results of all possible pairs of items and then calculate the average.
2. Split-half reliability: You randomly split a set of measures into two sets. After testing the entire set on the respondents, you calculate the correlation between the two sets of responses.

Internal consistency example

A group of respondents are presented with a set of statements designed to measure optimistic and pessimistic mindsets. They must rate their agreement with each statement on a scale from 1 to 5.

If the test is internally consistent, an optimistic respondent should generally give high ratings to optimism indicators and low ratings to pessimism indicators.

The correlation is calculated between all the responses to the “optimistic” statements, but the correlation is very weak. This suggests that the test has low internal consistency.

Improving internal consistency

Take care when devising questions or measures: those intended to reflect the same concept should be based on the same theory and carefully formulated.

Which type of reliability applies to my research?

- It’s important to consider reliability when planning your

Methodology?	Relevant reliability
Measuring a property that you expect to stay the	Test-retest
Multiple researchers making observations or	Inter-rater
Using two different tests to measure the same	Parallel forms
Using a multi-item test where all the items are	Internal consistency

intended to measure the same variable.

Constant Error (bias)

Constant error is an error that occurs in the same experimental condition every time the basic experiment is repeated – a systematic bias

Examples:

- Experimental groups always administered the treatment in the morning
- Control groups always in the afternoon
- Introduces an uncontrolled extraneous variable – time of day
- Hence, systematic or constant error

Extraneous Variables

- Variables other than the manipulated variables that affect the results of the experiment
- Can potentially invalidate the results

Sources of Constant Error

Demand Characteristics

Experimental design procedures or situational aspects of the experiment that provide unintentional hints to subjects about the experimenter's hypothesis

If this occurs, participants are likely to act in a manner consistent with the experimental treatment.

The most prominent demand characteristic is the person administering the experimental treatments.

Experimenter Bias

Effect on the subjects' behaviour caused by an experimenter's presence, actions, or comments.

Guinea Pig Effect

Effect on experimental results caused by subjects changing normal behaviour or attitudes to cooperate with the experimenter

Controlling Extraneous Variables

Blinding

- Technique used to control subjects' knowledge of whether or not they have been given the experimental treatment.
- Taste tests, placebos (chemically inert pills), etc.

Constancy of Conditions

- Subjects in experimental & control groups are exposed to identical situations except for differing conditions of the independent variable.

Order of Presentation

- If the experimental method requires that the same subjects be exposed to 2 or more experimental treatments, error may occur due to the order in which the treatments are presented
- Counterbalancing: ½ the subjects exposed to Treatment A first, then to Treatment B. Other ½ exposed to Treatment B first, then to Treatment A. Eliminates the effects of order of presentation.

Characteristics of Good Research Design

Generally, a good research design minimizes bias and maximizes the reliability of the data collected and analyzed.

The design with the minor experimental error is reported to be the best design in a scientific investigation.

Similarly, the most appropriate and efficient design yields maximum information and allows for considering different aspects of a problem.

A good research design possesses the following characteristics:

1. Objectivity

It refers to the findings related to the data collection method and the responses' scoring.

The research design should permit the measuring instruments to be fairly objective, and every observer or judge scoring the performance must precisely give the same report.

In other words, the objectivity of the procedure may be judged by the degree of agreement between the final scores assigned to different individuals by more than one independent observer.

This ensures the objectivity of the collected data, which shall be capable of analysis and interpretation.

2. Reliability and Validity

Reliability

It refers to consistency throughout a series of measurements.

For example, if a respondent gives out a response to a particular item, he is expected to give the same reaction to that item even if he is asked repeatedly.

If he changes his response to the same item, the consistency will be lost.

So, the researcher should frame the items in a questionnaire in such a way that it provides consistency or reliability.

Validity

Any measuring device or instrument is valid when it measures what it is expected to measure.

For example, an intelligence test to measure IQ should measure only intelligence and nothing else, and the questionnaire shall be framed accordingly.

3. Generalizability – 1

It means how best the data collected from the samples can be utilized for drawing certain generalizations applicable to a large group from which the sample is drawn.

Thus, a research design helps an investigator generalize his findings provided he has defined the population, selected the sample, derived

appropriate statistical analysis, etc., while preparing the research design.

Thus, a good research design is methodologically prepared and should ensure that generalization is possible.

To ensure the generalization, we should confirm that our research problem has the following characteristics:

3. Generalizability – 2

1. The problem is formulated.
2. The population is clearly defined.
3. Most appropriate sample selection techniques are used to form a proper sample.
4. Appropriate statistical analysis has been carried out.
5. The findings of the study are capable of generalizations.

Other features:

Some other essential features of a good research design are flexibility, adaptability, efficiency, economics and so on.

A good research design should minimise bias and maximise reliability and generalization.

4. Adequate Information

The most essential requirement of good research design is to provide adequate information to analyse the research problem from a broad perspective.

An ideal design should take into account important factors like:

a) Identifying the exact research problem to be studied
b) The objective of the research
c) The process of obtaining information
d) The availability of an adequate and skilled workforce and
e) The availability of sufficient financial resources for carrying out research.

Questions a critical reader should ask

1. Is the study design appropriate for the research question?
2. Is the sample size adequate and evenly distributed across groups and covariates?
3. Are the groups comparable aside from exposure status?
4. Do estimates appear precise?
5. Is causality appropriately attributed?
6. Is there potential bias due to differential loss to follow-up?
7. Is there a potential recall or interviewer bias?
8. Have alternative explanations been considered and ruled out?
9. Are results placed in the context of existing literature and inconsistencies explained?
10. . Is there external validity?

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Sample Size Calculation

- Read the following article to learn more about the methods for sample size calculation based on the epidemiological design:
- https://doi.org/10.4103/0253-7176.116232

Sample Size Determination

- In planning any investigation, we must decide how many people need to be studied to answer the study objectives. http://www.custominsight.com/articles/random-sample-calculator.asp
  - If the study is too small, we may fail to detect important effects or may estimate effects too imprecisely.
  - If the study is too large, then we will waste resources.
- In general, it is much better to increase the accuracy of data collection (by improving the training of data collectors and data collection tools) than to increase the sample size after a certain point.
- The eventual sample size is usually a compromise between what is desirable and what is feasible

Important facts to calculate the sample size

1. The reasonable estimate of the key proportion to be studied. If you cannot guess the proportion, take it as 50%.
2. The degree of accuracy required.
   - The allowed deviation from the proper proportion of the population as a whole. It can be within 1% or 5%, etc.
3. The confidence level required is usually specified as 95%.
4. The size of the population that the sample is to represent.
   - If it is more than 10,000, the precise magnitude is not likely to be very important; however, if the population is less than 10,000, then a smaller sample size may be required.
5. The difference between the two sub-groups and the value of the likelihood or the power that helps in finding a statistically significant difference.
   - It is required when there are two population groups and the interest is to compare between two means or proportions.

Response rate

- Refers to the number of people who answered the survey divided by the number of people in the sample.
- It is usually expressed in the form of a percentage.
- Example:
  - For a study, you contacted 100 people, and 40 responded.
  - You would then have 40/100 = 0.4 ≈ [40%] response rate
- Acceptable response rates vary by how the survey is administered:
  - Mail: 50% adequate, 60% good, 70% very good
  - Phone: 80% good
  - Email: 40% average, 50% good, 60% very good
  - Online: 30% average
  - Classroom paper: > 50% = good
  - Face-to-face: 80-85% good

Guidelines for maximizing response rates – 1

- Request participation from respondents in advance (when possible) and provide information about the purpose of the survey, how the results will be used, and the terms of anonymity and confidentiality.
- Give respondents a sufficient amount of time to complete the survey ….
  - For online surveys, 7-10 days is sufficient.
  - For mail surveys, provide a self-addressed stamped envelope and allow for transit time.
  - For in-class surveys, provide 1 minute per question and allow a longer response time for open-ended questions.

Guidelines for maximizing response rates – 2

- Provide clear instructions on how to complete and submit the survey when it is administered
- For mail or online surveys, send reminders during the survey period thanking the respondents who have completed the survey, while reminding others about the deadline for completing the survey. For online surveys, always provide a link to the survey and send a reminder a day before closing the survey.
- Offer an incentive for participating.

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Plan for data collection

Why should you develop a plan for data collection?

A plan for data collection should be developed so that:

- You will have a clear overview of what tasks have to be carried out, who should perform them, and the duration of these tasks;
- You can organize both human and material resources for data collection in the most efficient way and
- You can minimize errors and delays resulting from lack of planning (for example, the population not being available or data forms being misplaced).

It is likely that while developing a plan for data collection, you will identify problems (such as limited manpower) that will require modification of the proposal.

Such modifications might include adjusting the sample size or extending the period for data collection.

Stages in the data collection process

Stage 1: Permission to proceed

Stage 2–A: Logistics of data collection

Stage 2-B: Quality control

Stage 3: Data handling

Stage 1: Permission to proceed

Consent must be obtained from the relevant authorities, individuals and the community where the project will occur.

This may involve organizing meetings at the national or provincial levels, district and village levels.

For clinical studies, this may also involve obtaining written informed consent.

Stage 2: Data collection

When collecting data, you have to consider:

A. Logistics: who will collect what, when and with what resources

B. Ensuring quality: Quality control.

Stage 2-A: Logistics of data collection

WHO will collect WHAT data?

When allocating tasks for data collection, it is recommended to list them.

Then, identify who could best implement each of the tasks.

If it is clear beforehand that the research team will not be able to carry out the entire study by itself, plan to look for research assistants to assist in relatively simple but time-consuming tasks.

HOW LONG will it take to collect the data for each component of the study? – 1

Step 1: Consider;

- The time required to reach the study area (s);
- The time required to locate the study units (persons, groups, records); if you have to search for specific informants (e.g., users or defaulters of a specific service), it might take more time to locate informants than to interview them.
- The number of visits required per study unit. For some studies, it may be necessary to visit informants several times, for example, if the information needed is sensitive and can only be collected after informants are comfortable with the investigator or if observations have to be made more than once (for example, follow-up of pregnant mothers or malnourished children). The time needed for the follow-up of non-respondents should also be considered

HOW LONG will it take to collect the data for each component of the study? – 2

Step 2: Calculate the number of interviews that can be carried out per person per day

Step 3: Calculate the days needed to conduct the interviews. For example:

200 interviews are needed and 5 people in the research team,

Per day, research team can do 5 x 4 = 20 interviews,

200 ÷ 20 = 10 days for the interviews.

Step 4: Calculate the time needed for the other parts of the study (for example, 10 days)

Step 5: Determine how much time you must devote to the research.

If the team has fewer days for fieldwork than required, they would need additional research assistants to help complete this part of the study.

WHEN should the data be collected?

The type of data to be collected and the project's demands will determine the actual time needed to collect it.

Consideration should be given to:

- Availability of research team members and research assistants,
- The appropriate season (s) to conduct the fieldwork (if the problem is season-related or if data collection would be difficult during specific periods),
- Accessibility and availability of the sampled population, and
- Public holidays and vacation periods.

Stage 2–B: Ensuring quality – 1

It is essential that the data we collect are of good quality, reliable and valid.

Otherwise, we will come up with false or misleading conclusions.

Measures to help ensure good quality of data:

a) Prepare a fieldwork manual for the research team as a whole, including:

Guidelines on sampling procedures and what to do if respondents are not available or refuse to cooperate,

A clear explanation of the purpose and procedures of the study, which should be used to introduce each interview and

Instruction sheets on how to ask specific questions and record the answers.

Stage 2–B: Ensuring quality – 2

b) Select your research assistants, if required, with care. Choose assistants that are:

From the same educational level;

Knowledgeable concerning the topic and local conditions;

Not the object of study themselves; and

Not biased concerning the topic (for example, health staff are usually not the best possible interviewers for a study on alternative health practices)

c) Train research assistants carefully in all topics covered in the field work manual as well as in interview techniques and make sure that all members of the research team master interview techniques such as:

Stage 2–B: Ensuring quality – 3

neutrally asking questions;

Not showing by words or expression what answers one expects;

Not showing agreement, disagreement or surprise; and

Recording the answers precisely as they are provided, without sifting or interpreting them

d) Pre-test research instruments and procedures with the research team, including research assistants.

e) Take care that research assistants are not under too much stress (Requiring too many interviews a day; paying per interview instead of per day).

f) Arrange for ongoing supervision of research assistants. If special supervisors must be appointed in a more extensive survey, guidelines should be developed for supervisory tasks.

Stage 2–B: Ensuring quality – 4

g) Devise methods to ensure the quality of data collected by all members of the research team.

For example, quality can be assured by:

- Requiring interviewers to check whether the questionnaire is filled in completely before finishing each interview;
- Asking the supervisor to check at the end of each day during the data collection period whether the questionnaires are filled in completely and whether the recorded information makes sense and
- Having the researchers review the data during the data analysis stage to check whether the data are complete and consistent.

Stage 3: Data handling

Once the data have been collected and checked for completeness and accuracy, a straightforward procedure should be developed for handling and storing them.

1. Decide if the questionnaires are to be numbered;
2. Identify the person who will be responsible for storing the data and
3. How are they going to be stored?

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Methods of data collection

Methods of data collection

- The most commonly used methods of collecting information (quantitative data) are documentary sources, interviews and self-administered questionnaires.
- The choice of methods of data collection is based on:
  - The accuracy of the information they will yield
  - Practical considerations, such as the need for personnel, time, equipment and other facilities, about what is available
- Accuracy and “practicability” are often inversely correlated.
- A method providing more satisfactory information will often be more expensive or inconvenient.
- Therefore, accuracy must be balanced against practical considerations (resources and other practical limitations)

The use of documentary sources

- Clinical and other personal records, death certificates, published mortality statistics, census publications, etc.
- Advantages:
  - Documents can provide readymade information relatively easily
  - The best means of studying past events.
- Disadvantages:
  - Problems of reliability and validity (because the information is collected by several different persons who may have used different definitions or methods of obtaining data).
  - There is a possibility that errors may occur when the information is extracted from the records. (This may be an important source of unreliability if handwriting is difficult to read.
  - Since the records are maintained not for research purposes but for clinical, administrative or other purposes, the required information may not be recorded at all or only partly.

Interviews

- Interviews may be less or more structured.
- A public health worker conducting interviews may be armed with a checklist of topics but may not decide in advance precisely what questions he will ask.
- If his approach is flexible, the content of his questions, wording, and order vary from interview to interview. Hence, his interviews are relatively unstructured.
- On the other hand, if a more standardized technique where the wording and order of the questions are decided in advance is used, it may be a highly structured interview.

Advantages of interview

- A good interviewer can stimulate and maintain the respondent's interest by frankly answering questions.
- If anxiety is aroused (e.g., why am I being asked these questions?), the interviewer can allay it.
- An interviewer can repeat questions which are not understood and give standardized explanations where necessary.
- An interviewer can ask "follow-up" or "probing" questions to clarify a response.
- An interviewer can make observations during the interview, i.e., notes are taken on what the subject says and how he says it.

Interview: in general

- In general, apart from their expense, interviews are preferable to self-administered questionnaires, provided that skilled interviewers conduct them.
- While interviewing, the interviewer should take care not to influence the responses; he should ask his questions in a neutral manner.
- He should not show agreement, disagreement, or surprise, and should record the respondent's precise answers without shifting or interpreting them.

Self-administered questionnaires

- Self-administered questionnaire: the respondent reads the questions and fills in the answers by himself (sometimes in the presence of an interviewer who “stands by” to assist if necessary.
- Self-administered questionnaires are more straightforward and cheaper and can be administered to many persons simultaneously.
- Example:
  - To students, a school
  - They can also be sent by post, unlike interviews.
- However, they demand a certain level of education from the respondent.

Questionnaire

- A questionnaire elicits the feelings, beliefs, experiences, perceptions or attitudes of some individuals.
- It consists of questions in written form, usually sent by post to the respondents, who are expected to answer them and return them.
- Sometimes, it can be a walk-in exercise where the questionnaires are handed out in person.
- As a data collection instrument, a questionnaire can be structured, unstructured or semi-structured.

Types of Questionnaires

Structured

- It is one that has closed ended questions.
- It is restricted and calls for a “yes” or “no” answer.

Unstructured

- It is one that has open ended questions.
- It is unrestricted and calls for free response from the respondent.

Semi–structured

- It has both open and closed ended questions

Questionnaire Design – 1

- Questions may take two general forms: “Open-ended” questions, which the subject answers in his own words, or “closed” questions, which are answered by choosing from several fixed alternative responses.
- In questionnaire design, remember to:

1)Use familiar and appropriate language

2)Avoid abbreviations, double negatives, etc.

3)Avoid two elements to be collected through one question

4)Pre-code the responses to facilitate data processing

5)Avoid embarrassing and painful questions

6)Watch out for ambiguous wording

7)Avoid language that suggests a response

8)Start with more straightforward questions

Questionnaire Design – 2

9)Ask the same question to all respondents

10) Provide other or don't know options where appropriate

11) Provide the unit of measurement for continuous variables (years, months, kgs, etc.)

12) For open-ended questions, provide sufficient space for the response

13) Arrange questions in a logical sequence

14) Group questions by topic, and place a few sentences of transition between topics

15) Provide complete training for interviewers

16) Pretest the questionnaire on 20-50 respondents in the actual field situation

17) Check all filled questionnaires at the field level

18) Include "thank you" after the last question

Advantages

a) It has a low costs-even when the universe is large and widely spread geographically.

b) It is free from bias of the interviewer.

c) Respondents have adequate time to give all the answers.

d) Uniformity of the questions.

e) Standardized questions.

Disadvantages

a) Has a low rate of return.

b) Respondent's motivation is difficult to asses.

c) May present biased samples

d) It can only be used when respondents are educated and cooperative.

e) The control of the questionnaire may be lost once it is sent.

Factors affecting the rate of returned questionnaires

- Length of the questionnaire
- Reputation of the sponsoring agency
- Complexity of the questions asked
- Relative importance of the study
- Quality and design of the questionnaire
- Time of the year the questionnaire is distributed

Surveys

- Telephone Surveys
  - This is a direct talk with the interviewees over the phone
- Mail Surveys
  - This is the sending of questionnaires through the post office
- Computer Direct Surveys
  - These are interviews in which the interviewees enter their answers directly into a computer
- Email Surveys
  - Similar to computer direct surveys, email survey can be done by sending the questionnaire through the email

Pilot Testing

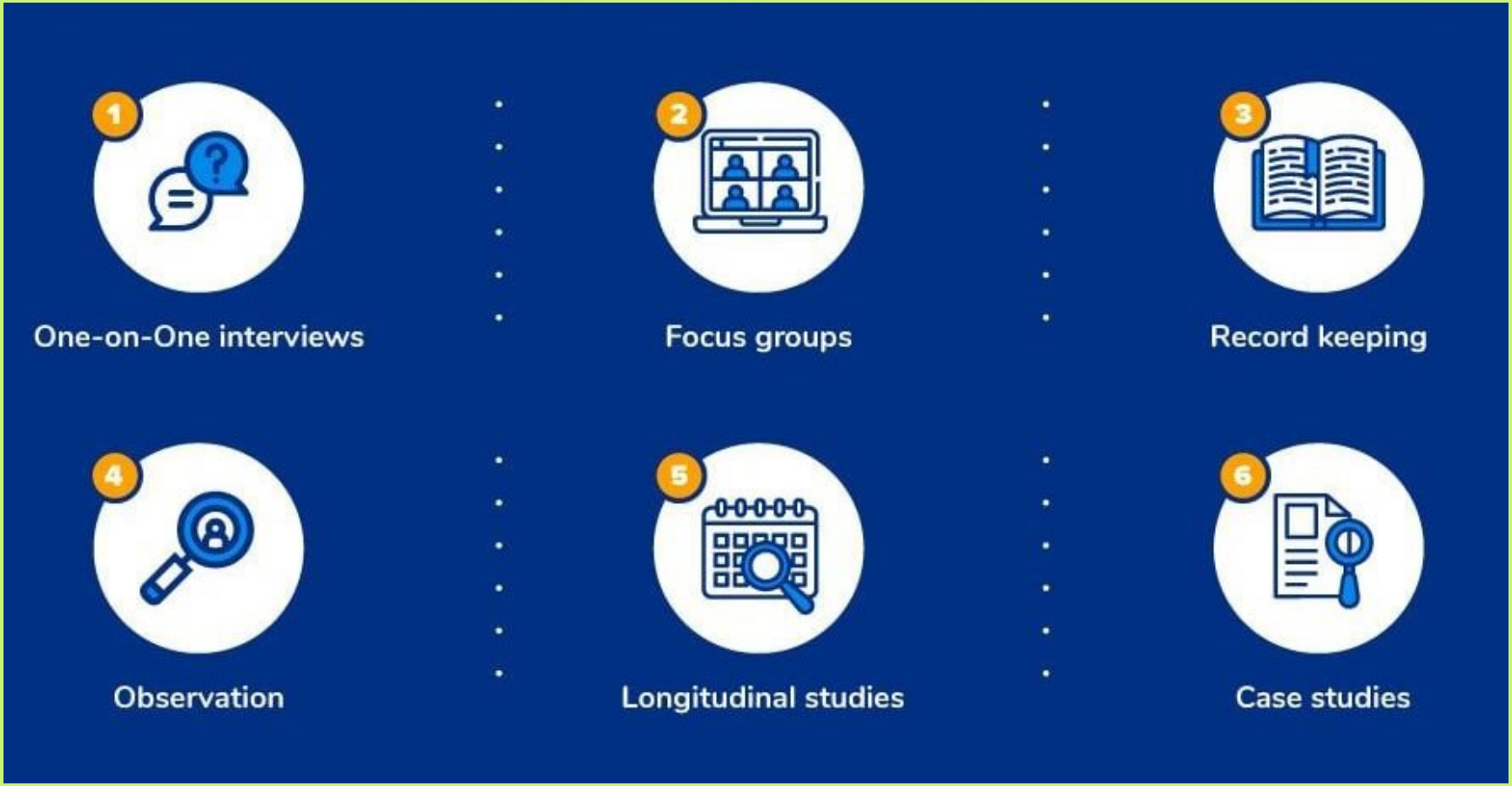
- It is conducted to detect design and instrumentation weaknesses and provide proxy data for selecting a probability sample.
- It should draw subjects from the target population and stimulate the procedures and protocols designated for data collection.
- Pretesting may rely on colleagues, respondent surrogates, or actual respondents to refine a measuring instrument.
- Pretesting may be repeated several times to refine questions, instruments or procedures.

Importance of combining different data-collection techniques

- A skilful use of a combination of different data-collection techniques can maximize the quality of the data collected and reduce the chance of bias.
- Investigators often use a combination of flexible and less flexible research techniques.
- Flexible techniques, such as loosely structured interviews using open-ended questions and focus group discussions, are called qualitative research techniques. They produce qualitative information, which is often recorded in narrative form.
- Structured questionnaires that enable the researcher to quantify pre- or post-categorized answers to questions are an example of quantitative research techniques.
- The answers to questions can be counted and expressed numerically.
- Both qualitative and quantitative research techniques are often used within a single study.

Methods of collecting qualitative data

Qualitative data can be observed and recorded (non-numerical)



Methods of collecting qualitative data

- Qualitative approaches to data collection usually involve direct interaction with individuals on a one-to-one basis or in a group setting.
- Data collection methods are time-consuming, and consequently, data is collected from smaller numbers of people than would usually be the case in quantitative approaches such as the questionnaire survey.
- The benefits of using these approaches include richness of data and deeper insight into the phenomena under study.
- Unlike quantitative data, raw qualitative data cannot be analyzed statistically.
- The data from qualitative studies often derives from face-to-face interviews, focus groups, or observation and so tends to be time-consuming to collect.
- Samples are usually smaller than with quantitative studies and are often locally based.
- Data analysis is also time-consuming and consequently expensive.

A. Qualitative interviews

- Qualitative interviews are semi-structured or unstructured.
- If the interview schedule is too tightly structured, this may prevent the phenomena under investigation from being explored in terms of either breadth or depth.
- Semi-structured interviews tend to work well when the interviewer has already identified several aspects he wants to be sure of addressing.
- The interviewer can decide what areas to cover in advance but is open and receptive to unexpected information from the interviewee.
- This can be particularly important if a limited time is available for each interview and the interviewer wants to be sure that the "key issues“ will be covered.

Semi-structured interviews

- Semi-structured interviews (sometimes called focused interviews) involve open-ended questions based on the topic areas the researcher wants to cover.
- The open-ended nature of the question defines the topic under investigation but provides opportunities for both interviewer and interviewee to discuss some issues in more detail.
- If the interviewee has difficulty answering a question or provides only a brief response, the interviewer can use cues or prompts to encourage the interviewee to consider the question further.
- In a semi-structured interview, the interviewer also has the freedom to probe the interviewee to elaborate on the original response or to follow a line of inquiry introduced by the interviewee.

Unstructured interviews – 1

- Unstructured interviews (also referred to as "depth" or "in-depth" interviews) have little structure.
- The interviewer begins by aiming to discuss a limited number of topics, sometimes as few as one or two, and frames the questions based on the interviewee's previous response.
- Although only one or two topics are discussed, they are covered in great detail.
- Subsequent questions would depend on how the interviewee responded.
- Unstructured interviews are exactly what they sound like—interviews where the interviewer wants to find out about a specific topic but has no structure, preconceived plan, or expectation as to how they will deal with the topic.

Unstructured interviews – 2

- The difference with semi-structured interviews is that in a semi-structured interview, the interviewer has a set of broad questions to ask and may have some prompts to help the interviewee. Still, the interviewer has the time and space to respond to the interviewees' responses.
- Qualitative interviews should be fairly informal. Interviewees should feel as though they are participating in a conversation or discussion rather than in a formal question-and-answer situation.
- However, achieving this informal style depends on careful planning and interview skills.

B. Focus group discussion

- Sometimes, collecting information from groups rather than from a series of individuals is preferable.
- Focus groups can be useful for obtaining certain types of information or when circumstances make it difficult to collect information using other data collection methods.
- Characteristics and uses of focus group discussions
- A focus group discussion (FGD) is a group discussion of 6-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic.
- The purpose of an FGD is to obtain in-depth information on the group's concepts, perceptions, and ideas. It aims to be more than a question-and-answer interaction.

FGD techniques can be used to:

a) Develop relevant research hypotheses by exploring in greater depth the problem to be investigated and its possible causes.

b) Formulate appropriate questions for more structured, larger-scale surveys.

c) Supplement information on community knowledge, beliefs, attitudes, and behavior already available but incomplete or unclear.

d) FGDs are not used to test hypotheses or to produce research findings that can be generalized.

Conducting a focus group discussion – 1

- Recruitment of participants:
  - Participants should be roughly of the same socioeconomic group or have a similar background about the issue under consideration.
  - The age and sex composition of the group should facilitate free discussion.
  - If we need to obtain information on a topic from several different categories of informants who are likely to discuss the issue from various perspectives, we should organize a focus group for each category.
  - For example, a group for men and a group for women.

Conducting a focus group discussion – 2

- Physical arrangements:
  - Communication and interaction during the FGD should be encouraged in every possible way.
  - Arrange the chairs in a circle.
  - Make sure the area will be quiet, adequately lighted, etc., and there will be no disturbances.
- Preparation of a discussion guide:
  - There should be a written list of topics to be covered.
  - It can be formulated as a series of open-ended questions.
- During the discussion:
  - One of the research team members should act as a "facilitator" for the focus group.
  - Other should serve as recorder.

Functions of the facilitator

1. Introduce the session
2. Encourage discussion
3. Encourage involvement
4. Listen carefully and move the discussion from topic to topic. Subtly control the time allocated to various topics to maintain interest.
5. Take time at the end of the meeting to summarize, check for agreement and thank the participants.

- In general, the facilitator should not act as an expert on the topic. His or her being there stimulates and supports discussion.

Report writing in focus group discussions – 1

- Start by describing the selection and composition of the group participants and commenting on the group process so the reader can assess the validity of the reported findings.
- Present your findings, following your list of topics and guided by the objective(s) of your FGD.
- Include questions whenever possible, particularly for key statements.
- The data collection method chosen for a study should be appropriate for the information required.
- Whether the required information is quantitative or qualitative in nature is the major consideration.

Report writing in focus group discussions – 2

- It would be time wasting to use unstructured interviews for essentially quantitative studies where information could be more efficiently collected through structured interviews or questionnaires.
- Conversely, self-completed questionnaires are generally unsuited to qualitative research: even when there is space for comments or for respondents to express ideas, the space is limited and requires respondents to have skills in articulation and literacy

C. Observation

- Not all qualitative data collection approaches require direct interaction with people.
- It is a technique that can be used when data collected through other means can be of limited value or is difficult to validate.
- For example, in interviews, participants may be asked about how they behave in certain situations, but there is no guarantee that they actually do what they say they do.
- Observing them in those situations is more reliable: it is possible to see how they behave.
- Observation can also serve as a technique for verifying or nullifying information in face-to-face encounters.

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Bias in Data Collection

BIAS in data collection is a distortion in the collected data so that it does not represent reality

1. Defective instruments

- Questionnaires with:
  - Fixed or closed questions on topics about which little is known (often asking the ‘wrong things’);
  - Open-ended questions without guidelines on how to ask (or to answer) them;
  - Vaguely phrased questions;
  - Leading questions that cause the respondent to believe one answer would be preferred over another or
  - Questions placed in an illogical order.
  - Weighing scales or other measuring equipment that are not standardized.
- These sources of bias can be prevented by carefully planning the data collection process and by pre-testing the data collection tools.

2. Observer bias

- Observer bias can quickly occur when observing or utilizing loosely structured group- or individual interviews.
- There is a risk that the data collector will only see or hear things in which (s) he is interested or will miss information critical to the research.
- Observation protocols and guidelines for conducting loosely structured interviews should be prepared, and training and practice should be provided to data collectors using both these tools.
- Moreover, it is highly recommended that data collectors work in pairs when using flexible research techniques and discuss and interpret the data immediately after collecting it.
- Another possibility—commonly used by anthropologists—is to use a tape recorder and transcribe the tape word by word.

3. Effect of the interview on the informant

- This is a possible factor in all interview situations.
- The informant may mistrust the intention of the interview and dodge certain questions or give misleading answers.
- For example, in a survey on alcoholism, you ask schoolchildren: 'Does your father sometimes get drunk?'
- Many will probably deny that he does, even if it is true.
- Such bias can be reduced by adequately introducing the purpose of the study to informants, phrasing questions on sensitive issues in a positive way, taking sufficient time for the interview, and assuring informants that the data collected will be confidential.

4. Information bias

- Sometimes, the information itself has weaknesses.
- Medical records may have many blanks or be unreadable.
- This tells something about the quality of the data and has to be recorded.
- For example, in a TB defaulter study, the percentage of defaulters with an incomplete or missing address should be calculated.
- Another common information bias is due to gaps in people's memory, called memory or recall bias.
- A mother may not remember all the details of her child's last diarrhoea episode and of the treatment she gave two or three months afterwards.
- For common diseases, it is advisable to limit the recall period, asking, for example, 'Has your child had diarrhoea over the past two weeks?

Data quality checks at the time of the interview

- There are several ways of checking the quality of data.
- Some are given below:
  - To examine the consistency of data, two or more questions are asked at the beginning and again at the end.
  - This helps to find consistency in the response.
  - Repeat the question in another form to avoid doubts
  - Use strict supervision checking on the spot, and re-interviewing
  - For data completeness, check right at the end of the interview
  - Check for odd answers given

Confidentiality of information

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Reporting of Randomized Controlled Trial

The Quality of RCT Reporting

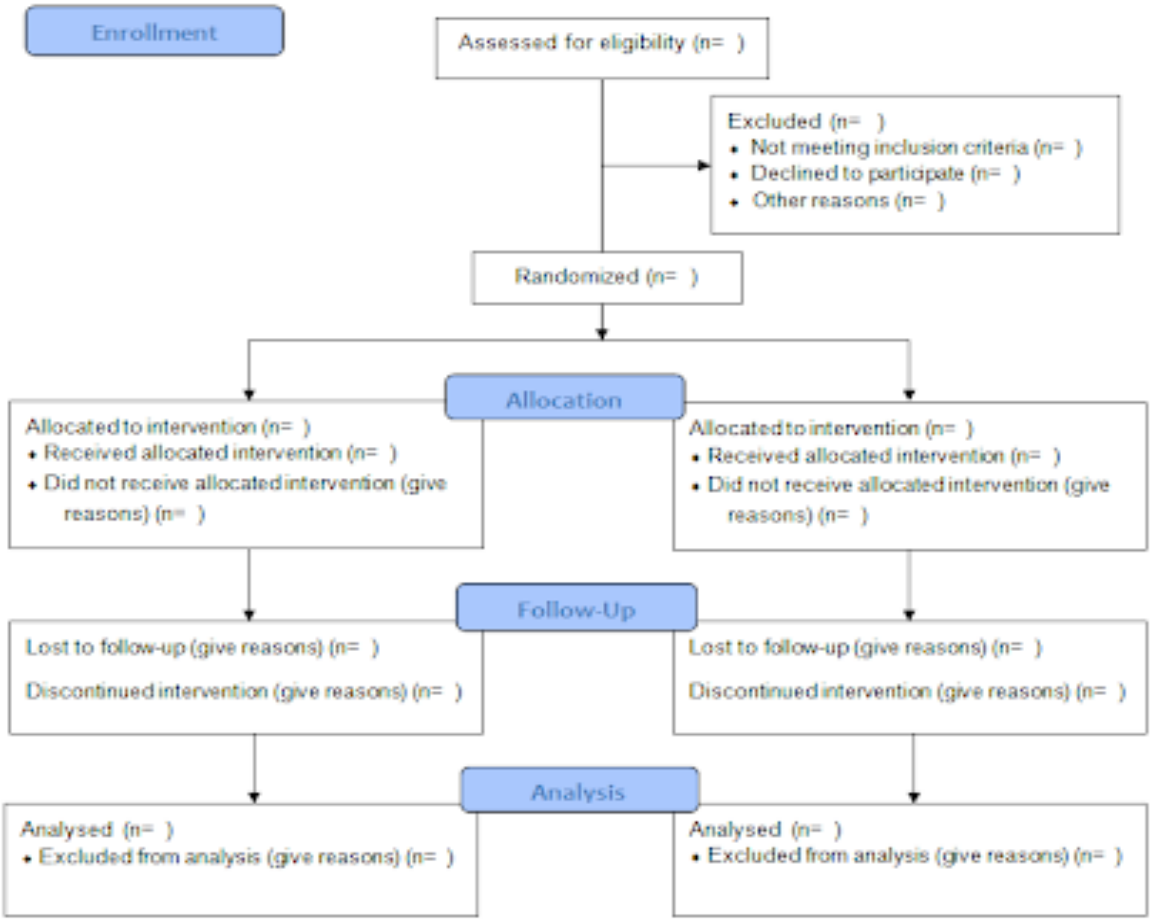
- Awareness concerning the quality of reporting randomized controlled trials and the limitations of the research methods of randomized controlled trials is growing.
- A major barrier to assessing trial quality is that, in most cases, we must rely on the information in the written report.
- A trial with a biased design, if well reported, could be judged high quality, whereas a well-designed but poorly reported trial could be considered low quality.
- Recently, efforts have been made to improve the quality of randomized controlled trials.
- In 1996, epidemiologists, biostatisticians, and journal editors published CONSORT (Consolidated Standards of Reporting Trials).

The CONSORT Statement

- CONSORT Statement is an evidence-based, minimum recommendation for reporting randomized trials.
- It offers a standard way for authors to prepare reports of trial findings, facilitating their complete and transparent reporting and aiding their critical appraisal and interpretation.
- The CONSORT Statement comprises a 25-item checklist and a flow diagram.
- The checklist items focus on reporting how the trial was designed, analyzed, and interpreted; the flow diagram displays the progress of all participants through the trial.
- The CONSORT "Explanation and Elaboration" document explains and illustrates the principles underlying the CONSORT Statement.
- In addition, extensions of the CONSORT Statement have been developed to give additional guidance for RCTs with specific designs, data and interventions.

CONSORT
TRANSPARENT REPORTING of TRIALS

CONSORT 2010 Flow Diagram



Limitations of the Research Methods Used in RCTs – 1

- The evaluation of the methodologic quality of RCTs is central to the appraisal of individual trials, the conduct of unbiased systematic reviews, and the performance of evidence-based health care.
- However, important methodologic details may be omitted from published reports, and the reporting quality is often used as a proxy measure for methodologic quality.
- High-quality reporting may hide essential differences in methodologic quality, and well-conducted trials may be poorly reported.
- As observed, “healthcare providers depend upon authors and editors to report essential methodological factors in RCTs to allow determination of trial validity (i.e., the likelihood that the trials’ results are unbiased).”

Limitations of the Research Methods Used in RCTs – 2

- The most critical limitations of research methods include the following:

1. Insufficient power: most of the trials were too small (i.e., had insufficient power to detect meaningful clinical differences), and the authors of those trials seemed unaware of these facts.
2. Poor reporting of randomization: one of the leading design features necessary to prevent bias in randomized controlled trials, was poorly reported.
3. Other limitations: inadequate randomization, failure to blind the assessors to the outcomes, and failure to follow up on all trial patients.

Websites and systems helping RCT reporting procedures

- Researchers have access to the following systems, including electronic case report forms (eCRFs), to help with all the procedures of a fully qualified RCT.
- iCReaT (clinical research and trial management system) in Korea Centers for Disease Control & Prevention (KCDC; http://icreat.nih.go.kr) is free for pre-educated and qualified researchers. There are regular education programs once a month, and some hospitals (for example, Severance Hospital) have their own educational programs. An English version is available.

Intent to Treat

- A method to correct for differential dropout rates between patients from one arm of the study and another is to analyze data by the intent to treat—that is, data are analyzed in the way patients were randomized, regardless of whether or not they received the intended intervention.
- The intent to treat correction is a form of protection against bias and strengthens the conclusions of a study.
- In the appraisal of RCTs, a clear distinction should be made between the quality of the reporting and the quality of the methodology of the trials.

Compliance – 1

- They respect participant effects and relate to compliance.
- Deviation from the protocol must be documented in all subjects, not just those on the treatment.
- It may be that a comparison or control group alters their behaviour to make them more like the treatment group in their exposure status.
- Perhaps more commonly, participants will forget or deliberately fail to take drugs, or if they have been placed on a dietary regime, they may occasionally 'break out' and deviate from the protocol.
- Measurement of compliance is essential in any clinical (dietary) trial.

Compliance – 2

- The study must be designed so that all variables of importance can be measured during the trial with sufficient precision to give a sensitive and specific (valid) indication of the level of each variable.
- Where possible, an independent measure of compliance should be used. For example, measuring changes in the levels of fatty acids in serum, red blood cells, or a fat biopsy enables the researcher to assess compliance with dietary advice to alter fat intake.
- If a dietary intervention aims to increase fiber intake, it may be possible to include a marker in the fiber diet that can subsequently be measured in fecal samples. The level in the fecal sample may give an indication of the amount of fiber supplement eaten. From our experience, it is helpful to tell participants that we are checking their compliance by taking blood or urine samples.

Compliance – 3

- It may be more difficult to measure individual compliance in a community trial, but by random sampling of subjects within each study community, it should be possible to measure at least whether subjects are aware of the community intervention and whether it has had any effect on their knowledge, attitudes, behaviour, or levels of some outcome variables.
- The efficiency of the treatment, as measured by changes in community disease rates, may be adequate. Not measuring the change in levels of the exposure that was supposed to be changed in the study may lead to a false impression of the effect of the exposure on the outcome (either positive or negative).

Compliance – 4

- The use of a run-in or familiarization period may improve compliance. It gives the subjects time to adjust to the rigours of the study protocol. However, the diet during this period should not affect the outcome measure. In theory, it should be similar to the subject's usual diet.
- If the trial is for a therapeutic agent, the run-in period should only use a placebo or usual care treatment. The run-in period should be before randomization so that any drop-outs during this period do not affect the study's internal validity.
- There are situations where it may not be possible or appropriate to have a run-in period. For example, where the experiment is assessing the effect of treatment following an acute event.

Concluding remarks – 1

- A proper RCT offers the best test of causality.
- If properly conducted, it is less likely to give a biased estimate of the effect of an exposure on an outcome.
  - Measurement error and the effects of confounding variables may still affect the outcome.
- A clearly defined aim for the research is essential and establishes the structure for the research protocol.
- The design used needs to be appropriate to the research question and population under study.

Concluding remarks – 2

- For clinical and field trials, once included in the study, all subjects should be observed and followed up in exactly the same way.
  - Poor compliance, subjects dropping out, and incomplete outcome ascertainment seriously affect the study's validity.
  - There should be sufficient subjects, observed for an adequate period of time, included in the study to allow appropriate analyses to be conducted.
- The design of community trials or community intervention studies must be paid close attention to.

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Plan for data processing and analysis

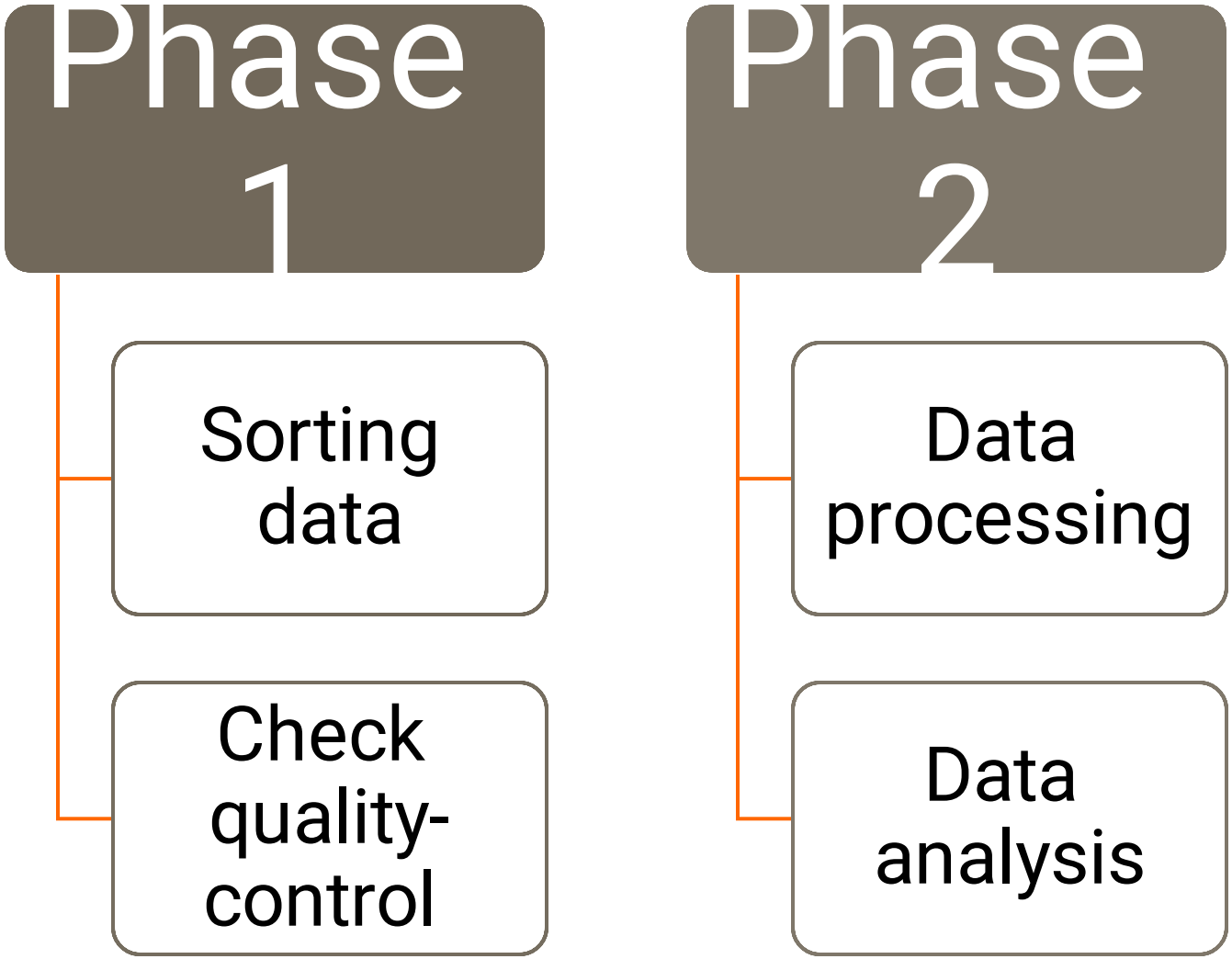
Why is it necessary to prepare a plan for processing and analysis of data? – 1

- Such a plan helps the researcher ensure that at the end of the study:
  1. all the information s/he needs has indeed been collected, and in a standardized way;
  2. s/he has not collected unnecessary data, which will never be analyzed.
- The plan for data processing and analysis must be made after careful consideration of the study's objectives and the tools developed to meet them.
- The procedures for analyzing data collected through qualitative and quantitative techniques are quite different.

Why is it necessary to prepare a plan for processing and analysis of data? – 2

- For quantitative data:
  - The starting point in the analysis is usually a description of the data for each variable for all the study units included in the sample.
  - Data processing may take place during data collection or after all data has been collected; description and analysis are usually carried out after the fieldwork has been completed.
- For qualitative data:
  - It is more a matter of describing, summarizing and interpreting the data obtained for each study unit (or each group of study units).
  - Here, the researcher starts analyzing while collecting the data so that questions that remain unanswered (or new questions that arise) can be addressed before data collection is over.

Planning for data processing and analysis



Performing quality control checks – 1

- Usually, the data have been checked in the field to ensure that all the information has been properly collected and recorded.
- However, before and during data processing, the information should be checked again for completeness and internal consistency.
- If a questionnaire has not been completed completely, you will have MISSING DATA for some of your variables.
- If there are many missing data in a particular questionnaire, you may decide to exclude the whole questionnaire from further analysis

Performing quality control checks – 2

- If an inconsistency is clearly due to a mistake made by the researcher/research assistant (for example, if a person in an earlier question is recorded as being a nonsmoker, whereas all other questions reveal that he is smoking), it may still be possible to check with the person who conducted the interview and to correct the answer.
- If the inconsistency is less clearly a mistake in recording, it may be possible (in a small-scale study) to return to the respondent and ask for clarification.
- If it is impossible to correct clearly inconsistent information, you may consider excluding this particular part of the data from further processing and analysis, as it will affect the study's validity.

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Data Management

Data Management

- Data management is an essential component of the scientific research.
- Basic knowledge of computers and statistics is required for the researcher to achieve the study's objectives.
- The availability of statistical software such as the Statistical Package of Social Science (SPSS) is of great help for researchers but should be used carefully.
- A common mistake is to call for computer commands without following a logical plan for data management. The plan includes:
  1. Coding and data entry
  2. Data processing
  3. Statistical examination

Importance of Statistics for Research

- Basic Knowledge of statistics is essential for all the research steps.
- It is not statistical analysis only but the art of data management, including three major items:

1. Study design and Data collection: This includes the sampling process, study procedures and forms preparation in the best possible way.
2. Description of the characteristics of the groups using reduction summary and presentations.
3. Data analysis: The basic procedure is to conduct cross-tabulations between two or more study variables and apply the relevant statistical tests.

Descriptive Analysis

1. Frequency and distribution (for categorical variables):
   - Describe the study variables and present your findings as:
   - Number and Percentage
   - Histogram, Charts or Pie
2. Measures of Central Tendency (for continuous variables):
   - Mean (Average)
   - Median
   - Standard deviation
   - Interquartile range
   - Confidence interval
   - Minimum and Maximum values

Cross tabulation

- Cross tabulation is a quantitative analysis of the relationship between multiple variables.
- Also known as contingency tables or cross tabs, cross-tabulation groups variables to understand the correlation between variables.
- It also shows how correlations change from one variable grouping to another.
- Cross tabulation is usually performed on categorical data, which can be divided into mutually exclusive groups.
- The region of sales for a product is an example of categorical data (as seen in the 2 x 2 Table).
- The important thing to remember about categorical data is that a categorical data point cannot belong to more than one category.

Statistical testing (t-test)

- A t-test is an inferential statistic used to determine if there is a significant difference between the means of two groups, which may be related in certain features.
- It is mainly used when the data sets, like those recorded as the outcome from flipping a coin 100 times, would follow a normal distribution and may have unknown variances.
- A t-test is used as a hypothesis testing tool, which allows testing of an assumption applicable to a population.
- A t-test looks at the t-statistic, the t-distribution values, and the degrees of freedom to determine the statistical significance.
- To conduct a test with three or more means, one must use an analysis of variance.

Explaining the t-test – 1

- Essentially, a t-test allows us to compare the average values of the two data sets and determine if they came from the same population.
- In the above examples, if we were to take a sample of students from class A and another from class B, we would not expect them to have the same mean and standard deviation.
- Similarly, samples from the placebo-fed control group and those from the drug-prescribed group should have a slightly different mean and standard deviation.
- Mathematically, the t-test takes a sample from each of the two sets and establishes the problem statement by assuming a null hypothesis that the two means are equal.

Explaining the t-test – 2

- Based on the applicable formulas, specific values are calculated and compared against the standard values, and the assumed null hypothesis is accepted or rejected accordingly.
- If the null hypothesis qualifies to be rejected, it indicates that data readings are strong and are probably not due to chance.
- The t-test is just one of many tests used for this purpose. Statisticians must additionally use tests other than the t-test to examine more variables and tests with larger sample sizes.
- For a large sample size, statisticians use a z-test.
- Other testing options include the chi-square test and the f-test.
- There are three types of t-tests, and they are categorized

Advanced Statistical Analysis

Independent	Dependent	Test
Discrete	Continuous	ANOVA, ANCOVA
Discrete	Discrete (0,1)	Logistic
Continuous	Discrete (count)	Logistic / Linear
Continuous	Discrete (Rate)	Logistic / Linear
Measurement of Risk:		

1. Relative Risk
2. Odds Ratio

Note: relative risk is not calculated in case control study.

What is Prevalence? How to Estimate?

- Prevalence is the proportion of a population with a specific characteristic in a given period.
- To estimate prevalence, researchers randomly select a sample (smaller group) from the entire population they want to describe. Using random selection methods increases the chances that the characteristics of the sample will be representative of (similar to) the characteristics of the population.
- For a representative sample, prevalence is the number of people with the characteristic of interest divided by the total number of people in the sample.
- To ensure a selected sample represents an entire population, statistical ‘weights’ may be applied. Weighting the sample mathematically adjusts its characteristics to match those of the target population.

2 x 2 Table: estimation of prevalence

	Outcome			
Exposure		Diseased	Not diseased	
	Yes	A	B	A+B
	No	C	D	C+D
		A+C	B+D	N

- Calculate the prevalence according to the formula:

$$\textbf{Prevalence} = \frac{\#\ of\ people\ in\ sample\ with\ characteristic}{Total\ \#\ of\ people\ in\ sample}$$

How is prevalence reported?

- Prevalence may be reported as a percentage (5%, or five people out of 100) or as the number of cases per 10,000 or 100,000 people. How prevalence is reported depends on how common the characteristic is in the population.
- There are several ways to measure and report prevalence depending on the timeframe of the estimate.
- Point prevalence is the proportion of a population with the characteristic at a specific point in time.
- Period prevalence is the proportion of a population with the characteristic at any point during a given period of interest. “Past 12 months” is a commonly used period.
- Lifetime prevalence is the proportion of a population who, at some point in life has ever had the characteristic.

How is prevalence different from incidence?

- Incidence measures the number of new cases of a characteristic that develop in a population in a specified time period. In contrast, prevalence is the proportion of a population with a specific characteristic in a given period, regardless of when they first developed the characteristic.
- Researchers may study incident (new) cases of illnesses to help identify causes and prevent additional cases. Incidence is often reported for infectious diseases.

How do methods impact prevalence estimates?

- A variety of methods are used to estimate the prevalence of nutrition disorders (or others).
- Differences in methodology may impact prevalence estimates. Some methodological differences that may affect comparisons between studies include but are not limited to, the populations covered, the timing of data collection, sample design, mode of data collection, instruments and surveys used, operational definitions, and estimation methods.

Calculation of Chi-square test (x^2)

- The distribution of a categorical variable in a sample often needs to be compared with the distribution of a categorical variable in another sample.
- The Chi-square Test for Independence is a statistical test commonly used to determine if there is a significant association between two variables.
- For example, a doctor might want to determine if two species of conditions are associated (are found together) in a community (obesity and type-2 diabetes).
- Just like other statistical tests, the Chi-square Test for Independence tests two hypotheses:

Null Hypotheses

- There is no significant association between variables; the variables are independent, and any association between variables is likely due to chance and sampling error.
- For example:
- There is no significant association between obesity and type-2 diabetes; the conditions are independent.

Alternative Hypothesis

- There is a significant (positive or negative) association between variables; the association between variables is likely not due to chance or sampling error.
- For example:
- There is a significant association between obesity and type-2 diabetes; the conditions are dependent.

2 x 2 Table: Chi-square test (x^2) observational values

		Type-2 diabetes		
Obesity		Diseased	Not diseased	
	Yes	40*	10	50
	No	20	30	50
		60	40	100

- Calculate the expected values according to the formula:

$$\textbf{Expected Value (*)} = \frac{Total\ Raw \times Total\ Column}{Ground\ Total}$$

$$\textbf{Expected Value (*)} = \frac{50 \times 60}{100} = 30$$

Complete the steps for all cells

2 x 2 Table: Chi-square test (x^2) expected values

		Type-2 diabetes		
Obesity		Diseased	Not diseased	
	Yes	30	20	50
	No	30	20	50
		60	40	100

- The Chi-square test statistic measured by:

$$x^2 = \sum \frac{(O-E)^2}{E}$$

$$x^2 = \frac{(40-30)^2}{30} + \frac{(10-20)^2}{20} + \frac{(20-30)^2}{30} + \frac{(30-20)^2}{20} = 16.67$$

2 x 2 Table: Chi-square test (x^2)

- Calculate the degree of freedom (DF) by this formula:

DF = (number of rows – 1) x (number of columns – 1)

- In the above mentioned example: DF = (2-1) x (2-1) = 1
- By using the Chi-square distribution table, we can get the P value.
- In this example, The p-value is significant; less than 0.001.
- The null hypothesis has been rejected, and the alternative hypothesis has been accepted.
- Comment: There is a high association between obesity and Type-2 diabetes in the selected population.
- The past example can be easily determined by SPSS, Excel, or Epi-Info software.

	Probability of exceeding the critical value						
d	0.05	0.01	0.001	d	0.05	0.01	0.001
1	3.841	6.635	10.828	11	19.675	24.725	31.264
2	5.991	9.210	13.816	12	21.026	26.217	32.910
3	7.815	11.345	16.266	13	22.362	27.688	34.528
4	9.488	13.277	18.467	14	23.685	29.141	36.123
5	11.070	15.086	20.515	15	24.996	30.578	37.697
6	12.592	16.812	22.458	16	26.296	32.000	39.252
7	14.067	18.475	24.322	17	27.587	33.409	40.790
8	15.507	20.090	26.125	18	28.869	34.805	42.312
9	16.919	21.666	27.877	19	30.144	36.191	43.820
10	18.307	23.209	29.588	20	31.410	37.566	45.315

Measurement of Risk – 1

- Risk:
  - Risk is the probability that an event will occur, e.g., an individual will become ill or die within a stated period of time or age.
  - The term is usually used concerning unfavourable events.
- Absolute Risk:
  - The absolute risk of an event is the likelihood of occurrence of that event in the population at risk.
  - Absolute Risk is measured by the incidence of an event (disease) in a population.
  - This can indicate the magnitude of the risk associated with a specific exposure.
  - The main implications of absolute risk are clinical and public health policies.

Measurement of Risk – 2

- Due to specific exposure, we can not measure the risk of the event; therefore, other measures are used.
- These measures depend on a comparison between the incidence of the event among those who are exposed to specific exposure and those who are not.

- Attributable Risk:
  - Attributable Risk (AR) is the proportion of disease incidence attributed to a specific exposure.
  - The incidence in the exposed group that is attributable to the exposure can be calculated by subtracting:

(Incidence in Exposed) – (Incidence among non-Exposed)

Measurement of Risk – 3

- The proportion of the total incidence in the exposed group that is attributable to the exposure can be calculated by:

$$\text{Proportion AR} = \frac{(Incidence\ in\ Exposed) - (Incidence\ among\ non{-}Exposed)}{(Incidence\ in\ Exposed)}$$

- **Relative Risk (RR):**
  - RR of an event is the likelihood of its occurrence after exposure to a risk variable compared to the possibility of its occurrence in a control or reference group.
  - Other Term: RR is the ratio of the incidence of an event in exposed individuals to the incidence of an event in non-exposed individuals.

Measurement of Risk – 4

$$\text{Relative Risk (RR)} = \frac{Incidence\ in\ Exposed\ population}{Incidence\ in\ Non\ Exposed\ population}$$

Relative risk score understanding:

- An RR of 1.00 means that the risk of the event is identical in the exposed and control samples.
- An RR less than 1.00 means the risk is lower in the exposed sample.
- An RR greater than 1.00 means the risk increases in the exposed sample.

Points to Note about RR – 1

- An RR of 0.3 can be expressed in plain English in many ways. It indicates that the risk is lowered to less than one-third, that the risk is reduced to 30%, that more than two-thirds decrease the risk, and that the risk is reduced by 70%. The reduction is from 1.00 (or from 100%, which is mathematically the same as 1.00).
- As a measure of effect size, an RR value is generally considered clinically significant if it is less than 0.50 or more than 2.00 if the risk is at least halved or more than doubled. However, RR values closer to 1.00 can also be considered clinically significant if the event is severe or essential to public health.

Points to Note about RR – 2

- The statistical significance of an RR value is usually provided. An RR value can be well below 1.00 or well above 1.00 and yet not statistically significant. It is possible for an RR value to be very close to 1.00 (i.e., probably not clinically significant) but statistically significant because the study was conducted on a large sample.
- RR values are accompanied by their 95% confidence intervals (CIs). The statistical significance of an RR value can be inferred from the 95% CI. If the CI includes 1.00, the RR is not statistically significant. For example, suppose the RR is 1.70 and the CI is 0.90–2.50. In that case, the elevation in risk is not statistically significant because the 1.00 (no difference in risk) lies within the confidence interval range.

Odds Ratio (OR) – 1

- Concept of odds:
  - The odds of an event is a ratio of the frequency (or likelihood) of its occurrence to the frequency (or likelihood) of its nonoccurrence.
  - Thus, the odds of rolling 6 with a die are 1 to 5 (i.e., 0.20); this contrasts with the risk of rolling 6, which is 1/6 (i.e., 0.17).
  - Similarly, the odds of tossing heads with a coin are 1 to 1 (or "50-50," or 1.00), whereas the risk of tossing heads is 1/2 or 0.5.
- Concept of odds ratio:
  - The OR is a comparison of the odds of an event after exposure to a risk factor with the odds of that event in a control or reference situation.

Odds Ratio (OR) – 2

- Calculation:
  - The OR is estimated as the odds of an event in the exposure group divided by the odds of that event in the control or reference group; the result is expressed as a ratio to denominator 1.
- When is it reasonable to compute odds ratios?
  - There are two principal situations in which the computation of ORs is justified:

1. In case-control studies, where the absolute risks and, hence, the relative risks cannot be estimated.
2. In logistic regression analyses, ORs are generated as part of the analysis.

Odds Ratio (OR) – 3

- The odds of an event can be defined as the ratio of the number of ways the event can occur to the number of ways the event can not happen (P/1-P).
- In prospective studies:
  - Odds Ratio is the ratio of the odds of exposed people developing the disease to those of non-exposed people developing the disease.
- In retrospective studies:
  - Odds Ratio is the ratio of the odds of the cases being exposed to the odds of the controls being exposed.

2 x 2 Table: Odds Ratio (OR)

		Outcome		
Exposure		Diseased	Not diseased	
	Yes	A	B	A+B
	No	C	D	C+D
		A+C	B+D	N

- **OR** can be calculated from 2 x 2 Table as the following:

$$\textbf{OR} = \frac{A}{B} \div \frac{C}{D} = \frac{A \times D}{B \times C} = \frac{AD}{BC}$$

- Note: Confidence interval is the best for interpreting the results of both RR and OR

Interpretation of OR results

- Remember the Calculation:
  - The OR is estimated as the odds of an event in the exposure group divided by the odds of that event in the control or reference group.
  - The result is expressed as a ratio to denominator 1.
- OR is a measure of the strength of association with an exposure and an outcome:
  - OR > 1 means greater odds of association with the exposure and outcome (the risk variable increases the risk of the event).
  - OR = 1 means there is no association between exposure and outcome (there is no increase or decrease in risk).
  - OR < 1 means a lower odds of association between the exposure and outcome (the risk is reduced).

Exercise

- In a prospective study, 1000 children were followed for exposure to radiation therapy.
- 400 were exposed, and 600 were not exposed.
- Among exposed children, 12 children developed blood diseases, and 6 cases developed the same diseases among non-exposed.
- Questions:
- Construct a 2x2 table and calculate:

1. Absolute Risk:
   - Incidence among all the children
   - Incidence among the Exposed children
   - Incidence among the Non-exposed children.
2. Attribute Risk; Relative Risk; Odds Ratio

Points to Note about OR – 1

- The OR is numerically different from the RR, even though both seek to compare the same risk between the groups. This is because the two statistics are based on various underlying concepts.
- ORs and RRs are similar when the event being assessed is rare in the control group (or population).
- The relationship between the OR and RR is nonlinear, but mathematical methods exist to convert ORs to RRs. ORs are interpreted in the same way as RRs.
- The statistical significance of an OR is stated along with the OR and its 95% CI. If the 95% CI for the OR includes 1.00, the OR is not statistically significant.

Points to Note about OR – 2

- How does one express an OR of 0.15 in plain English?
  - Had this been an RR, we would have said that the intervention reduced the risk by 85%.
  - Because it is an OR, we must say that for every 0.15 (or 15) persons who experience the event in the experimental group, one person (or 100 persons) will experience it in the control group.
  - That is, the odds are 15 to 100.
- Parting Note:
  - The risk of an illness is the chance, or the likelihood, that a person will develop that illness.
  - A conditional risk, in contrast, is the chance, or the likelihood, that a person who fulfils some condition will develop an illness.

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Writing Research Proposal

Definition of a Proposal

- A research proposal is also known as a work plan, an outline, a draft plan, a statement of intent or a prospectus.
- A proposal tells us what, why, how, where, and whom the
- study will be done. It must also show the benefits of the research/study.
- A research proposal is essentially a road map showing clearly the location from which the journey begins, the destination to be reached, and the methods to reach that destination.
- Every research must have a sponsor; thus, the sponsor must understand the research/project.
- The sponsor may be the lecturer for the students, firms, universities, Government, etc.

Significance of a Research Proposal

1. It serves as a summary of significant decisions in a research process
2. Serves as a written agreement between the researcher and the sponsor, e.g. on geographical scope and time scope
3. It is used to solicit funds
4. It helps in thinking through the project
5. Resolves any future misunderstandings

Characteristics of a Good Proposal

Clear/neat	Convincing	Detailed
Reliable	Data based	Realistic
Systematic in focus	Time bound	Awareness of alternative proposals

Types of Proposals

1. Academic proposal:
   - This type of proposal is found in academic institutions
2. Funding proposal:
   - This is commonly used by NGOs, to solicit funds
3. Business bidding proposal:
   - Business people use this to bid for tenders
4. Business proposal:
   - This type of proposal is also known as a business plan. Potential investors or public institutions use it to get approval

Contents of a Proposal

Preliminary Pages – 1

- 1. Title page
- 2. Declaration
- 3. Dedication
- 4. Acknowledgement
- 5. Abstract

Preliminary Pages – 2

- 6. Table of Contents
- 7. List of Tables
- 8. List of Figures
- 9. Abbreviations
- 10. Definition of Terms

Main Text of proposal

1. Chapter One: This contains the Introduction to the Study and should have the Introduction, Background of the Study, Statement of the Problem, Objectives of the Study (General and Specific), Research Questions, Significance/Importance of the Study, Limitations of the Study and Scope of the Study
2. Chapter Two: This deals with Literature Review and should have an Introduction, Review of Theoretical Literature, Critical Review, Summary and Conceptual/Theoretical Framework
3. Chapter Three: This captures the Research Design and Methodology and should give the Introduction, Research Design, Target Population, Sampling Technique, Data Collection Methods and Data Analysis/Presentation Methods
4. References: All the authors and books used in the text should appear in the references. Use the recommended style of referencing, but usually, it is the APA style of referencing

Appendices

- Letter of Introduction
- Questionnaire
- Budget
- Time Frame

Composition of The Research Project

Full Research Content		
Chapter	Description	
1	Introduction	+ References The components of a Proposal
2	Literature Review	
3	Methodology	
4	Results	
5	Discussion	
6	Conclusion and Recommendations	

Main Components of the Proposal

Chapter 1: Introduction	
Section	Description
1.1	Background (Based on Literature Review)
1.2	Statement of the Problem (Justification)
1.3	Significance (Importance of the study)
1.4	Research Question
1.5	Objectives; Main and Specific
1.6	Hypotheses
1.7	Framework of the study (Optional)
1.8	Operational definitions (Preferred)

Main Components of the Proposal

Chapter 3: Methodology	
Section	Description
3.1	Study design
3.2	Sample and sampling
3.3	Ethical consideration
3.4	Study instruments (Study Tools)

- Sampling method
- Questionnaire
- Physical measurements
- Biochemical measures (If any)

Main Components of the Proposal

Chapter 3: Methodology	
Section	Description
3.5	Quality Control
3.6	Data collection • Data handling
3.7	Statistical analysis
3.8	Research flow chart • (Optional)

As much as possible, improve your statistics skills



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Department of Nursing
Subject: Research Methodology

The End

Thank You

Read about Research Methodology in Nursing at the website of International Journal of Nursing Practice: https://onlinelibrary.wiley.com/page/journal/1440172x/homepage/research_methodology_papers.htm

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